Explicit Transcendence Criteria for Beta-Type Integrals

Integral Quotients, Cyclotomic Fields, Statistical Quantiles, and Transfers to the Gamma Function

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Abstract

We consolidate a sequence of explicit transcendence arguments for algebraic integrals, normalized hypergeometric values, Gamma quotients, and inverse distribution functions. The arithmetic inputs are classical theorems on logarithms, beta values, algebraic groups, and 1-periods. Applications include all forty-four selected entries of the DBKZ Log2 catalogue, twenty-two additional Log1 entries, six double-integral constants, and seven triple-integral quotients. Uniform families are obtained from generalized Jacobians, cyclotomic eigenspaces, and torsion-packet classifications. Polynomially weighted incomplete beta integrals admit an exact exceptional hyperplane, while related arguments classify several statistical quantiles and probability intervals. Balanced Gamma quotients at denominators 21 and 33 yield explicit number fields and counterexamples to proposed dimension bounds. We then audit transfers to individual Gamma values and Euler's constant. A positive-polynomial kernel yields an explicit fourth-power degree obstruction, and a rapidly convergent algebraically independent sequence shows why finite-point transcendence does not pass to a limit. The remaining exponential-period and mixed-Gevrey assertions are stated separately as unproved conditions. Neither the irrationality of Euler's constant nor the transcendence of $\Gamma(1/5)$ is proved.

Keywords. Transcendence; beta integrals; generalized Jacobians; hypergeometric functions; torsion; Gamma values; quantiles; Euler's constant.

Scope and organization

This manuscript organizes the mathematical content of a research dialogue into one proof document. Earlier special cases are consolidated under later uniform results where possible. Deep external theorems are cited as inputs; their application here is not presented as their invention. A public output labelled "unidentified" does not establish a literature-wide open problem. No historical-priority or external-peer-review claim is made.

Counts in the catalogues refer to exact parameter tuples, unless explicitly stated otherwise. They do not assert that all resulting real numbers are distinct or algebraically independent. Algebraic cases, endpoint exceptions, and hypotheses on positivity and denominator structure are retained. The quantitative irrationality measures and divisibility conjectures attached to the original DBKZ constructions are not established by qualitative transcendence alone.

All real powers in real integrals are taken on their positive branches. The symbols $\Gamma(z)$, $\gamma = -\Gamma'(1)$, and $\delta = \int_0^\infty e^{-t}/(1+t)\,dt$ denote different objects. The functional transcendence of Γ must not be confused with arithmetic transcendence of one value or derivative.

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Part I
One-dimensional integral quotients

1 The one-dimensional DBKZ family and all forty-four entries

For rational $a, b > -1$, real algebraic $c > -1$, and rational d, put

$$C(a,b,c,d) = \frac{\int_0^1 x^a(1-x)^b(1+cx)^{-d-1}\,dx}{\int_0^1 x^a(1-x)^b(1+cx)^{-d}\,dx}.$$

Positive real branches are understood. Both integrals converge and the denominator is positive. The final unidentified block of the DBKZ `Log2` output contains the forty-four tuples in Table 1 [1]. In the proofs below every one of these forty-four values is transcendental. "Unidentified" refers to that output, and is not a literature-wide priority claim. The count is of parameter tuples; pairwise distinctness and algebraic independence of the resulting values are not asserted.

The earlier computational status was 26 completed and 18 outstanding. The latter were rows $6, 7, 29, 30, \dots, 44$. The present organization uses the strongest common arguments, so it supersedes some earlier isolated elliptic and real-multiplication proofs. The method codes are: L, logarithms; I, incomplete second-kind periods; D, the uniform diagonal theorem; E, an explicit elliptic quotient; B, a beta quotient; and S, a simple genus-four Jacobian. All rows have final status O, meaning a proof is supplied, rather than a numerical identification.

Table 1: The exact forty-four tuples, in the source order.

Row	a	b	c	d	Method	Status
1	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	D	O
2	0	$\frac{1}{3}$	1	$\frac{2}{3}$	I	O
3	0	$\frac{1}{3}$	$\frac{1}{3}$	$\frac{2}{3}$	I	O
4	0	$\frac{2}{3}$	1	$\frac{1}{3}$	I	O
5	0	$\frac{2}{3}$	$\frac{1}{3}$	$\frac{1}{3}$	I	O
6	$\frac{1}{3}$	$\frac{1}{3}$	$\frac{1}{3}$	$\frac{1}{3}$	D	O
7	$\frac{2}{3}$	$\frac{2}{3}$	$\frac{1}{3}$	$\frac{2}{3}$	D	O
8	0	$\frac{1}{2}$	1	$\frac{1}{4}$	I	O
9	0	$\frac{1}{2}$	1	$\frac{3}{4}$	I	O
10	0	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{4}$	I	O
11	0	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{3}{4}$	I	O
12	0	$\frac{1}{2}$	$\frac{1}{4}$	0	L	O
13	0	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{4}$	I	O
14	0	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{3}{4}$	I	O
15	0	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{2}$	I	O
16	0	$\frac{3}{4}$	$\frac{1}{4}$	$\frac{1}{2}$	I	O
17	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{2}$	D	O
18	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{4}$	E	O
19	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	D	O
20	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{1}{4}$	$\frac{3}{4}$	D	O
21	0	$\frac{1}{5}$	$\frac{1}{5}$	0	L	O
22	0	$\frac{1}{5}$	$\frac{1}{5}$	$\frac{3}{5}$	I	O
23	0	$\frac{2}{5}$	$\frac{1}{5}$	0	L	O
24	0	$\frac{2}{5}$	$\frac{1}{5}$	$\frac{1}{5}$	I	O
25	0	$\frac{3}{5}$	$\frac{1}{5}$	0	L	O
26	0	$\frac{3}{5}$	$\frac{1}{5}$	$\frac{4}{5}$	I	O
27	0	$\frac{4}{5}$	$\frac{1}{5}$	0	L	O
28	0	$\frac{4}{5}$	$\frac{1}{5}$	$\frac{2}{5}$	I	O
29	$\frac{1}{5}$	$\frac{1}{5}$	1	$\frac{3}{5}$	B	O

Row	a	b	c	d	Method	Status
30	$\frac{1}{5}$	$\frac{1}{5}$	$\frac{1}{5}$	$\frac{1}{5}$	D	O
31	$\frac{1}{5}$	$\frac{1}{5}$	$\frac{1}{5}$	$\frac{3}{5}$	S	O
32	$\frac{1}{5}$	$\frac{1}{5}$	$\frac{2}{5}$	$\frac{3}{5}$	S	O
33	$\frac{1}{5}$	$\frac{2}{5}$	1	$\frac{4}{5}$	B	O
34	$\frac{1}{5}$	$\frac{2}{5}$	$\frac{1}{5}$	$\frac{4}{5}$	S	O
35	$\frac{1}{5}$	$\frac{2}{5}$	$\frac{2}{5}$	$\frac{4}{5}$	S	O
36	$\frac{1}{5}$	$\frac{3}{5}$	1	$\frac{2}{5}$	B	O
37	$\frac{1}{5}$	$\frac{3}{5}$	$\frac{1}{5}$	$\frac{2}{5}$	S	O
38	$\frac{1}{5}$	$\frac{3}{5}$	$\frac{2}{5}$	$\frac{2}{5}$	S	O
39	$\frac{2}{5}$	$\frac{2}{5}$	$\frac{1}{5}$	$\frac{2}{5}$	D	O
40	$\frac{2}{5}$	$\frac{4}{5}$	1	$\frac{3}{5}$	B	O
41	$\frac{2}{5}$	$\frac{4}{5}$	$\frac{1}{5}$	$\frac{3}{5}$	S	O
42	$\frac{2}{5}$	$\frac{4}{5}$	$\frac{2}{5}$	$\frac{3}{5}$	S	O
43	$\frac{3}{5}$	$\frac{3}{5}$	$\frac{1}{5}$	$\frac{3}{5}$	D	O
44	$\frac{4}{5}$	$\frac{4}{5}$	$\frac{1}{5}$	$\frac{4}{5}$	D	O

1.1 The fixed-cycle period input

For an abelian variety $A/\overline{\mathbb{Q}}$ and a rational cycle γ, let B_γ be the smallest abelian subvariety with $\gamma \in H_1(B_\gamma, \mathbb{Q})$, in the rational isogeny category. The subgroup theorem for 1-periods implies

$$\int_\gamma \alpha = 0 \quad \Longleftrightarrow \quad \alpha|_{B_\gamma} = 0, \qquad \alpha \in H^1_{\mathrm{dR}}(A/\overline{\mathbb{Q}}). \tag{1.1}$$

This is the pure-abelian specialization of [2, Theorem 9.7]. In particular, if A is simple and $\gamma \neq 0$, two independent algebraic de Rham classes have $\overline{\mathbb{Q}}$-linearly independent periods on γ, so their ratio is transcendental. The corresponding fixed-cycle statement also appears in [3, Satz 1].

A useful refinement avoids assuming simplicity. Let ρ be a finite-order automorphism such that every nonzero eigenspace of $H^0(A, \Omega^1)$ is one-dimensional. Suppose a rational cycle γ has nonzero period against every holomorphic eigenform. Then evaluation on γ is injective on each eigenspace of $H^1_{\mathrm{dR}}(A)$. Indeed, $B = \sum_j \rho^j B_\gamma$ is ρ-stable. The kernel of restriction of holomorphic forms to B is stable and cannot contain an eigenform, by the nonzero-period hypothesis. Thus $B = A$. An eigenclass with zero γ-period vanishes on each $\rho^j B_\gamma$ by (1.1); the surjective addition map from their product to A then forces that class to vanish. This criterion concerns the specified cycle, not arbitrary periods of a reducible Jacobian.

1.2 All twenty-one entries with first parameter zero

Suppose first that $0 < b, d < 1$ and $c > 0$ is algebraic. Set

$$m = \frac{c}{1+c}, \quad T = B_m(b, 1-d), \quad H = m^b(1-m)^{1-d}.$$

Two integrations of derivatives give

$$I := \int_0^m t^b(1-t)^{-d}\,dt = \frac{bT - H}{b+1-d}, \quad J := \int_0^m t^b(1-t)^{-d-1}\,dt = \frac{(1+c)H - bT}{d}.$$

For example, differentiate $t^b(1-t)^{1-d}$ for the first identity and $t^{b+1}(1-t)^{-d}$ for the second. Substituting $t = c(1-x)/(1+c)$ in the original integrals yields

$$C(0, b, c, d) = \frac{b+1-d}{d(1+c)} \frac{(1+c)H - bT}{bT - H}. \tag{1.2}$$

The denominator is positive up to a positive factor, and the inner fractional-linear determinant is $-bcH \neq 0$. Consequently the target is algebraic if and only if T is algebraic.

We use the following precise incomplete-period consequence of [2, Theorem 13.9, Summaries 14.11 and 14.16, Corollary 14.22]: on a curve with simple Jacobian, the integral of a nonexact algebraic differential of the second kind along one path between distinct algebraic points is transcendental. The second-kind hypothesis matters: it excludes the third-kind exception. One way to obtain the statement is that an algebraic integral would, after subtracting an exact differential, give a nonexact second-kind differential with zero path integral. The cited vanishing criterion forces the Abel–Jacobi logarithm of the path to be zero. The Abel–Jacobi map is injective on points of a positive-genus curve, contradicting the distinct endpoints.

Write $b = p/q$, $1 - d = r/q$ and normalize

$$w^q = t^p(1-t)^r, \qquad \omega = \frac{w\,dt}{t(1-t)}.$$

With $g_0 = \gcd(q,p)$, $g_1 = \gcd(q,r)$ and $g_\infty = \gcd(q,p+r)$, the genus and local orders are

$$2g - 2 = q - g_0 - g_1 - g_\infty, \qquad \operatorname{ord}_{0,1,\infty}\omega = \left(\frac{p}{g_0} - 1, \frac{r}{g_1} - 1, \frac{q-p-r}{g_\infty} - 1\right).$$

The ten beta types occurring here are listed below; repeated values of c account for the sixteen $d > 0$ entries. In this table $g_\infty = 1$. Deck symmetry kills all residues at infinity. If $k = \operatorname{lcm}(g_0, g_1)$, the positive lift δ of $[0,1]$ satisfies $\int_{\delta - \tau^k\delta}\omega = (1-\zeta_q^k)B(b,1-d) \neq 0$. Hence ω is nonexact. For $q = 4$ one must use $k = 2$, not $k = 1$, in the four displayed types.

b	d	$(q;p,r)$	g	$\operatorname{ord}_\infty\omega$	k
1/3	2/3	$(3;1,1)$	1	0	1
2/3	1/3	$(3;2,2)$	1	-2	1
1/2	1/4	$(4;2,3)$	1	-2	2
1/2	3/4	$(4;2,1)$	1	0	2
1/4	1/2	$(4;1,2)$	1	0	2
3/4	1/2	$(4;3,2)$	1	-2	2
1/5	3/5	$(5;1,2)$	2	1	1
2/5	1/5	$(5;2,4)$	2	-2	1
3/5	4/5	$(5;3,1)$	2	0	1
4/5	2/5	$(5;4,3)$	2	-3	1

The genus-one Jacobians are simple. The genus-two Jacobians are simple by the three-point prime-cover criterion [2, Corollary 19.16], since $5 \not\equiv 1 \pmod 3$. The lifted path $[0,m]$ has distinct algebraic endpoints. The incomplete-period theorem and (1.2) prove all sixteen cases.

For the other five entries $d = 0$, write $b = p/q \in (0,1)$ and $A = (1+c)/c > 1$. Substitution $1 - x = u^q$ and polynomial division give

$$C(0,p/q,c,0) = \frac{p+q}{c}\left(AL - \frac{1}{p}\right), \quad L = \int_0^1 \frac{u^{p-1}}{A-u^q}\,du > 0.$$

Partial fractions at the roots $\rho^q = A$ yield

$$L = -\frac{1}{q}\sum_{\rho^q = A}\rho^{p-q}\operatorname{Log}(1 - 1/\rho).$$

All coefficients and logarithm arguments are algebraic, and the continued branches are unambiguous along the integration path. Baker's theorem makes this nonzero logarithmic linear form transcendental; its affine transformation is transcendental as well. For instance, $C(0,1/2,1/4,0) = 12\sqrt{5}\,\operatorname{atanh}(1/\sqrt{5}) - 12$.

The same proof yields useful infinite extensions. The $a = 0$ result holds for every positive algebraic c when $d = 0$, $0 < b < 1$ is rational; when $0 < b = d < 1$ is rational; when $b, d \in$

$\{1/4, 1/2, 3/4\}$; and when $b = r/p, d = v/p$ with $1 \le r, v < p$ and an odd prime $p \not\equiv 1 \pmod 3$. For $b = d = r/q$, substitute $t = u^q/(1+u^q)$ to reduce T to $q\int_0^{c^{1/q}} u^{r-1}/(1+u^q)\,du$, again a positive logarithmic linear form. Off-diagonal prime cases use the same simplicity criterion; the remaining quarter cases are genus one by the displayed genus formula.

1.3 The complete rational diagonal

Theorem 1.1. *For every rational $h > -1$ and real algebraic $c > -1$, $C(h, h, 0, h) = 1$, while $C(h, h, c, h)$ is transcendental whenever $c \neq 0$.*

Proof. First let $c > 0$ and $h = k/n$ in lowest terms, with $n \ge 2$. Put $A = 2 + 4/c$ and consider the genus-$n-1$ curve

$$X_A:\ v^2 = t^{2n} - At^n + 1, \qquad t^n = \frac{cx(1-x)}{1+cx}, \quad v = 2x + t^n - 1.$$

The deck map $\rho(t, v) = (\zeta_n t, v)$ has distinct eigenvalues on the holomorphic basis $\omega_j = t^{j-1}dt/v$, $1 \le j < n$. The positive lift δ of $0 \le x \le 1$ runs from $(0, -1)$ to $(0, 1)$, through the first positive branch point. The identity $dx/[x(1-x)] = -n\,dt/(tv)$ gives

$$-nc^{-j/n}\int_\delta \omega_j = \int_0^1 [x(1-x)]^{j/n-1}(1+cx)^{-j/n}\,dx > 0.$$

Thus the closed cycle $\gamma = \delta - \rho\delta$ has nonzero period against every ω_j. The character-support criterion following (1.1) applies even to reducible or CM fibers.

Set $g = c^{-h}t^k$, $\alpha = g\,dx$, and $\beta = g\,dx/(1+cx)$. Both have the nontrivial deck character ζ_n^k. Their possible poles lie at the four deck-fixed branch points $0, 1, -1/c, \infty$, so invariance of residues makes them second-kind differentials. They are regular at $0, 1$, since the local order of $g\,dx$ there is $n(h+1) - 1 \ge 0$. If $a\alpha + b\beta$ is exact, character averaging gives a primitive $gR(x)$ with $R \in \overline{\mathbb{Q}}(x)$, satisfying

$$R' + h\left(\frac{1}{x} - \frac{1}{1-x} - \frac{c}{1+cx}\right)R = a + \frac{b}{1+cx}. \tag{1.3}$$

No finite pole of R is possible: at a branch point the leading coefficient involves a nonintegral residue, and at an ordinary point differentiation raises the pole order. Consequently R is polynomial, and regularity on the right at $0, 1$ forces $R(0) = R(1) = 0$. A nonzero R of degree $M \ge 2$ gives a leading term of degree $M-1$ with coefficient $(M+h)\,\mathrm{lc}(R) \neq 0$, whereas the right side is bounded at infinity. Thus $R = 0$ and $a = b = 0$. The two de Rham classes are independent in one character. Their γ-periods are $(1 - \zeta_n^k)$ times the positive-path integrals. Characterwise injectivity proves that their ratio is transcendental.

For integral $h = m \ge 0$, $C_0(c) = \log(1+c)/c$. If $m \ge 1$, put $z = (c+2)/c > 1$. Rodrigues' formula and integration by parts for the Legendre functions give

$$C_m(c) = \frac{2(m+1)}{mc}\frac{Q_m(z)}{Q_{m-1}(z) - zQ_m(z)}.$$

Here $Q_0(z) = \frac{1}{2}\log(1+c)$, $Q_m = P_mQ_0 - W_{m-1}$ with $P_m, W_{m-1} \in \mathbb{Q}[z]$, and the common three-term recurrence gives $P_mQ_{m-1} - P_{m-1}Q_m = 1/m$. Hence the displayed ratio is a nonconstant fractional-linear function of the transcendental $\log(1+c)$ over $\overline{\mathbb{Q}}$. Positivity of the original denominator excludes a pole. This proves the integer case. Finally, for $-1 < c < 0$, reflection $x \mapsto 1 - x$ gives

$$C_h(c) = \frac{1}{1+c}C_h\left(-\frac{c}{1+c}\right),$$

reducing to positive algebraic parameters. □

There is also an inverse-value consequence. For every real $h > -1$, $C_h(c)$ decreases strictly from 1 to 0 on $(0, \infty)$. Indeed, with expectation for density proportional to $[x(1-x)/(1+cx)]^h$ and $Y = (1+cx)^{-1}$, the involution $x \mapsto (1-x)/(1+cx)$ gives $\mathbb{E}(Y^2) = 1/(1+c)$. Differentiation then yields

$$cC_h' = \frac{h+1}{1+c} - C_h - hC_h^2, \qquad C_h'(0) = -\frac{1}{2}.$$

At a first zero of C_h' this equation would imply $C_h'' = -(h+1)/[c(1+c)^2] < 0$, a contradiction. Moreover $C_h^2 \le \mathbb{E}(Y^2) \to 0$. Consequently the unique positive solution of $C_h(c) = r$, for rational $h > -1$ and algebraic $0 < r < 1$, is transcendental. Equivalently, the diagonal theorem classifies the contiguous quotient

$$\frac{{}_2F_1(h+1, h+1; 2h+2; -c)}{{}_2F_1(h, h+1; 2h+2; -c)}.$$

1.4 The additional genus-two entry has an elliptic quotient

For every positive algebraic c, $Q(c) = C(1/2, 1/2, c, 1/4)$ is transcendental; row 18 is $c = 1/4$. Here is an explicit reduction which avoids any simplicity assertion for the genus-two Jacobian. Put

$$s = (1+c)^{1/4}, \quad A = s^2 + s^{-2}, \quad B = A + 2 > 4, \quad g = \sqrt{B}, \quad W(t) = \sqrt{t(At^2 - t^4 - 1)}.$$

Substitutions $U = \sqrt{1+cx}$ and $t = U/s$ transform the denominator and numerator into

$$D = \frac{2s^{7/2}}{c^2} \int_{1/s}^{s} W(t)\, dt, \qquad N = \frac{2s^{3/2}}{c^2} \int_{1/s}^{s} W(t) t^{-2}\, dt.$$

Use $W(1/t) = W(t)/t^3$, then $u = t + t^{-1}$ and $V = W(t)(t-1)/t^2$ on $1 < t < s$. The elliptic equation and transformed integrals are

$$V^2 = (u-2)(B-u^2), \quad D_t = \int_2^g (B-u^2)(u^2-u-1)\frac{du}{V}, \quad N_t = \int_2^g (B-u^2)\frac{du}{V}.$$

Set $\Omega = \int_2^g du/V > 0$, $H = \int_2^g u\, du/V$ and $R = H/\Omega$. For $f = (u-2)(B-u^2)$, direct polynomial identities give

$$(B-u^2)(u^2-u-1) = \tfrac{2}{21}\{B(B-5) + (B-2)u\} + 2r_D' f + r_D f',$$
$$B - u^2 = \tfrac{2}{3}(B - 2u) + 2r_N' f + r_N f',$$

where $r_D = (1-2B)/21 + u/7 + u^2/7$ and $r_N = 1/3$. The exact terms are $2d(r_D V)$ and $2d(r_N V)$ and vanish on integration because V vanishes at both endpoints. Hence

$$Q(c) = \frac{7}{s^2} \frac{B - 2R}{B(B-5) + (B-2)R}. \tag{1.4}$$

The elliptic de Rham basis $du/V, u\, du/V$ evaluates on the nonzero cycle obtained by returning on the other sheet to $2\Omega, 2H$. The fixed-cycle theorem makes R transcendental. The determinant of (1.4), before the nonzero scalar, is $3B(4-B) \neq 0$; its denominator is a nonzero multiple of $D_t > 0$. This proves the claim. If desired, the conventional parameter-form elliptic integrals give $R = (g+2)E(\mu)/K(\mu) - g$, $\mu = -(g-2)/(g+2)$.

1.5 Four complete-beta quotients

For rows 29, 33, 36 and 40 the following exact identities hold:

row	P	Q	C
29	$B(3/5, 1/5)$	$B(11/10, 1/5)$	$(2P - 2Q)/(-2P + 3Q)$
33	$B(7/10, 1/5)$	$B(6/5, 1/5)$	$(2P - 2Q)/(-3P + 4Q)$
36	$B(3/5, 3/5)$	$B(11/10, 3/5)$	$(-4P + 7Q)/(2P - 2Q)$
40	$B(9/10, 7/5)$	$B(7/5, 7/5)$	$(-104P + 192Q)/(65P - 96Q)$

Their fractional-linear determinants are $2, 2, -6, -2496$. Positivity of the original denominator excludes poles. For verification, define

$$J_k(u,v) = \int_0^1 t^v (1-t^2)^u (1+t)^{-k}\,dt, \quad P_0 = \tfrac{1}{2}B((v+1)/2, u), \quad Q_0 = \tfrac{1}{2}B((v+2)/2, u).$$

Rationalization and integration of $[t^{v+1}(1-t)(1-t^2)^{u-1}]'$ give

$$J_1 = P_0 - Q_0, \qquad J_2 = \frac{u+v}{u-1}P_0 - \frac{2u+v}{u-1}Q_0.$$

For rows 29 and 33 use $x = (1-t)/(1+t)$, obtaining $D = 2^{b-d+1}J_2(a,b)$, $N = 2^{b-d}J_1(a,b)$. Row 36 directly gives $D = J_1(b,a)$, $N = J_2(b,a)$. For row 40 the substitution gives $D = 2^{b-d+1}J_3(a,b)$, $N = 2^{b-d}J_2(a,b)$. A certificate avoiding any analytic-continuation issue in the last case uses $h = t^{4/5}(1-t^2)^{2/5}$ and

$$R_2 = \frac{5t(1-t)(4t+5)}{6(1+t)}, \qquad R_3 = -\frac{5t(1-t)(16t^2+33t+11)}{96(1+t)^2}.$$

Direct differentiation yields $h/(1+t)^2 = h(-13/2 + 12t) + (hR_2)'$ and $h/(1+t)^3 = h(65/32 - 3t) + (hR_3)'$. The primitives vanish at both endpoints and establish the listed beta formulas.

For rational arguments, put $\epsilon_k(r,s) = \{kr\} + \{ks\} - \{k(r+s)\}$. The beta-quotient theorem of Wolfart–Wüstholz [3, Satz 4] says that unequal profiles, with k coprime to a common denominator and nonintegral arguments and sums, force the beta quotient to be transcendental. Integer shifts only multiply a beta value by a nonzero algebraic number. The following single profile discrepancies therefore suffice:

row	$(r,s)_P$	$(r,s)_Q$	k	$(\epsilon_k(P), \epsilon_k(Q))$
29	$(3/5, 1/5)$	$(1/10, 1/5)$	3	$(1,0)$
33	$(7/10, 1/5)$	$(1/5, 1/5)$	3	$(0,1)$
36	$(3/5, 3/5)$	$(1/10, 3/5)$	1	$(1,0)$
40	$(9/10, 2/5)$	$(2/5, 2/5)$	1	$(1,0)$

This proves all four assertions through the nondegenerate fractional-linear transformations.

1.6 Eight simple genus-four Jacobians: exact certificates

The eight remaining rows arise from the normalizations

$$X_{r,s,t,c}: \quad y^5 = x^r(1-x)^s(1+cx)^t,$$
$$(r,s,t) \in \{(1,1,2),(1,3,3),(2,4,2),(1,2,1)\}, \qquad c \in \{1/5, 2/5\}.$$

All four branch exponents, including infinity, are nonzero modulo five; the curves have genus four. Good reductions at 31 for $c = 1/5$ and at 11 for $c = 2/5$ have the following exact point counts:

c	(r,s,t)	p	N_1	N_2	N_3	N_4
1/5	all four	31	19	949	29434	923449
2/5	$(1,1,2),(1,3,3)$	11	9	69	1164	14369
2/5	$(2,4,2),(1,2,1)$	11	9	129	1374	14369

These integers can be obtained without period approximations. In $\mathbb{F}_{p^k}$, there is one point above each of the three finite branch points and one at infinity. Every other fiber contains five points precisely when the nonzero radicand is a fifth power. Thus

$$N_k = 4 + 5\#\{x \in \mathbb{F}_{p^k} : x \notin \{0, 1, -1/c\},\ [x^r(1-x)^s(1+cx)^t]^{(p^k-1)/5} = 1\}. \tag{1.5}$$

Suitable irreducible presentations, for degrees $2, 3, 4$ respectively, are

p	$f_2(T)$	$f_3(T)$	$f_4(T)$
11	T^2+1	T^3+4T^2+1	T^4+4T^3+1
31	T^2+1	T^3+3T^2+1	T^4+T^3+1

Enumeration using these fields and (1.5) completely specifies the finite count certificate.

Newton's identities and reciprocity give, in table order,

$$\begin{aligned}
P_1(X) &= X^8 - 13X^7 + 78X^6 - 401X^5 + 2195X^4 \\
&\quad - 12431X^3 + 74958X^2 - 387283X + 923521, \\
P_2(X) &= X^8 - 3X^7 - 22X^6 + 19X^5 + 335X^4 \\
&\quad + 209X^3 - 2662X^2 - 3993X + 14641, \\
P_3(X) &= X^8 - 3X^7 + 8X^6 - X^5 - 85X^4 \\
&\quad - 11X^3 + 968X^2 - 3993X + 14641.
\end{aligned}$$

Writing $P_i(X) = X^4 F_i(X + p/X)$ gives

$$\begin{aligned}
F_1(U) &= U^4 - 13U^3 - 46U^2 + 808U - 719, \\
F_2(U) &= U^4 - 3U^3 - 66U^2 + 118U + 1061, \\
F_3(U) &= U^4 - 3U^3 - 36U^2 + 98U - 19.
\end{aligned}$$

Each reduces modulo two to the irreducible polynomial $U^4 + U^3 + 1$. The Weil bounds make its four roots real. None equals $\pm 2\sqrt{p}$, as it has degree four. For a root u, the corresponding Frobenius root π is nonreal and solves $X^2 - uX + p = 0$; hence $[\mathbb{Q}(\pi) : \mathbb{Q}] = 8$. Each P_i is irreducible.

To certify absolute simplicity, let

$$R_i(Z) = \mathrm{Res}_X(P_i(X), P_i(ZX)), \qquad Q_i(Z) = R_i(Z)/(Z-1)^8.$$

The roots of R_i are ratios of Frobenius roots. There is no nontrivial root-of-unity ratio: Q_i has degree 56 and is divisible by none of Φ_n with $2 \le n \le 6272$, $\varphi(n) \le 56$. This finite check is exhaustive, since $\varphi(n) \ge \sqrt{n/2}$. The latter bound follows by multiplying $p^{2-a}/(p-1)^2$ over $p^a \| n$: each odd-prime factor is at most one and the factor at two is at most two.

For a short reproducible certificate, all the relevant gcds are already one after reduction modulo 10007. The following complete SageMath fragment performs the certificate directly from the three displayed polynomials; it needs no external table or data file.

```
ZX.<X> = PolynomialRing(ZZ)
polys = [
 X^8-13*X^7+78*X^6-401*X^5+2195*X^4
   -12431*X^3+74958*X^2-387283*X+923521,
 X^8-3*X^7-22*X^6+19*X^5+335*X^4
   +209*X^3-2662*X^2-3993*X+14641,
 X^8-3*X^7+8*X^6-X^5-85*X^4
   -11*X^3+968*X^2-3993*X+14641]
FZ.<z> = PolynomialRing(GF(10007))
RX.<x> = PolynomialRing(FZ)
orders = [n for n in range(2,6273) if euler_phi(n)<=56]
for P in polys:
    p = RX(P.list())
```

```
r = FZ(p.resultant(p(z*x)))
q, rem = r.quo_rem((z-1)^8)
assert rem == 0 and q.degree() == 56 and q(1) != 0
for n in orders:
    assert q.gcd(FZ(cyclotomic_polynomial(n).list())) == 1
```

The same assertions were checked with exact integer resultants and rational cyclotomic division. This modular version is sufficient: a monic integral cyclotomic divisor over $\mathbb{Q}$ would remain a divisor modulo 10007.

Thus every positive power of π retains eight distinct conjugates. The absolute-simplicity criterion [4, Proposition 3 and Lemma 4] applies. A nontrivial geometric idempotent in the characteristic-zero endomorphism algebra would specialize to one in the geometric special fiber, so these characteristic-zero Jacobians are absolutely simple as well.

For the eight targets, write $g = x^a(1-x)^b(1+cx)^{-d}$ and consider $g\,dx$ and $g\,dx/(1+cx)$ on the corresponding cover. Their branch signatures are

(a,b,d)	(r,s,t)	rows	$(c = 1/5, 2/5)$
$(1/5,1/5,3/5)$	$(1,1,2)$		$31, 32$
$(1/5,2/5,4/5)$	$(1,2,1)$		$34, 35$
$(1/5,3/5,2/5)$	$(1,3,3)$		$37, 38$
$(2/5,4/5,3/5)$	$(2,4,2)$		$41, 42$

Deck symmetry kills residues. Exactness of a linear combination would require a rational S solving

$$S' + \left(\frac{a}{x} - \frac{b}{1-x} - \frac{dc}{1+cx}\right) S = A + \frac{B}{1+cx}.$$

As before, nonintegral residues force S to be polynomial and divisible by $x(1-x)$. A nonzero polynomial of degree $k \geq 2$ has a nonzero leading term of degree $k-1$, since $k + a + b - d > 0$, contrary to the bounded right-hand side. Thus the classes are independent. The closed cycle obtained by subtracting a deck translate of the positive interval has nonzero denominator period. Simplicity and the fixed-cycle period theorem prove all eight ratios transcendental. Together with the previous subsections this completes every row of Table 1.

1.7 A congruence family and a rational-weight classifier

The finite certificates yield an infinite family and an exact decision procedure for normalized rational weights. Let $a, b, d \in \frac{1}{5}\mathbb{Z}$, $a, b > -1$, and $c > 0$ algebraic. Define

$$g = x^a(1-x)^b(1+cx)^{-d}, \quad D = \int_0^1 g\,dx, \quad M = \frac{\int_0^1 xg\,dx}{D}, \quad \mathcal{L}S = S' + \left(\frac{a}{x} - \frac{b}{1-x} - \frac{dc}{1+cx}\right) S.$$

Assume that the ordered residues

$$(5a, 5b, -5d, 5(d-a-b)) \pmod 5 \tag{1.6}$$

are a permutation of $(h, h, h, 2h)$ for some $h \in \{1, 2, 3, 4\}$. Assume also that a number field containing c has a finite place with one of the following residue conditions:

$$\operatorname{char} k(v) = 31,\ \bar{c} = 25 \in \mathbb{F}_{31}; \qquad \text{or} \qquad \operatorname{char} k(v) = 11,\ \bar{c} = 7 \in \mathbb{F}_{11}. \tag{1.7}$$

For rational $c = m/n$ in lowest terms these are respectively $31 \nmid n$, $m \equiv 25n \pmod{31}$, or $11 \nmid n$, $m \equiv 7n \pmod{11}$.

Theorem 1.2. *Under (1.6)–(1.7), both $C(a,b,c,d)$ and M are transcendental. If $R \in \overline{\mathbb{Q}}(x)$ has poles only in $\{0,1,-1/c,\infty\}$ and gR is integrable on $(0,1)$, there are unique $u,v \in \overline{\mathbb{Q}}$ and $S \in \overline{\mathbb{Q}}(x)$ such that*

$$R = u + vx + \mathcal{L}S. \tag{1.8}$$

Moreover

$$\frac{\int_0^1 gR\,dx}{D} = u + vM,$$

which is algebraic if and only if $v = 0$. All coefficients and the rational primitive can be found by a finite linear system over an explicit coefficient number field. For $H(c) = {}_2F_1(d, a+1; a+b+2; -c)$, the three numbers $1, H(c), H'(c)$ are linearly independent over $\overline{\mathbb{Q}}$.

Proof. The curve with function field $\overline{\mathbb{Q}}(x,g)$ is a cyclic fivefold cover with four fully ramified branch points. Integer shifts of exponents and common multiplication of the residues by a unit modulo five do not change its geometric function field, up to a twist. Twelve of the sixteen signatures in (1.6) reduce to the four triples of Section 1.6. For the four with the exceptional exponent at infinity, $x \mapsto -1/(cx)$ swaps $0 \leftrightarrow \infty$ and $1 \leftrightarrow -1/c$, moving that exponent to zero; they then reduce to $(1,3,3,3)$ by a scalar modulo five. Condition (1.7) makes the geometric special fiber one of the certified absolutely simple curves. The four branch sections remain distinct and the characteristic is not five, so specialization proves that the characteristic-zero Jacobian A is simple of dimension four.

Every $gR\,dx$ with the stated pole support is of the second kind by deck symmetry. Its nontrivial deck eigenspace in de Rham cohomology has dimension two: rational cohomology has no invariant part, since the quotient has genus zero, and its characteristic polynomial is Φ_5^2. The forms $g\,dx$ and $xg\,dx$ are independent. Indeed, a rational solution of $\mathcal{L}S = A_0 + A_1 x$ must be polynomial and vanish at all three finite branch points. If nonzero it has degree $k \geq 3$, while its leading image has degree $k-1 \geq 2$ and coefficient proportional to $k + a + b - d \neq 0$. Nonvanishing follows from the nonintegral residue at infinity. This contradicts the right-hand side. The same argument with right-hand side $A_0 + A_1/(1+cx)$ proves independence of $g\,dx$ and $g\,dx/(1+cx)$.

The first independent pair forms a basis of the eigenspace, proving existence and uniqueness of u, v in (1.8). An exact difference has a rational primitive gS by character averaging. The primitive S is unique, since $\mathcal{L}S = 0$ would make gS constant, contradicting the nontrivial fifth-degree function-field extension unless $S = 0$.

The closed cycle $\gamma = \sigma - \tau\sigma$, where σ is the positive lift of $[0,1]$, has nonzero denominator period $(1-\zeta_5)D$. The simple fixed-cycle theorem makes C and M transcendental. Integrating (1.8) around this cycle proves the integral formula, with no separate assumption on endpoint values of the rational primitive.

For the last assertion put $B = B(a+1, b+1) > 0$, so $D = BH(c)$. The signature condition implies $a + b \notin \mathbb{Z}$. The value B, up to a nonzero algebraic deck factor, is a second-kind period of the genus-two three-point cover with function field $\overline{\mathbb{Q}}(x, x^a(1-x)^b)$. Its Jacobian has no simple factor isogenous to the simple fourfold A. On their product the subgroup theorem separates the two isogeny components. Consequently B, D, N are linearly independent over $\overline{\mathbb{Q}}$, where $N = \int_0^1 g/(1+cx)\,dx$. Finally

$$N = D + \frac{c}{d}D', \qquad C = 1 + \frac{c}{d}\frac{H'}{H}$$

transfers that independence to $1, H, H'$. The nonintegrality of d ensures $d \neq 0$. This conclusion is linear, not algebraic, independence. □

To implement the finite algorithm, let m_j denote the pole order of R at each finite branch point, and put $M_\infty = \max(1, \deg_\infty R)$. Every possible S in (1.8) has pole order at most $\max(m_j - 1, 0)$ there and polynomial part of degree at most $M_\infty + 1$. These bounds follow from the nonintegral

local residues and the nonzero leading coefficient at infinity. Substitute the corresponding partial-fraction ansatz and solve for its coefficients and u, v. Existence follows from the cohomology argument, rather than from an experimental fit.

The theorem applies, for example, to $(a, b, d) = (1/5, 1/5, 3/5)$ and

$$c = 1/8, \quad c = 7 + 11n, \quad c = 25 + 31n \quad (n \geq 0), \quad c = \sqrt{5}.$$

For the last value use the place $(11, \sqrt{5} - 7)$. The classifier also detects genuine algebraic exceptions among nonconstant weights:

$$\mathcal{L}[x(1-x)(1+cx)] = (1+a) + \{(a+2)(c-1) - b - dc\}x - c(a+b+3-d)x^2.$$

Its normalized integral is zero. Thus the theorem does not claim transcendence for every nonconstant rational weight.

Part II
Third-kind periods and uniform families

2 Third-kind periods and twenty-two values in the Log1 output

This section concerns the slice with fourth parameter zero,

$$F_h(c) = C(h,h,c,0) = \frac{\displaystyle\int_0^1 [x(1-x)]^h (1+cx)^{-1}\,dx}{B(h+1,h+1)} = {}_2F_1(1,h+1;2h+2;-c), \qquad h,c>-1. \tag{2.1}$$

All real fractional powers are positive. This slice differs from $C(h,h,c,h)$ considered elsewhere in the paper. The results below are applications of existing period and torsion theorems, supplemented by explicit algebraic and local calculations. They are not assertions of historical priority for the corresponding special-function values.

2.1 An exact catalogue of the public entries

Table 2 lists the two initial third-kind targets and the twenty subsequent targets in the public DBKZ `Log1` output [5]. Each row is specified exactly by (h,c); decimal approximations are included only to identify the source entry. Source lines refer to the text snapshot examined in this work. All twenty-two rows are proved transcendental below. Only rows 3 and 4 are asserted here to be algebraically independent of one another. None is a duplicate of a fourth-parameter-nonzero entry from the earlier `Log2` catalogue. The absence of an identification in a computer-generated list does not establish that the transcendence of an entry was previously unknown.

Table 2: Twenty-two catalogued values $F_h(c) = C(h,h,c,0)$ in Log1.

No.	h	c	Approximate value	Source line
1	1/3	1/3	0.86196583755412212…	124
2	2/3	1/3	0.86121914488406966…	194
3	1/3	1	0.68815073461701577…	133
4	2/3	1	0.68472526272182017…	203
5	1/4	1/4	0.89204558868041487…	232
6	1/4	1/2	0.80934781900111164…	238
7	1/4	1	0.68921689098230447…	244
8	3/4	1/4	0.89134153873904941…	338
9	3/4	1/2	0.80724556697156396…	344
10	3/4	1	0.68403133820765647…	350
11	1/5	1/5	0.91131086128452211…	416
12	2/5	1/5	0.91107655008701731…	469
13	3/5	1/5	0.91088694358067085…	575
14	4/5	1/5	0.91073036736539899…	628
15	1/7	1/7	0.93459900758455854…	955
16	2/7	1/7	0.93449760261342181…	979
17	3/7	1/7	0.93441123426490072…	1003
18	4/7	1/7	0.93433678973658805…	1045
19	5/7	1/7	0.93427195982569432…	1069
20	6/7	1/7	0.93421499535486150…	1093
21	1/4	1/8	0.94210870681369684…	1176
22	3/4	1/8	0.94190133226732776…	1336

2.2 The period-theoretic input

We use the following consequence of the Huber–Wüstholz theorem on linear relations among 1-periods [2, Theorem 9.7]. If G is a semi-abelian variety over $\overline{\mathbb{Q}}$ and $v \in H_1(G(\mathbb{C}), \mathbb{Q})$, there is a connected semi-abelian subgroup $B \subseteq G$, understood up to isogeny, such that v comes from $H_1(B, \mathbb{Q})$ and the annihilator of v in $H^1_{\mathrm{dR}}(G)$ is the kernel of restriction to B. This is a statement about the full algebraic de Rham realization, including second-kind classes; it is not restricted to invariant differentials on the semi-abelian variety.

Lemma 2.1. *Let $0 \to T \to G \to A \to 0$ have a finite-order automorphism s. Assume A has no proper nonzero s-stable abelian subvariety, $X^*(T) \otimes \mathbb{Q}$ is an irreducible s-representation, and the extension does not split up to isogeny. If v has nonzero projection to $H_1(A, \mathbb{Q})$, evaluation at v is injective on every eigenspace of s^* in $H^1_{\mathrm{dR}}(G)$.*

Proof. If an eigenclass ω annihilates v, the subgroup B just described annihilates its restriction. The connected sum $H = \sum_j s^j B$ also annihilates ω. Its nonzero s-stable image in A is all of A. Irreducibility of the torus representation gives $(H \cap T)^0 = 0$ or T. The former case makes H abelian and $H \to A$ an isogeny, splitting the pullback extension, contrary to the hypothesis. The latter makes $H = G$, whence $\omega = 0$. In this last step restriction to the factors of the surjection $\prod_j s^j B \to G$ detects first de Rham cohomology. □

For a curve X punctured at a reduced set S, the generalized Jacobian has torus character group $\mathbb{Z}[S]^0$, and its extension class is the map

$$\mathbb{Z}[S]^0 \longrightarrow \mathrm{Jac}(X)(\overline{\mathbb{Q}}), \qquad D \longmapsto [D] \tag{2.2}$$

[2, Lemma 4.24]. A non-torsion value of this map obstructs splitting up to isogeny. When the map is injective after tensoring with $\mathbb{Q}$, G has no nonzero algebraic character, up to isogeny. If its abelian quotient is elliptic, no proper connected subgroup can then surject onto that quotient: its quotient in G would otherwise be a positive-dimensional torus and provide such a character.

2.3 Prime denominators, including the exceptional fifth-order case

Theorem 2.2. *Let $h = m/p > -1$ be in lowest terms, where $p \geq 5$ is prime. Then $F_h(c)$ is transcendental for every real algebraic $c > -1$, $c \neq 0$. At $c = 0$ it equals 1.*

Proof. First take $c > 0$. On the smooth projective curve

$$X : \; w^p = x(1-x), \qquad \rho(x, w) = (x, \zeta_p w),$$

the points above $0, 1, \infty$ are totally ramified, so $g(X) = (p-1)/2$. The quotient by ρ is the projective line. Consequently $H_1(J, \mathbb{Q})$, for $J = \mathrm{Jac}(X)$, is the irreducible $(p-1)$-dimensional Φ_p representation. In particular, J has no proper nonzero ρ-stable abelian subvariety, and $1 - \rho$ is an isogeny. This reasoning does not require absolute simplicity of J.

Let S be the p points above $x = -1/c$, and set

$$\alpha = w^m \, dx, \qquad \beta = \frac{w^m \, dx}{1 + cx}, \qquad G = J(X \setminus S).$$

At $0, 1$, both differentials have order $m + p - 1 \geq 0$. At infinity, their residues vanish because that point is fixed by ρ and their eigenvalue is $\zeta_p^m \neq 1$. Thus α is of the second kind, whereas β has nonzero residues $w(P)^m/c$ at every $P \in S$. Both represent classes in $H^1_{\mathrm{dR}}(G)$; the meromorphic description of this cohomology permits second-kind poles away from S. The rational character group of its torus is $\mathbb{Q}[S]^0$, again the irreducible Φ_p representation.

Let σ be the positive-real-branch path from $x = 0$ to 1 and $v = \sigma - \rho\sigma$. Its endpoints are fixed by ρ, so it is a closed cycle, and

$$\int_v \alpha = (1 - \zeta_p^m)B(h+1, h+1), \qquad F_h(c) = \frac{\int_v \beta}{\int_v \alpha}.$$

Its image in $H_1(J, \mathbb{Q})$ is nonzero: the holomorphic form dx/w^{p-1} has period $(1 - \zeta_p)$ times a positive integral on σ. If $F_h(c) = a \in \overline{\mathbb{Q}}$, the nonzero eigenclass $\beta - a\alpha$ annihilates v. The subgroup argument in Lemma 2.1 forces either a contradiction or an abelian subgroup $H \subset G$ isogenous onto J, containing v in rational homology.

Use infinity as the Abel–Jacobi base point and write $\iota(P) = [P - \infty]$. For $P \in S$, $[P - \rho P] = (1 - \rho)\iota(P)$. Hence non-torsion of $\iota(P)$ rules out the abelian-subgroup alternative via (2.2). For $p \geq 7$, Grant–Shaulis [6, Theorem 1.1] classify the entire geometric torsion packet on this curve: its points have $x = 0, 1, 1/2, \infty$. None has $x = -1/c < 0$. For $p = 5$, Boxall–Grant [7, Theorem 0.2] add the points with $x = (1 \pm \sqrt{5})/2$. The sole remaining positive parameter is $c = \varphi = (1 + \sqrt{5})/2$.

For this exception put $t = -w$, $s = (1 - \sqrt{5})/2$, and

$$q(t) = -\sqrt{5}\, t^2 + \sqrt{5}\, t + s, \qquad q(t)^2 - q(t) - t^5 = -(t-1)^5.$$

On $t^5 = x^2 - x$, define, with indices modulo 5,

$$f_j = x - q(\zeta_5^{-j} t), \qquad P_j = (\zeta_5^j, s).$$

The identity $(x - q(t))(x + q(t) - 1) = (t-1)^5$ shows that $\operatorname{div}(f_j) = 5(P_j - \infty)$: the second factor is $-\sqrt{5}$ at P_0, and x has a pole of order 5 at infinity. Thus $\chi = f_2/f_1$, after multiplication by a nonzero constant, defines an algebraic character of G.

We compute its period without numerical winding tests. On σ, write $u = [x(1-x)]^{1/5}$, so

$$f_j = x - s + \sqrt{5}(u^2 \zeta_5^{-2j} + u\zeta_5^{-j}).$$

Every f_j starts and ends on the positive real axis. Let n_j be its continuous argument change divided by 2π. Since $f_0 > 0$ and $\operatorname{Im} f_1 < 0$ in the interior, $n_0 = n_1 = 0$. Moreover,

$$\operatorname{Im} f_2 = \sqrt{5}\, u \sin(\pi/5)(\varphi u - 1).$$

There are two interior real-axis crossings, at $x_- < 1/2 < x_+$ with $x(1-x) = \varphi^{-5}$. At these crossings $\operatorname{Re} f_2 = x - k$, where $k = \sqrt{5} - 2$ and $k(1-k) = 2\varphi^{-5}$. Therefore $x_- < k < x_+$: the first crossing passes from below to above the negative real axis; the second crosses the positive axis. It follows that $n_2 = -1$. Since $\rho^* f_j = f_{j-1}$, the endpoint logarithms cancel and

$$\int_v d\log\chi = 2\pi i(n_2 - 2n_1 + n_0) = -2\pi i \neq 0. \tag{2.3}$$

Every character restricts trivially to an abelian variety. Thus an abelian H containing v would force the left side of (2.3) to vanish. This excludes the last alternative.

Finally substitution $x \mapsto 1 - x$ gives, for all $h, c > -1$,

$$F_h(c) = \frac{1}{1+c} F_h\!\left(-\frac{c}{1+c}\right). \tag{2.4}$$

For $-1 < c < 0$ the new parameter is positive algebraic, which completes the proof. □

An inverse consequence also follows. For an exponent in Theorem 2.2 and algebraic $r \in (0, 1)$, the unique positive solution of $F_h(c) = r$ is transcendental. Indeed, F_h decreases strictly from 1 to 0 on $[0, \infty)$, since

$$F_h'(c) = -\frac{1}{B(h+1, h+1)} \int_0^1 \frac{x^{h+1}(1-x)^h}{(1+cx)^2}\, dx < 0,$$

and dominated convergence gives its limit at infinity. An algebraic solution would contradict the theorem.

2.4 Quarter exponents and all rational parameters

Theorem 2.3. *If $h > -1$ and $4h$ is odd, then $F_h(c)$ is transcendental for every rational $c > -1$, $c \neq 0$.*

Proof. For $c > 0$ use the elliptic curve $w^4 = x(1-x)$ with $\rho(x,w) = (x, iw)$. The coordinates $U = x/w^2$, $V = x/w^3$ give $V^2 = U^3 + U$ and $\rho(U,V) = (-U, iV)$. The four points above $x = -1/c$ are $P, iP, -P, -iP$, where

$$U^2 = -\frac{1}{c+1}, \qquad V^2 = \frac{c}{c+1}U, \qquad V^4 = -\frac{c^2}{(c+1)^3}.$$

We first prove uniformly that P is non-torsion. Exact duplication gives

$$2P = (AU, BV), \qquad A = -\frac{(c+2)^2}{4c}, \qquad B = \frac{(c+2)(c^2-4c-4)}{8c^2}. \tag{2.5}$$

Write $c = m/n$ in lowest positive terms. An odd prime ℓ dividing $(m+2n)(m^2-4mn-4n^2)$ divides none of $m, n, m+n$, as reduction modulo each shows. Over $\mathbb{Q}_\ell$, the polynomial

$$Z^4 + \frac{m^2 n}{(m+n)^3}$$

has unit discriminant. Hence every completion of $\mathbb{Q}(V)$ above ℓ is unramified, and U, V are units. The elliptic model has good reduction, since its discriminant is -64. If $r = v_\ell(B) > 0$, then $2P$ reduces to a nonzero point of order two, while duplication yields

$$v_\ell(U(4P)) = -2r, \qquad v_\ell(V(4P)) = -3r.$$

Thus $4P \neq O$ lies in the formal subgroup. This subgroup is torsion-free over an unramified extension of $\mathbb{Q}_\ell$ for odd ℓ: the formal logarithm has expansion $z + \sum_{j\geq 2} a_j z^j/j$, with integral a_j, and for $v(z) \geq 1$ every higher term has valuation greater than $v(z)$. Its kernel is zero and its target is an additive characteristic-zero field. This proves non-torsion whenever such an odd prime exists.

If both numerator factors have no odd prime divisor, m must be even; write $m = 2r$ with n odd. The absence of odd divisors in $r+n$ forces r odd and $r+n = 2^s$. Also $v_2(r^2 - 2rn - n^2) = 1$, so $r^2 - 2rn - n^2 = \pm 2$. Substitution yields $r^2 = 2^{2s-1} \pm 1$. For $s = 1$ this gives $r = n = 1$. For $s \geq 2$, one sign is impossible modulo 8; factoring the other as $(r-1)(r+1) = 2^{2s-1}$ gives only $r = 3$, $s = 2$, $n = 1$. The exceptional parameters are therefore $c = 2, 6$. They admit the following exact triplication certificates:

c	$U(3P)/U$	$V(3P)/V$	ℓ
2	1/9	−11/27	11
6	169/121	−1443/1331	13

At the indicated prime, $3P$ reduces to a nonzero point of order two and $6P \neq O$ lies in the torsion-free formal subgroup. Thus P is non-torsion for every positive rational c.

There is a small representation-theoretic point in applying the period theorem. The torus of $G = J(E \setminus \{P, iP, -P, -iP\})$ has rank three, with rational character types $\Phi_2 \oplus \Phi_4$. The connected image T_2 of $1 + \rho^2$ on G lies in the torus, because $\rho^2 = [-1]$ on E; it is one-dimensional of type Φ_2. The quotient $\overline{G} = G/T_2$ has a rank-two torus of type Φ_4. Its classifying map sends the anti-pair divisors $[P] - [-P]$ and $[iP] - [-iP]$ to $2P$ and $2iP$. This map is injective over $\mathbb{Q}$: any nonzero $a + b\rho \in \mathbb{Q}(i)$ is an isogeny, since its product with $a - b\rho$ is $[a^2 + b^2]$, and cannot send P to torsion. Hence $\overline{G}$ has no nonzero character, and no proper connected subgroup of it surjects onto E.

Put $k = 4h$ and use $\alpha = w^k dx$, $\beta = w^k dx/(1+cx)$. At $0, 1$ their order is $k + 3 \geq 0$. The two points at infinity are fixed by ρ^2, whereas the eigenvalue of these forms under ρ^2 is -1, so both

residues vanish there. Their cohomology classes have eigenvalue $i^k \in \{i, -i\}$ and descend uniquely to $H^1_{\mathrm{dR}}(\overline{G})$, because the removed torus has only the eigenvalue -1. For $v = \sigma - \rho\sigma$ on the positive branch,

$$\int_v \alpha = (1 - i^k)B(h+1, h+1) \neq 0.$$

The class of α comes from E, so the projected cycle is nonzero. The nonzero residues of β at the punctures prove that the two classes are independent. The subgroup supplied by the Huber–Wüstholz theorem must surject onto E, hence must be all of $\overline{G}$. Its period-evaluation map is therefore injective. An algebraic ratio $F_h(c)$ would contradict this injectivity. Equation (2.4) handles the negative rational parameters. □

For the eight public quarter-denominator rows, the duplication factors in (2.5) give particularly short certificates:

c	A	B	ℓ
1/8	$-289/32$	$-4879/64$	41
1/4	$-81/16$	$-711/32$	79
1/2	$-25/8$	$-115/16$	23
1	$-9/4$	$-21/8$	7

The indicated valuation of B is one. For each fixed positive rational c, the four unnormalized integrals $D_{1/4}, N_{1/4}(c), D_{3/4}, N_{3/4}(c)$, where $D_h = B(h+1, h+1)$ and $N_h(c) = D_h F_h(c)$, are linearly independent over $\overline{\mathbb{Q}}$. They form a de Rham basis, with two independent classes in each of the i and $-i$ eigenspaces, and the whole period-evaluation map just used is injective. This conclusion does not assert algebraic independence of the two ratios.

2.5 The two initial third-kind values at $c = 1/3$

Theorem 2.4. *The numbers $F_{1/3}(1/3)$ and $F_{2/3}(1/3)$ are transcendental. The corresponding four unnormalized numerator and denominator integrals are linearly independent over $\overline{\mathbb{Q}}$.*

Proof. The curve $w^3 = x(1-x)$ becomes $V^2 = U^3 + 16$ under $U = -4w$, $V = 8x - 4$. Write $\tau(w) = \zeta_3 w$. The three punctures above $x = -3$ are $P, \tau P, \tau^2 P$, with

$$P = (U_0, -28), \quad U_0^3 = 768, \qquad 2P = \left(\frac{10}{49}U_0, -\frac{1628}{343}\right).$$

At every place above 7, the field $\mathbb{Q}(U_0)$ is unramified, because $X^3 - 768$ has discriminant prime to 7; the elliptic model also has good reduction. The coordinates of $2P$ have valuations $-2, -3$. The same formal-logarithm argument as above proves that P is non-torsion.

The generalized Jacobian has rank-two torus with character group $\mathbb{Q}[\{P, \tau P, \tau^2 P\}]^0$. Its classifying map is injective: for rational a, b, the divisor $a(\tau P - P) + b(\tau^2 P - P)$ maps to

$$\big((-a - 2b) + (a - b)\tau\big)P.$$

Since $1 + \tau + \tau^2 = 0$, this is a nonzero rational CM endomorphism unless $a = b = 0$, and cannot send the non-torsion P to torsion. Thus the generalized Jacobian has no nonzero character. The cycle $v = \sigma - \tau\sigma$ has nonzero elliptic projection, as its integral against $w^k dx$ is $(1 - \zeta_3^k)B(1 + k/3, 1 + k/3)$ for $k = 1, 2$. Its period-evaluation map is injective by the subgroup theorem. For each k, the classes $w^k dx$ and $w^k dx/(1 + x/3)$ are independent: the first has zero residues, the second has residues $3w(Q)^k$ at the punctures. Infinity has zero residue by its nontrivial τ-character. The two pairs lie in distinct eigenspaces and form a basis of the four-dimensional de Rham cohomology. The two ratio statements and the four-period linear independence follow. □

2.6 Explicit evaluations and independence at $c = 1$

Theorem 2.5. *Put $r = \sqrt[3]{2}$, $\Omega = B(1/3, 1/3)$, and $\Psi = B(2/3, 2/3)$. Then*

$$L_1 := F_{1/3}(1) = 5 - \frac{10}{3r^2}\Psi, \tag{2.6}$$

$$L_2 := F_{2/3}(1) = \frac{21}{2} - \frac{7}{3r}\Omega. \tag{2.7}$$

The pair L_1, L_2 is algebraically independent over $\overline{\mathbb{Q}}$, and

$$\overline{\mathbb{Q}}(L_1, L_2) = \overline{\mathbb{Q}}(\pi, \Gamma(1/3)^3). \tag{2.8}$$

Proof. For $w = [x(1-x)]^{1/3}$ substitute $y = (1-x)/w$. This decreases from ∞ to 0 as x increases from 0 to 1, and direct differentiation gives

$$y^3 + 4 = \frac{(1+x)^2}{x}, \qquad \frac{dx}{(1+x)w} = -\frac{3\,dy}{y^3+4}, \qquad \left(\frac{1}{1+x} - \frac{1}{2}\right)\frac{dx}{w^2} = -\frac{3}{2}\frac{y\,dy}{y^3+4}.$$

Using $\int_0^\infty dt/(1+t^3) = \int_0^\infty t\,dt/(1+t^3) = 2\pi/(3\sqrt{3})$ and scaling $y = r^2 t$ yields

$$T_2 := \int_0^1 \frac{dx}{(1+x)w} = \frac{\pi\sqrt{3}}{3r}, \qquad T_1 := \int_0^1 \frac{dx}{(1+x)w^2} = \frac{\Omega}{2} + \frac{\pi\sqrt{3}}{3r^2}.$$

The beta recurrence gives $D_{1/3} = \Omega/10$ and $D_{2/3} = \Psi/7$. Polynomial division $x(1-x)/(1+x) = -x + 2 - 2/(1+x)$ and symmetry about $x = 1/2$ give

$$N_{1/3}(1) = \tfrac{3}{2}\Omega - 2T_1, \qquad N_{2/3}(1) = \tfrac{3}{2}\Psi - 2T_2.$$

Finally $\Omega\Psi = 2\pi\sqrt{3}$ proves (2.6)–(2.7). Writing $g = \Gamma(1/3)$, reflection also gives

$$\Omega = \frac{\sqrt{3}\,g^3}{2\pi}, \qquad \Psi = \frac{4\pi^2}{g^3}.$$

The affine transformations are invertible over $\overline{\mathbb{Q}}$, so they imply (2.8). Chudnovsky's established theorem states that π and $\Gamma(1/3)$ are algebraically independent [8, p. 344, equation (9)]. Passing from g to g^3 does not change transcendence degree. This proves the assertion. □

For example, the exact reconstruction formula

$$\pi = \frac{9}{70\sqrt{3}}(5 - L_1)\left(\frac{21}{2} - L_2\right)$$

is consistent with independence, since it is not a relation with algebraic coefficients between L_1, L_2 alone. Every nonconstant polynomial in this pair with algebraic coefficients is transcendental. The explicit evaluations can also be approached through classical hypergeometric transformations; their derivation here is not a claim of a historically new special-function identity.

These evaluations extend to all third-integer exponents:

$$F_h(1) \text{ and } F_h(-1/2) \text{ are transcendental} \qquad (h \in \tfrac{1}{3}\mathbb{Z},\ h > -1). \tag{2.9}$$

Indeed write $h = h_0 + j$ with $j \geq 0$ and $h_0 \in \{-2/3, -1/3, 0\}$. Put $P_j(x) = [x(1-x)]^j$ and $Q_j = (P_j - (-2)^j)/(1+x) \in \mathbb{Q}[x]$. Under the beta probability measure with exponent h_0, all polynomial moments are rational: $\mathbb{E}_{h_0}(x^k) = (h_0+1)_k/(2h_0+2)_k$. Therefore

$$F_h(1) = \frac{q_j + (-2)^j F_{h_0}(1)}{d_j}, \qquad q_j = \mathbb{E}_{h_0}(Q_j) \in \mathbb{Q}, \quad d_j = \frac{(h_0+1)_j^2}{(2h_0+2)_{2j}} > 0.$$

The three base values are

$$F_{-2/3}(1) = \frac{1}{2} + \frac{\Psi}{6r^2}, \qquad F_{-1/3}(1) = \frac{\Omega}{6r}, \qquad F_0(1) = \log 2.$$

The first two follow from the preceding substitutions; the last is transcendental by Hermite–Lindemann. The transfer coefficient is nonzero, and (2.4) gives $F_h(-1/2) = 2F_h(1)$. Moreover, any value in the class $h \equiv 1/3 \pmod 1$ and any value in $h \equiv 2/3 \pmod 1$, at either of these two parameters, are algebraically independent. This makes no independence assertion involving the integer-exponent class and $\log 2$.

2.7 An algebraic complementary family

Not all unidentified entries of this type are transcendental. For rational $0 < a < 1$ and algebraic $c > 0$, one has

$$C(a, 1-a, c, 0) = \frac{2\{1 + (1-a)c - (1+c)^{1-a}\}}{a(1-a)c^2} \in \overline{\mathbb{Q}}. \tag{2.10}$$

The left side is ${}_2F_1(1, a+1; 3; -c)$. Expanding the binomial on the right at zero gives its hypergeometric series, and analytic continuation on $c > 0$ proves the identity. For example, $C(1/3, 2/3, 1, 0) = 15 - 9\,2^{2/3}$. This example explains why selection by proved geometric hypotheses, rather than a missing numerical identification, is essential.

The proofs in this section use exact divisor identities, local valuations, and established transcendence theorems. Numerical agreement with the source entries serves only identification and arithmetic checking. A general algorithmic treatment of linear relations among 1-periods is itself already part of the literature [9]; the contribution recorded here is the explicit application and its stated parameter ranges, not the invention of that general theory.

Part III
Multiple-integral evaluations

3 Six double-integral and two denominator-six triple-integral values

The following reductions preserve the boundary terms of the hypergeometric summations and Barnes contours. They are applications of classical summation and beta-transcendence theorems, with explicit identities supplied here.

3.1 Definitions and scope

We use

$$B(r,s)=\frac{\Gamma(r)\Gamma(s)}{\Gamma(r+s)},\qquad {}_3F_2\left(\begin{matrix}a,b,c\\ d,e\end{matrix};1\right)=\sum_{n\ge 0}\frac{(a)_n(b)_n(c)_n}{(d)_n(e)_n n!}.$$

The six double-integral constants are exactly the six entries under "Denominator 7 with not yet identified constants" on p. 14 of [10]. Their common definition is

$$\begin{aligned}C_2(a_1,a_2,b_1,b_2)&=\frac{\displaystyle\int_0^1\int_0^1\frac{x^{-a_1}(1-x)^{-a_2}y^{-b_1}(1-y)^{-b_2}}{1-xy}\,dx\,dy}{B(1-a_1,1-a_2)B(1-b_1,1-b_2)}\\&={}_3F_2\left(\begin{matrix}1,1-a_1,1-b_1\\ 2-a_1-a_2,2-b_1-b_2\end{matrix};1\right).\end{aligned}$$

For the triple-integral constants define

$$\begin{aligned}J(a,b,c,d,e)&=\iiint_{[0,1]^3}\frac{x^b(1-x)^c y^e(1-y)^a z^a(1-z)^c}{(1-z+xyz)^d}\,dx\,dy\,dz,\\K(a,b,c,d,e)&=\frac{J(a,b,c,d+1,e)}{J(a,b,c,d,e)}.\end{aligned}$$

The two denominator-six entries treated here were left in apparently unreduced hypergeometric form in the 2023 revision of [11]. Their evaluation is established below rather than inferred from that description. A statement about these dated sources does not establish that no later or differently phrased proof exists.

3.2 The arithmetic input

Schneider's beta theorem [12], also stated in [13, Theorem 12] and in the introduction of the research article [14], says that for positive rational numbers r,s, if $r,s,r+s\notin\mathbb{Z}$, then $B(r,s)$ is transcendental. The condition does not require $r+s<1$. Thus every beta value used below is transcendental. If α,β are algebraic with $\alpha\neq 0$, then $\alpha B(r,s)+\beta$ is transcendental. More generally, an invertible fractional-linear transformation over $\overline{\mathbb{Q}}$ preserves transcendence. These elementary transfer steps do not assume algebraic independence of arbitrary gamma values.

3.3 A common residue lemma

For later use set $S=A+B+C+E-2$, and let

$$I(A,B,C,E)=\frac{1}{2\pi i}\int_{\Re s=-1/12}\frac{\Gamma(A+s)\Gamma(B+s)\Gamma(C+s)\Gamma(-2-s)\Gamma(E-s)}{\Gamma(S+s)}\,ds.$$

On $\Re A, \Re B, \Re C > 1/12$ and $\Re E > -1/12$, this integral is holomorphic, with exponential decay on the vertical line locally uniform in the parameters. Barnes' second lemma [15, Eq. (5.6)] and two residues give the meromorphic identity

$$\begin{aligned} I(A,B,C,E) &= G + R_{-2} + R_{-1}, \\ G &= \frac{\Gamma(A-2)\Gamma(B-2)\Gamma(C-2)\Gamma(A+E)\Gamma(B+E)\Gamma(C+E)}{\Gamma(A+B+E-2)\Gamma(A+C+E-2)\Gamma(B+C+E-2)}, \\ R_{-2} &= -\frac{\Gamma(A-2)\Gamma(B-2)\Gamma(C-2)\Gamma(E+2)}{\Gamma(S-2)}, \\ R_{-1} &= \frac{\Gamma(A-1)\Gamma(B-1)\Gamma(C-1)\Gamma(E+1)}{\Gamma(S-1)}. \end{aligned}$$

For a rigorous starting domain take $\Re A, \Re B, \Re C > 9/4$, $\Re E > 0$. The line $\Re s = -9/4$ separates the pole families, so Barnes' lemma applies there. Moving this line right to $-1/12$ crosses precisely the poles $-2, -1$ of $\Gamma(-2-s)$. With upward orientation on both lines the right-line integral equals the left-line integral plus these two residues. Horizontal segments vanish by Stirling's estimate. Meromorphic continuation gives the displayed identity throughout the connected domain stated above. At the specializations below no displayed gamma factor has a pole. In particular this argument does not apply a separating-contour formula directly to a nonseparating vertical line.

3.4 Reduction and transcendence of the first denominator-six triple-integral target

Define

$$I_d = \iiint_{[0,1]^3} \frac{x^{1/2}(1-x)^{1/2}y^{1/6}(1-z)^{1/2}}{(1-z+xyz)^d}\,dx\,dy\,dz, \qquad d \in \left\{\frac{1}{3}, \frac{4}{3}\right\},$$

and

$$C_6 = \frac{I_{4/3}}{I_{1/3}} = K\left(0, \frac{1}{2}, \frac{1}{2}, \frac{1}{3}, \frac{1}{6}\right).$$

This is the parameter convention of [11]:

$$K(a,b,c,d,e) = \frac{J_3(a,b,c,d+1,e;0)}{J_3(a,b,c,d,e;0)}.$$

The integrals are finite and positive. Indeed, $1 - z + xyz \geq 1 - z$, and even for $d = 4/3$ the remaining power $(1-z)^{-5/6}$ is integrable.

Evaluation and result. Set

$$V = B\left(\frac{1}{3}, \frac{1}{2}\right) = \frac{\sqrt{\pi}\,\Gamma(1/3)}{\Gamma(5/6)}.$$

We will prove

$$\boxed{C_6 = \frac{35(17-4V)}{2(32V-133)}.} \tag{3.1}$$

In particular C_6 is transcendental. For orientation only,

$$V = 4.20654631597636278352505723 7\ldots, \qquad C_6 = 1.88989853345080685014367397 4\ldots.$$

The original Mellin–Barnes formula. Specializing the formula of [11] gives

$$I_d = P_d \frac{1}{2\pi i} \int_{\Re s=-1/12} \frac{\Gamma(d+s)\Gamma(1+s)\Gamma(3/2+s)\Gamma(7/6+s)\Gamma(-s)\Gamma(3/2-d-s)}{\Gamma(13/6+s)\Gamma(3+s)}\, ds,$$

where

$$P_d = \frac{\Gamma(3/2)}{\Gamma(d)\Gamma(5/2-d)}.$$

The line is admissible for both values of d: it lies to the right of all the left pole families and to the left of both right pole families in this original representation. The identity

$$\frac{\Gamma(1+s)\Gamma(-s)}{\Gamma(3+s)} = \Gamma(-2-s)$$

reduces the integrand to

$$F_d(s) = \frac{\Gamma(d+s)\Gamma(3/2+s)\Gamma(7/6+s)\Gamma(-2-s)\Gamma(3/2-d-s)}{\Gamma(13/6+s)}.$$

The five numerator parameters now sum to the denominator parameter:

$$d + \frac{3}{2} + \frac{7}{6} - 2 + \left(\frac{3}{2} - d\right) = \frac{13}{6}.$$

This is precisely the parameter relation in Barnes' second lemma. The two newly exposed poles at $s = -2, -1$ require residue corrections.

Corrected Barnes evaluation. Apply the residue identity of Section 3.3 with

$$(A, B, C, E) = (d, 3/2, 7/6, 3/2 - d).$$

Both $d = 1/3$ and $d = 4/3$ satisfy its hypotheses. The two residues are retained; no separating-contour formula is used on a nonseparating line.

Exact gamma reduction. At the required parameters the three terms are

$$G_d = \frac{\Gamma(d-2)\Gamma(-1/2)\Gamma(-5/6)\Gamma(3/2)\Gamma(3-d)\Gamma(8/3-d)}{\Gamma(1)\Gamma(2/3)\Gamma(13/6-d)},$$

$$R_{-2,d} = -\frac{\Gamma(d-2)\Gamma(-1/2)\Gamma(-5/6)\Gamma(7/2-d)}{\Gamma(1/6)},$$

$$R_{-1,d} = \frac{\Gamma(d-1)\Gamma(1/2)\Gamma(1/6)\Gamma(5/2-d)}{\Gamma(7/6)}.$$

For example, write

$$Q_d = \frac{\sqrt{\pi}\,\Gamma(1/6)\Gamma(3-d)\Gamma(8/3-d)}{\Gamma(5/2-d)\Gamma(2/3)\Gamma(13/6-d)}.$$

Gamma shifts alone show $Q_{1/3} = (64/21)V$ and $Q_{4/3} = (4/3)V$, and

$$\frac{P_d G_d}{\pi} = -\frac{Q_d}{2(d-1)(d-2)(-5/6)},$$

$$\frac{P_d R_{-2,d}}{\pi} = \frac{5/2-d}{(d-1)(d-2)(-5/6)}, \qquad \frac{P_d R_{-1,d}}{\pi} = \frac{3}{d-1}.$$

Thus the complete exact coefficient table is

	$d=1/3$	$d=4/3$
P_dG_d/π	$\frac{288}{175}V$	$-\frac{18}{5}V$
$P_dR_{-2,d}/\pi$	$-\frac{117}{50}$	$\frac{63}{10}$
$P_dR_{-1,d}/\pi$	$-\frac{9}{2}$	9

Consequently

$$\frac{I_{1/3}}{\pi}=\frac{9(32V-133)}{175},\qquad \frac{I_{4/3}}{\pi}=\frac{9(17-4V)}{10},$$

which proves (3.1). Positivity also proves $133/32<V<17/4$, so none of the displayed denominators vanish.

Transcendence and scope. Schneider's beta-value theorem [12], explicitly recalled in the primary research article [14], states that $B(a,b)$ is transcendental for positive rational nonintegral a,b with $a+b$ nonintegral. It applies to $a=1/3$, $b=1/2$, $a+b=5/6$, proving that V is transcendental. Solving (3.1) for V gives

$$V=\frac{266C_6+595}{64C_6+140}.$$

The denominator is positive. If C_6 were algebraic then V would be algebraic, a contradiction. Hence C_6 is transcendental.

The 2023 version of [11], page 3, lists this precise target among expressions involving ${}_3F_2$ values that seem not reducible to a simpler form. The present proof supplies such a reduction and transfers a classical transcendence theorem to the integral ratio. Historical priority of this specific evaluation has not been established. This does not prove any proposed irrationality measure or divisibility assertion associated with the original construction.

3.5 The second denominator-six triple-integral quotient

We continue to use the normalization in Zudilin's note [11], so $J_3(a,b,c,d,e;0)=J(a,b,c,d,e)$ as defined above. All powers in the real integrals use their positive real values. In this subsection only, C_2 denotes the scalar below, rather than the four-argument double-integral function $C_2(a_1,a_2,b_1,b_2)$.

Theorem 3.1. *Let*

$$C_2=2^{-1/3}\frac{\Gamma(1/3)^3}{\pi^2}.$$

Then

$$\boxed{K\left(0,\frac12,\frac12,\frac16,\frac13\right)=\frac{1001C_2-1540}{275-175C_2}.}$$

In particular this constant is transcendental. Its numerical value is

$$1.73295989536011838\ldots.$$

Barnes representation and exact cancellation

Put $J_d=J_3(0,1/2,1/2,d,1/3;0)$, where $d\in\{1/6,7/6\}$. Zudilin's Barnes representation gives

$$J_d=P_dI_d,\qquad P_d=\frac{\Gamma(3/2)}{\Gamma(d)\Gamma(5/2-d)},$$

$$I_d = \frac{1}{2\pi i}\int_{\Re s=-1/12} \frac{\Gamma(d+s)\Gamma(3/2+s)\Gamma(4/3+s)\Gamma(-2-s)\Gamma(3/2-d-s)}{\Gamma(7/3+s)}\,ds.$$

Indeed, the original expression has factors $\Gamma(1+s)\Gamma(-s)/\Gamma(3+s)$, and the gamma recurrence gives the exact identity

$$\frac{\Gamma(1+s)\Gamma(-s)}{\Gamma(3+s)} = \Gamma(-2-s).$$

The original contour condition is

$$-\min\{1, 3/2, d, 4/3\} < -1/12 < 3/2 - d,$$

which holds for both values of d. In particular the common vertical contour is legitimate before making any analytic-continuation argument.

Barnes lemma and the contour correction

Apply the common residue lemma in Section 3.3 with

$$A = d, \quad B = 3/2, \quad C = 4/3, \quad E = 3/2 - d, \qquad S = A + B + C + E - 2 = 7/3.$$

For both values of d, the hypotheses $A, B, C > 1/12$ and $E > -1/12$ hold. The analytic-continuation argument of that section begins with a genuine separating line and keeps the two residues at $s = -2, -1$. Thus no separating-contour formula is being applied without correction to the nonseparating line $\Re s = -1/12$. None of the gamma factors below has a pole at either specialization. Explicitly,

$$I_d = C_d + R_{-2,d} + R_{-1,d},$$

where

$$\begin{aligned} C_d &= \frac{\Gamma(d-2)\Gamma(-1/2)\Gamma(-2/3)\Gamma(3/2)\Gamma(3-d)\Gamma(17/6-d)}{\Gamma(5/6)\Gamma(7/3-d)}, \\ R_{-2,d} &= -\frac{\Gamma(d-2)\Gamma(-1/2)\Gamma(-2/3)\Gamma(7/2-d)}{\Gamma(1/3)}, \\ R_{-1,d} &= \frac{\Gamma(d-1)\Gamma(1/2)\Gamma(1/3)\Gamma(5/2-d)}{\Gamma(4/3)}. \end{aligned}$$

The signs follow from $\operatorname{Res}_{s=-2}\Gamma(-2-s) = -1$ and $\operatorname{Res}_{s=-1}\Gamma(-2-s) = 1$.

Gamma recurrence simplification

Write

$$H = \sqrt{\pi}\,\frac{\Gamma(2/3)}{\Gamma(1/6)}.$$

Gamma recurrence alone gives

$$P_d(R_{-2,d} + R_{-1,d}) = \frac{3\pi(4d-9)}{4(d-1)(d-2)}.$$

For clarity, the individual exact terms are

d	P_dC_d	$P_dR_{-2,d}$	$P_dR_{-1,d}$
$1/6$	$(135/14)\pi H$	$-(126/55)\pi$	$-(9/5)\pi$
$7/6$	$-54\pi H$	$(72/5)\pi$	9π

and hence

$$J_{1/6} = \pi\left(\frac{135}{14}H - \frac{45}{11}\right), \qquad J_{7/6} = \pi\left(-54H + \frac{117}{5}\right).$$

Taking the quotient yields

$$K = \frac{2002 - 4620H}{825H - 350}.$$

Reflection and duplication give

$$\Gamma(1/3)\Gamma(2/3) = \frac{2\pi}{\sqrt{3}}, \qquad \Gamma(1/6)\Gamma(2/3) = 2^{2/3}\sqrt{\pi}\,\Gamma(1/3),$$

so

$$H = \frac{B(2/3, 2/3)}{3\,2^{2/3}} = \frac{2}{3C_2}.$$

Substitution proves the asserted exact evaluation.

Transcendence and verification limits

The numbers $2/3, 2/3, 4/3$ are rational and nonintegral. Schneider's beta-value theorem [12] therefore makes $B(2/3, 2/3)$ transcendental, even though its two arguments sum to more than one. The identity $H = B(2/3, 2/3)/(3\,2^{2/3})$ makes H transcendental. The fractional-linear map in H has determinant

$$(-4620)(-350) - 2002 \cdot 825 = -34650 \neq 0,$$

and inverse $H = (350K + 2002)/(825K + 4620)$. Since $K > 0$, the inverse has nonzero denominator; hence K is transcendental.

As an optional alternative, the transcendence of C_2 is explicitly recalled in [11, p. 2]. The rational fractional-linear map in C_2 has determinant

$$1001 \cdot 275 - 1540 \cdot 175 = 5775 \neq 0,$$

and its inverse is

$$C_2 = \frac{275K + 1540}{175K + 1001}.$$

Since the original integrals are positive, $K > 0$, and this inverse has nonzero denominator. Algebraicity of K would contradict transcendence of C_2. The beta-value argument above does not require an algebraic-independence theorem for gamma values.

3.6 The first two double-integral candidates

Put $s_j = \sin(j\pi/7)$ and $g_j = \Gamma(j/7)$. For the first two denominator-seven double-integral candidates, the hypergeometric forms and evaluations are

$$F_1 = {}_3F_2\left(\begin{matrix}1, 13/7, 11/7\\ 26/7, 23/7\end{matrix}; 1\right) = \frac{418}{7}\frac{s_1}{s_2}B(2/7, 4/7) - \frac{437}{3}, \tag{3.2}$$

$$F_2 = {}_3F_2\left(\begin{matrix}1, 13/7, 10/7\\ 25/7, 22/7\end{matrix}; 1\right) = \frac{198}{7}\frac{s_2}{s_3}B(1/7, 4/7) - 180. \tag{3.3}$$

These correspond respectively to $C_2(-6/7, -6/7, -4/7, -5/7)$ and $C_2(-6/7, -5/7, -3/7, -5/7)$ in the DBKZ list, with the hypergeometric normalization used there. Both are transcendental. The proof below gives explicit finite telescoping certificates; it does not rely on numerical recognition and makes no priority claim.

3.6.1 Classical identities used

Watson's summation, in the normalization of DLMF 16.4.6, is

$$ {}_3F_2\left(\begin{matrix}a,b,c\\(a+b+1)/2,2c\end{matrix};1\right)=\frac{\Gamma(1/2)\Gamma(c+1/2)\Gamma((a+b+1)/2)\Gamma(c+(1-a-b)/2)}{\Gamma((a+1)/2)\Gamma((b+1)/2)\Gamma(c+(1-a)/2)\Gamma(c+(1-b)/2)}. \tag{3.4}$$

All applications below satisfy its convergence condition $\mathrm{Re}(2c-a-b)>-1$. We also use Euler's transformation of ${}_2F_1$ inside its beta integral. For the positive real parameters appearing here this gives directly

$$ {}_3F_2\left(\begin{matrix}a,b,1\\e,f\end{matrix};1\right)=\frac{f-1}{e+f-a-b-1}\,{}_3F_2\left(\begin{matrix}e-a,e-b,1\\e,e+f-a-b\end{matrix};1\right). \tag{3.5}$$

To see the prefactor explicitly, write the left side as $(f-1)\int_0^1(1-t)^{f-2}\,{}_2F_1(a,b;e;t)\,dt$ and use ${}_2F_1(a,b;e;t)=(1-t)^{e-a-b}\,{}_2F_1(e-a,e-b;e;t)$. This is a special case of a Thomae transformation.

3.6.2 First candidate

Set

$$ W_1={}_3F_2\left(\begin{matrix}1,4/7,13/7\\9/7,26/7\end{matrix};1\right),$$

$$ h_n=\frac{(4/7)_n(13/7)_n}{(23/7)_n(26/7)_n},\qquad P_1(n)=\frac{(7n+16)(7n+19)(21n+23)}{48}.$$

The summands of F_1 and W_1, respectively, are

$$ h_n\frac{n+4/7}{4/7},\qquad h_n\frac{(n+9/7)(n+16/7)}{(9/7)(16/7)}.$$

Direct multiplication, using $h_{n+1}/h_n=(n+4/7)(n+13/7)/((n+23/7)(n+26/7))$, proves

$$ P_1(n+1)h_{n+1}-P_1(n)h_n=h_n\left\{\frac{n+4/7}{4/7}-99\frac{(n+9/7)(n+16/7)}{(9/7)(16/7)}\right\}. \tag{3.6}$$

Here $h_n=O(n^{-32/7})$, so $P_1(n)h_n\to 0$, and $P_1(0)=437/3$. Summing proves

$$ F_1=99W_1-437/3. \tag{3.7}$$

Watson's formula with $(a,b,c)=(1,4/7,13/7)$ gives

$$ W_1=\frac{\sqrt{\pi}\,\Gamma(33/14)\Gamma(9/7)\Gamma(11/7)}{\Gamma(11/14)\Gamma(13/7)\Gamma(29/14)}=\frac{38}{63}\frac{g_2^2g_4g_5}{g_1g_6^2}=\frac{38}{63}\frac{s_1}{s_2}B(2/7,4/7).$$

The middle equality uses only the gamma shift and duplication formulas; the last uses $g_jg_{7-j}=\pi/s_j$. Combining this with (3.7) proves (3.2).

3.6.3 Second candidate

Taking $(a,b,e,f)=(13/7,10/7,22/7,25/7)$ in (3.5) gives

$$ F_2=\frac{18}{17}V_2,\qquad V_2={}_3F_2\left(\begin{matrix}1,9/7,12/7\\22/7,24/7\end{matrix};1\right).$$

Now set

$$ W_2={}_3F_2\left(\begin{matrix}1,2/7,12/7\\8/7,24/7\end{matrix};1\right),$$

$$h_n = \frac{(2/7)_n(12/7)_n}{(22/7)_n(24/7)_n}, \qquad P_2(n) = \frac{(7n+15)(7n+17)(35n+32)}{48}.$$

The exact difference certificate is

$$P_2(n+1)h_{n+1} - P_2(n)h_n = h_n\left\{\frac{n+2/7}{2/7} - \frac{275}{2}\frac{(n+8/7)(n+15/7)}{(8/7)(15/7)}\right\}. \tag{3.8}$$

Again $h_n = O(n^{-32/7})$, so the upper boundary vanishes. Since $P_2(0) = 170$, this proves

$$V_2 = \frac{275}{2}W_2 - 170, \qquad F_2 = \frac{2475}{17}W_2 - 180.$$

Watson's formula with $(a, b, c) = (1, 2/7, 12/7)$ evaluates the remaining series:

$$W_2 = \frac{\sqrt{\pi}\,\Gamma(31/14)\Gamma(8/7)\Gamma(11/7)}{\Gamma(9/14)\Gamma(12/7)\Gamma(29/14)} = \frac{34}{175}\frac{g_1g_3g_4^2}{g_2g_5^2} = \frac{34}{175}\frac{s_2}{s_3}B(1/7, 4/7).$$

This proves (3.3).

3.6.4 Transcendence conclusion and scope of verification

Schneider's beta-value transcendence theorem for rational arguments [12, 13] applies to $B(2/7, 4/7)$ and $B(1/7, 4/7)$: their arguments and their sums are nonintegral. Both sine ratios in (3.2)–(3.3) are nonzero algebraic numbers. A nonconstant affine transformation with algebraic coefficients preserves transcendence. Therefore both existing DBKZ candidates F_1 and F_2 are transcendental.

3.7 The third and fourth double-integral candidates

The third and fourth denominator-seven entries in the list of unidentified double-integral constants in [10, p. 14] are

$$F_3 = {}_3F_2\left(\begin{matrix}1, 13/7, 9/7\\ 25/7, 17/7\end{matrix}; 1\right), \qquad F_4 = {}_3F_2\left(\begin{matrix}1, 13/7, 8/7\\ 24/7, 16/7\end{matrix}; 1\right).$$

The source has 17/7, not 15/7, as the last lower parameter of F_3. The parameter tuple gives the lower parameter

$$2 - (-2/7) - (-1/7) = 17/7.$$

The excesses of these two series are 13/7 and 12/7, respectively. Thus both series converge.

Put

$$s_j = \sin(j\pi/7), \qquad T = \frac{\pi\Gamma(3/7)}{\Gamma(1/7)\Gamma(2/7)} = \frac{s_2}{7}B\left(\frac{3}{7}, \frac{5}{7}\right).$$

The beta identity follows from $\Gamma(2/7)\Gamma(5/7) = \pi/s_2$ and $\Gamma(8/7) = \Gamma(1/7)/7$.

Exact evaluations and transcendence. We have

$$F_3 = 40 - \frac{66s_2}{s_1s_3}T = 40 - \frac{66s_2^2}{7s_1s_3}B\left(\frac{3}{7}, \frac{5}{7}\right), \tag{3.9}$$

$$F_4 = \frac{323}{24} - \frac{425s_1}{8s_2^2}T = \frac{323}{24} - \frac{425s_1}{56s_2}B\left(\frac{3}{7}, \frac{5}{7}\right). \tag{3.10}$$

Schneider's beta-value theorem [12], in the form stated in [13, Theorem 12], implies that $B(3/7, 5/7)$ is transcendental: 3/7, 5/7, and their sum 8/7 are all nonintegers. The coefficients of this beta value in (3.9) and (3.10) are nonzero algebraic numbers. Consequently both F_3 and F_4 are transcendental. This conclusion uses the transcendence of a single beta value; no conjecture about gamma values or their algebraic independence is needed.

Watson sums. Define the two convergent Watson series

$$W_3 = {}_3F_2\left(\begin{matrix}1, 6/7, 9/7\\ 10/7, 18/7\end{matrix}; 1\right), \qquad W_4 = {}_3F_2\left(\begin{matrix}1, 6/7, 8/7\\ 10/7, 16/7\end{matrix}; 1\right).$$

Watson's formula [16, Eq. 16.4.6] yields

$$W_3 = \frac{\sqrt{\pi}\,\Gamma(25/14)\Gamma(10/7)\Gamma(6/7)}{\Gamma(13/14)\Gamma(9/7)\Gamma(19/14)} = \frac{33 s_2}{10 s_1 s_3}\,T, \tag{3.11}$$

$$W_4 = \frac{\sqrt{\pi}\,\Gamma(23/14)\Gamma(10/7)\Gamma(5/7)}{\Gamma(13/14)\Gamma(8/7)\Gamma(17/14)} = \frac{9 s_1}{s_2^2}\,T. \tag{3.12}$$

The last equalities use only the gamma recurrence, reflection, and duplication formulas. The convergence condition of Watson's formula, $\mathrm{Re}(2c-a-b) > -1$, gives $5/7 > -1$ and $3/7 > -1$ here.

Exact telescoping certificates. The remaining relations are

$$F_3 + 20 W_3 = 40, \qquad F_4 + \frac{425}{72} W_4 = \frac{323}{24}. \tag{3.13}$$

They can be checked term by term without a generalized Watson table. Write $\Delta g_n = g_{n+1} - g_n$. First put

$$h_n = \frac{(1)_n (6/7)_n (9/7)_n}{(17/7)_n (25/7)_n n!}, \qquad P_3(n) = -\frac{(7n+10)(7n+12)(7n+18)}{54}.$$

If $f_{3,n}, w_{3,n}$ are the defining summands of F_3, W_3, then

$$\frac{f_{3,n}}{h_n} = \frac{7n+6}{6}, \qquad \frac{w_{3,n}}{h_n} = \frac{(7n+10)(7n+18)}{180},$$

and direct rational-function cancellation gives

$$\Delta\big(h_n P_3(n)\big) = f_{3,n} + 20 w_{3,n}.$$

The initial boundary is $h_0 P_3(0) = -40$; since $h_n = O(n^{-27/7})$ and $P_3(n) = O(n^3)$, the upper boundary is zero. This proves the first identity in (3.13).

For the second identity put

$$k_n = \frac{(1)_n (6/7)_n (8/7)_n}{(24/7)_n (16/7)_n n!}, \qquad P_4(n) = -\frac{(7n+9)(7n+17)(21n+38)}{432}.$$

Writing $f_{4,n}, w_{4,n}$ for the summands of F_4, W_4 gives

$$\frac{f_{4,n}}{k_n} = \frac{7n+6}{6}, \qquad \frac{w_{4,n}}{k_n} = \frac{(7n+10)(7n+17)}{170},$$

and

$$\Delta\big(k_n P_4(n)\big) = f_{4,n} + \frac{425}{72} w_{4,n}.$$

Here $k_0 P_4(0) = -323/24$, and the upper boundary is zero because $k_n = O(n^{-26/7})$ and $P_4(n) = O(n^3)$. Combining the contiguous relations with the Watson sums proves the exact evaluations.

The argument proves the qualitative transcendence assertions. It does not prove the quantitative irrationality bounds conjectured for the original rational-approximation constructions.

3.8 The fifth and sixth double-integral targets

The last two entries in the denominator-seven list of unidentified double-integral constants in [10, p. 14] are exactly

$$F_5 = C_2(-6/7,-3/7,-5/7,-3/7) = {}_3F_2\left(\begin{matrix}1,13/7,12/7\\23/7,22/7\end{matrix};1\right),$$
$$F_6 = C_2(-5/7,-3/7,-4/7,-2/7) = {}_3F_2\left(\begin{matrix}1,12/7,11/7\\22/7,20/7\end{matrix};1\right).$$

Let $s = \sin(3\pi/7)/\sin(\pi/7)$. We prove

$$\boxed{\begin{aligned} F_5 &= -152 + \frac{48}{7}s\,B(1/7,2/7),\\ F_6 &= -\frac{195}{2} + \frac{195}{7}s\,B(5/7,6/7).\end{aligned}} \tag{3.14}$$

In particular, both specified pre-existing constants are transcendental.

3.8.1 An elementary transformation and exact certificates

Euler's transformation of ${}_2F_1$, followed by beta integration, yields

$${}_3F_2\left(\begin{matrix}1,a,b\\e,f\end{matrix};1\right) = \frac{f-1}{e+f-a-b-1}{}_3F_2\left(\begin{matrix}1,e-a,e-b\\e,e+f-a-b\end{matrix};1\right). \tag{3.15}$$

For the present positive parameters both beta integrals converge; this identity also follows from [16, Eq. 16.4.11]. Explicitly, the integral representation on the left is

$$(f-1)\int_0^1 (1-t)^{f-2}{}_2F_1(a,b;e;t)\,dt,$$

and Euler's factor $(1-t)^{e-a-b}$ gives the right side of (3.15). Thus

$$F_5 = \frac{16}{13}G_5, \qquad G_5 = {}_3F_2\left(\begin{matrix}1,9/7,10/7\\22/7,20/7\end{matrix};1\right).$$

For F_5, set

$$h_n = \frac{(2/7)_n(10/7)_n}{(22/7)_n(20/7)_n}, \quad U(n) = \frac{7n+2}{2}, \quad V(n) = \frac{(7n+8)(7n+15)}{120},$$

and

$$Q(n) = \frac{(7n+13)(7n+15)(77n+76)}{120}.$$

The following is an identity of rational functions of n:

$$Q(n+1)\frac{(n+2/7)(n+10/7)}{(n+22/7)(n+20/7)} - Q(n) = U(n) - 99V(n). \tag{3.16}$$

Multiplication by h_n and summation telescope. Since $Q(n)h_n = O(n^{-9/7}) \longrightarrow 0$ and $Q(0) = 247/2$, this proves

$$G_5 = 99W_5 - \frac{247}{2}, \qquad W_5 = {}_3F_2\left(\begin{matrix}1,2/7,10/7\\8/7,20/7\end{matrix};1\right).$$

Indeed $U(n)h_n$ and $V(n)h_n$ are respectively the summands of G_5 and W_5.

For F_6, instead put

$$h_n = \frac{(12/7)_n(11/7)_n}{(20/7)_n(22/7)_n}, \qquad V(n) = \frac{7n+13}{13}, \qquad Q(n) = \frac{(7n+13)(7n+15)}{2}.$$

The exact certificate is

$$Q(n+1)\frac{(n+12/7)(n+11/7)}{(n+20/7)(n+22/7)} - Q(n) = 1 - \frac{65}{2}V(n). \tag{3.17}$$

Here $Q(n)h_n = O(n^{-5/7}) \longrightarrow 0$ and $Q(0) = 195/2$; hence

$$F_6 = \frac{65}{2}W_6 - \frac{195}{2}, \qquad W_6 = {}_3F_2\left(\begin{matrix}1, 12/7, 11/7\\ 13/7, 22/7\end{matrix}; 1\right).$$

All series used in these two telescoping arguments converge absolutely. Both displayed certificates can be verified simply by clearing their quadratic denominators and expanding over $\mathbb{Q}$.

3.8.2 Watson summation and beta reduction

Watson's sum [16, Eq. 16.4.6] applies to W_5 and W_6 with, respectively,

$$(a,b,c) = (1, 2/7, 10/7), \qquad (a,b,c) = (1, 12/7, 11/7).$$

The convergence quantities $c + (1-a-b)/2$ are $9/7$ and $5/7$, both strictly positive. The resulting exact values are

$$W_5 = \frac{\sqrt{\pi}\,\Gamma(8/7)\Gamma(27/14)\Gamma(9/7)}{\Gamma(9/14)\Gamma(10/7)\Gamma(25/14)},$$
$$W_6 = \frac{\sqrt{\pi}\,\Gamma(13/7)\Gamma(29/14)\Gamma(5/7)}{\Gamma(19/14)\Gamma(11/7)\Gamma(17/14)}.$$

Writing $g_j = \Gamma(j/7)$, gamma translation and duplication give

$$W_5 = \frac{13}{231}\frac{g_1^2 g_2 g_6}{g_3^2 g_4}, \qquad W_6 = \frac{3}{2}\frac{g_1 g_5 g_6^2}{g_4^2 g_3}.$$

Reflection gives

$$\frac{g_1 g_6}{g_3 g_4} = \frac{\sin(3\pi/7)}{\sin(\pi/7)} = s.$$

Using $B(1/7, 2/7) = g_1 g_2/g_3$ and $B(5/7, 6/7) = 7g_5 g_6/(4g_4)$, we therefore obtain

$$W_5 = \frac{13}{231}s\,B(1/7, 2/7), \qquad W_6 = \frac{6}{7}s\,B(5/7, 6/7).$$

Substituting these into the two proved contiguous relations establishes (3.14).

3.8.3 Transcendence

Schneider's theorem [12, p. 114] states that $B(r,t)$ is transcendental when $r, t \in \mathbb{Q}$ and none of $r, t, r+t$ is an integer. Its hypotheses hold for $(r,t) = (1/7, 2/7)$ and $(5/7, 6/7)$. The coefficient s is a nonzero algebraic number, because its sine values are formed from roots of unity. Both expressions in (3.14) are therefore nonzero algebraic multiples of transcendental numbers plus rational numbers, and both are transcendental.

The displayed finite telescoping identities and subsequent gamma algebra are exact. The numerical values

$$F_5 = 1.67234841025118232998438522 99\ldots, \qquad F_6 = 1.66449355925700954891900725 27\ldots$$

are included only for identifying the constants, not as evidence for the identities. This argument uses classical transformations and transcendence theorems with explicit certificates; it does not establish a historical priority claim for these evaluations or their transcendence consequences.

4 The five denominator-seven triple-integral quotients

These five entries are separate from the six double integrals above. The first evaluation is already explicit in Zudilin's note; the remaining formulas are retained as proof records without a historical-priority claim.

4.1 Definitions and the five conclusions

Set

$$J(a,b,c,d,e) = \iiint_{[0,1]^3} \frac{x^b(1-x)^c y^e (1-y)^a z^a (1-z)^c}{(1-z+xyz)^d}\,dx\,dy\,dz,$$
$$K(a,b,c,d,e) = \frac{J(a,b,c,d+1,e)}{J(a,b,c,d,e)}.$$

All integrals used below converge and are positive. Write $g_j = \Gamma(j/7)$, $t = 2\cos(2\pi/7)$, and

$$U = \frac{2g_2^2 g_5}{g_1 g_4^2}, \quad H = \frac{g_1^2 g_6}{g_2^2 g_4}, \quad L = \frac{t^2}{1+t}\frac{g_1 g_4}{g_2 g_3}, \quad W = \frac{g_4^2 g_3}{g_2 g_1^2}, \quad T = \frac{g_3^2 g_4}{g_5 g_6^2}.$$

The exact evaluations are

$$K_1 = K(1/7,0,2/7,3/7,4/7) = \frac{945U-1512}{832-525U},$$
$$K_2 = K(1/7,0,2/7,5/7,3/7) = \frac{15H-48}{70-23H},$$
$$K_3 = K(1/7,0,3/7,4/7,5/7) = \frac{525L-567}{462-425L},$$
$$K_4 = K(1/7,0,4/7,2/7,5/7) = \frac{9702W-363}{101-2744W},$$
$$K_5 = K(2/7,0,3/7,4/7,5/7) = \frac{520-120T}{88T-373}.$$

Their decimal values, for identification only, are

$$K_1 = 2.56248224568\ldots, \qquad K_2 = 3.59939534133\ldots,$$
$$K_3 = 2.52360910522\ldots, \qquad K_4 = 1.99773527607\ldots,$$
$$K_5 = 2.72543856463\ldots.$$

These are the five entries under "Denominator 7" in [10]. The first formula is printed in [11]. The same source states that similar gamma evaluations exist for the next three, without displaying them, and lists the fifth among the expressions not apparently reducible beyond ${}_3F_2$. Accordingly, the explicit formulas for the middle three should not be described as the first discovery of their gamma evaluability.

4.2 Arithmetic step: a finite certificate for each value

For positive rational arguments with none of $a, b, a+b, c, d, c+d$ integral, define

$$h_{a,b}(u) = \{ua\} + \{ub\} - \{u(a+b)\}.$$

The beta-value theorem of Wolfart–Wüstholz, Satz 4, together with the classical-relation criterion on pp. 6–8 of [3], implies

$$h_{a,b}(u) \neq h_{c,d}(u) \text{ for some } (u,N)=1 \quad \Longrightarrow \quad \frac{B(a,b)}{B(c,d)} \notin \overline{\mathbb{Q}},$$

where N is a common denominator. The theorem includes both first-kind and second-kind beta periods; the sums of the arguments need not both be less than one. This is a theorem about ratios of *two beta values*, not an unconditional classification of arbitrary products of gamma values.

Reflection, duplication and gamma shifts give

$$U = \frac{2\sin(3\pi/7)}{\sin(2\pi/7)}\frac{B(2/7,3/7)}{B(1/7,4/7)},$$
$$H = \frac{\sin(3\pi/7)}{\sin(\pi/7)}\frac{B(1/7,3/7)}{B(2/7,2/7)},$$
$$L = \frac{t^2}{1+t}\frac{B(1/7,4/7)}{B(2/7,3/7)},$$
$$W = \frac{1}{7}\frac{B(4/7,4/7)}{B(1/7,2/7)}, \qquad T = \frac{7}{4}\frac{B(3/7,3/7)}{B(5/7,6/7)}.$$

All prefactors are nonzero algebraic numbers. The following table supplies exact witnesses; its two middle columns are sums of fractional parts.

Parameter	witness u	numerator beta	denominator beta	carries
U	2	10/7	3/7	1, 0
H	3	5/7	12/7	0, 1
L	2	3/7	10/7	0, 1
W	1	8/7	3/7	1, 0
T	1	6/7	11/7	0, 1

Thus all five parameters are transcendental. In the order $K_1, \dots, K_5$, the fractional-linear determinants are

$$-7560, \quad -54, \quad 1575, \quad -16170, \quad -1000.$$

They are nonzero. An algebraic value of any K_i would therefore imply an algebraic value of its parameter, a contradiction.

Theorem 4.1. *All five specified denominator-seven DBKZ integral quotients are transcendental.*

4.3 Common analytic reduction

We use the Barnes representation in [11]:

$$J(a,b,c,d,e) = \frac{\Gamma(1+a)\Gamma(1+c)}{\Gamma(d)\Gamma(2+a+c-d)}\frac{1}{2\pi i}\int_{\Re s=\sigma} \frac{\Gamma(d+s)\Gamma(1+a+s)\Gamma(1+b+s)\Gamma(1+e+s)}{\Gamma(2+a+e+s)\Gamma(2+b+c+s)} \cdot \Gamma(-s)\Gamma(1+c-d-s)\,ds,$$

where $-\min(a+1, b+1, d, e+1) < \sigma < c-d+1$. The usual Mellin–Barnes separation also requires $\sigma < 0$; the line $\sigma = -1/2$ used for the fifth case satisfies all these bounds. For the other cases the two-residue-family expression is

$$J = \frac{\pi}{\sin\pi(d-c)}(A_d F_d - B_d G_d),$$

with

$$A_d = \frac{\Gamma(a+1)^2\Gamma(b+1)\Gamma(c+1)\Gamma(e+1)}{\Gamma(d-c)\Gamma(a+e+2)\Gamma(b+c+2)\Gamma(a+c-d+2)},$$
$$B_d = \frac{\Gamma(a+1)\Gamma(c+1)^2\Gamma(b+c-d+2)\Gamma(e+c-d+2)}{\Gamma(c-d+2)\Gamma(d)\Gamma(a+e+c-d+3)\Gamma(b+2c-d+3)},$$
$$F_d = {}_4F_3\left(\begin{matrix} a+1, b+1, d, e+1 \\ d-c, a+e+2, b+c+2 \end{matrix}; 1\right),$$
$$G_d = {}_4F_3\left(\begin{matrix} a+c-d+2, b+c-d+2, c+1, e+c-d+2 \\ c-d+2, a+e+c-d+3, b+2c-d+3 \end{matrix}; 1\right).$$

In the first case the printed formula in [11] uses $2^{2/7}\sqrt{\pi}\Gamma(9/14)/[\cos(3\pi/14)g_4^2]$. Duplication and reflection identify this with U, proving the first row. The following sections give exact certificates for the four remaining rows.

4.4 The second denominator-seven candidate

Write $g_k = \Gamma(k/7)$ for $1 \le k \le 6$. The evaluation proved in this section is

$$K\left(\frac{1}{7}, 0, \frac{2}{7}, \frac{5}{7}, \frac{3}{7}\right) = \frac{15H - 48}{70 - 23H}, \qquad H = \frac{g_1^2 g_6}{g_2^2 g_4}. \tag{4.1}$$

Together with the beta-period criterion of Wolfart and Wüstholz [3], this proves that this K is transcendental. The argument uses the hypergeometric representation in [11] and explicit telescoping identities. No claim of historical priority is made.

For clarity, the conventions for the integrals are

$$J(a,b,c,d,e) = \int_0^1\int_0^1\int_0^1 \frac{x^b(1-x)^c y^e(1-y)^a z^a(1-z)^c}{(1-z+xyz)^d}\,dx\,dy\,dz,$$
$$K(a,b,c,d,e) = \frac{J(a,b,c,d+1,e)}{J(a,b,c,d,e)}.$$

The particular integrals used below converge and are strictly positive.

4.4.1 Two elementary balanced summations

Let

$$t_n = \frac{(A)_n(B)_n}{(C)_n(D)_n}, \qquad \delta = 1 - C, \quad p = A + \delta, \quad q = B + \delta, \quad R = \frac{\Gamma(C)\Gamma(D)}{\Gamma(A)\Gamma(B)}.$$

The following identities are used only at positive rational parameters with nonzero displayed denominators; they also hold for generic complex parameters for which the series and expressions are defined.

First suppose $C + D = A + B + 2$. Then

$${}_3F_2\left(\begin{matrix} 1, A, B \\ C, D \end{matrix}; 1\right) = \frac{\delta(D-1) + R}{pq}. \tag{4.2}$$

Indeed, with $Q(n) = (n + C - 1)(n + D - 1)$, direct multiplication gives

$$Q(n+1)t_{n+1} - Q(n)t_n = pq\,t_n.$$

The gamma-ratio asymptotic gives $n^2 t_n \longrightarrow R$. Summing the difference from $n = 0$ to N and then taking the limit yields $pq\sum_{n\ge 0} t_n = R - (C-1)(D-1)$, proving (4.2). In particular, the boundary at infinity must not be discarded.

Next suppose $C + D = A + B + 3$. Put

$$E = p(p+1)q(q+1), \qquad k = \frac{pq + 1 + \delta}{E}, \qquad \ell = \frac{(1+\delta)(D-2)}{E},$$
$$P(n) = (n + C - 1)(n + D - 1)(kn + \ell).$$

Another polynomial identity is

$$P(n+1)t_{n+1} - P(n)t_n = (n+1)t_n.$$

Here $n^3 t_n \longrightarrow R$, so the upper boundary is kR. Since $(2)_n/n! = n + 1$, telescoping proves

$${}_3F_2\left(\begin{matrix}2, A, B\\ C, D\end{matrix}; 1\right) = \frac{\delta(1+\delta)(D-1)(D-2) + R(pq+1+\delta)}{p(p+1)q(q+1)}. \tag{4.3}$$

Thus both summations have finite algebraic difference certificates.

4.4.2 Reduction of the four hypergeometric series

Specialize $a = 1/7$, $b = 0$, $c = 2/7$, and $e = 3/7$, and abbreviate $J_d = J(a, b, c, d, e)$. The Barnes-residue representation from [11], specialized to these parameters, is

$$J_d = \frac{\pi}{\sin \pi(d - 2/7)}(A_d F_d - B_d G_d), \qquad d \in \{5/7, 12/7\}, \tag{4.4}$$

where the prefactors are

d	A_d	B_d
$5/7$	$\dfrac{7g_1^2}{660 g_4 g_5}$	$\dfrac{49 g_2^2}{2340 g_5 g_6}$
$12/7$	$\dfrac{7g_1^2}{396 g_4 g_5}$	$\dfrac{7 g_2^2}{60 g_5 g_6}$

and, after cancelling equal upper and lower parameters where possible,

$$F_{5/7} = {}_4F_3\left(\begin{matrix}8/7, 1, 5/7, 10/7\\ 3/7, 18/7, 16/7\end{matrix}; 1\right),$$
$$G_{5/7} = {}_3F_2\left(\begin{matrix}2, 9/7, 12/7\\ 20/7, 22/7\end{matrix}; 1\right),$$
$$F_{12/7} = {}_3F_2\left(\begin{matrix}1, 8/7, 12/7\\ 16/7, 18/7\end{matrix}; 1\right),$$
$$G_{12/7} = {}_3F_2\left(\begin{matrix}1, 5/7, 9/7\\ 13/7, 15/7\end{matrix}; 1\right).$$

All four series converge absolutely. Formulas (4.2)–(4.3) immediately evaluate the last three series.

To reduce $F_{5/7}$ as well, set

$$U = {}_3F_2\left(\begin{matrix}1, 5/7, 8/7\\ 16/7, 18/7\end{matrix}; 1\right), \qquad V = F_{12/7}.$$

For the summands t_n of U, the corresponding summands of V are $t_n(1 + 7n/5)$. Also $(10/7)_n/(3/7)_n = 1 + 7n/3$, whence

$$F_{5/7} = -\frac{2}{3}U + \frac{5}{3}V.$$

To eliminate U, use $A = 5/7$, $B = 8/7$, $C = 16/7$, $D = 18/7$ and $Q(n) = (n + C - 1)(n + D - 1)$. Here $C + D = A + B + 3$, and

$$Q(n+1)t_{n+1} - Q(n)t_n = (AB - (C-1)(D-1) - n)\, t_n.$$

This time $Q(n)t_n \longrightarrow 0$. Consequently

$$\sum_{n\geq 0} nt_n = -\frac{59}{49}U + \frac{99}{49}, \qquad V = -\frac{24}{35}U + \frac{99}{35}.$$

It follows that

$$U = \frac{99 - 35V}{24}, \qquad F_{5/7} = -\frac{11}{4} + \frac{95}{36}V.$$

Using only $\Gamma(x+1) = x\Gamma(x)$ in the balanced summations gives

$$\begin{aligned} F_{5/7} &= \frac{11(23g_1g_5 - 38g_2g_4)}{3g_1g_5}, \\ G_{5/7} &= -\frac{26(19g_1g_6 - 35g_2g_5)}{7g_2g_5}, \\ F_{12/7} &= \frac{33(5g_1g_5 - 8g_2g_4)}{5g_1g_5}, \\ G_{12/7} &= -\frac{8(g_1g_6 - 2g_2g_5)}{g_2g_5}. \end{aligned}$$

Substituting in (4.4), and noting that $\sin(10\pi/7) = -\sin(3\pi/7)$, proves the two exact integral values

$$J_{5/7} = \frac{\pi}{\sin(3\pi/7)} \frac{7(23g_1^2g_6 - 70g_2^2g_4)}{180g_4g_5g_6}, \tag{4.5}$$

$$J_{12/7} = \frac{\pi}{\sin(3\pi/7)} \frac{-7(5g_1^2g_6 - 16g_2^2g_4)}{60g_4g_5g_6}. \tag{4.6}$$

Their ratio proves (4.1). In particular the denominator in that equation is nonzero, since $J_{5/7} > 0$.

4.4.3 The beta-period obstruction

Euler's reflection formula yields

$$H = \frac{\sin(3\pi/7)}{\sin(\pi/7)} \frac{g_1g_3}{g_2^2} = \frac{\sin(3\pi/7)}{\sin(\pi/7)} \frac{B(1/7, 3/7)}{B(2/7, 2/7)}.$$

The sine ratio is a nonzero algebraic number. For rational beta arguments the criterion of [3] tests algebraic proportionality using the profiles

$$h_{r,s}(u) = \{ur\} + \{us\} - \{u(r+s)\}, \qquad u \in (\mathbb{Z}/7\mathbb{Z})^\times.$$

The two profiles differ already at $u = 3$:

$$h_{1/7,3/7}(3) = \frac{3}{7} + \frac{2}{7} - \frac{5}{7} = 0, \qquad h_{2/7,2/7}(3) = \frac{6}{7} + \frac{6}{7} - \frac{5}{7} = 1.$$

Thus the beta ratio, and hence H, is transcendental. Finally, the rational fractional linear transformation in (4.1) has determinant $15 \cdot 70 - (-48)(-23) = -54 \neq 0$. Such a transformation preserves transcendence at every point where its denominator is nonzero. This completes the proof of the transcendence of the second candidate.

4.5 The third denominator-seven candidate

Put

$$\begin{aligned} J(d) &= J_3\left(\frac{1}{7}, 0, \frac{3}{7}, d, \frac{5}{7}; 0\right) \\ &= \int_{[0,1]^3} \frac{(1-x)^{3/7} y^{5/7} (1-y)^{1/7} z^{1/7} (1-z)^{3/7}}{(1-z+xyz)^d}\, dx\, dy\, dz, \\ K_3 &= \frac{J(11/7)}{J(4/7)}. \end{aligned}$$

These are the parameters of the third denominator-seven candidate; the notation and the hypergeometric reduction of J_3 are those of Zudilin [11]. Define

$$t = 2\cos\frac{2\pi}{7}, \qquad u = 2\cos\frac{\pi}{7} = t^2 + t - 1, \qquad R = \frac{\Gamma(1/7)\Gamma(4/7)}{\Gamma(2/7)\Gamma(3/7)},$$

$$L = \frac{t^2}{1+t} R = (1 + t - t^2)R, \qquad D = \frac{u\Gamma(1/7)\Gamma(3/7)^2}{180}.$$

The equalities involving t follow from $t^3 + t^2 - 2t - 1 = 0$.

Exact evaluation. The following identities hold:

$$J(4/7) = D(462 - 425L), \qquad J(11/7) = D(525L - 567), \qquad K_3 = \frac{525L - 567}{462 - 425L}. \tag{4.7}$$

We give certificates for the evaluations, rather than relying on a numerical identification.

All hypergeometric series below have argument 1. Set

$$C = \frac{\Gamma(20/7)\Gamma(17/7)}{\Gamma(4/7)\Gamma(12/7)}, \qquad C' = \frac{\Gamma(20/7)\Gamma(17/7)}{\Gamma(11/7)\Gamma(12/7)}.$$

The four required series are

$$F_A = {}_4F_3\left(\begin{matrix}8/7, 1, 4/7, 12/7 \\ 1/7, 20/7, 17/7\end{matrix}; 1\right) = \frac{8575}{36} C - \frac{1105}{3}, \tag{4.8}$$

$$F_B = {}_3F_2\left(\begin{matrix}1, 11/7, 12/7 \\ 20/7, 17/7\end{matrix}; 1\right) = \frac{49}{2} C' - 65, \tag{4.9}$$

$$F_C = {}_3F_2\left(\begin{matrix}2, 10/7, 18/7 \\ 26/7, 23/7\end{matrix}; 1\right) = \frac{38000 W_2 - 42408}{77}, \tag{4.10}$$

$$F_E = {}_3F_2\left(\begin{matrix}1, 10/7, 11/7 \\ 19/7, 16/7\end{matrix}; 1\right) = 9(W_1 - 1), \tag{4.11}$$

where Whipple's summation formula [17, Eq. 16.4.7], followed by the gamma recurrence, reflection, and duplication formulas, gives

$$W_2 = {}_3F_2\left(\begin{matrix}2, 3/7, 4/7 \\ 19/7, 16/7\end{matrix}; 1\right) = \frac{\sin(3\pi/7)}{2\sin(\pi/7)}, \tag{4.12}$$

$$W_1 = {}_3F_2\left(\begin{matrix}1, 3/7, 4/7 \\ 12/7, 9/7\end{matrix}; 1\right) = \frac{5\sin^2(\pi/7)}{\sin(2\pi/7)\sin(3\pi/7)}. \tag{4.13}$$

In particular the use of Whipple's formula is inside its convergent range: its parameter c equals 2 and 1, respectively.

Four telescoping certificates. Write $\Delta g_n = g_{n+1} - g_n$, and use $(a)_n$ for the rising factorial. For (4.8), let

$$a_n = \frac{(8/7)_n(1)_n(4/7)_n(12/7)_n}{(1/7)_n(20/7)_n(17/7)_n n!}, \qquad r_A(n) = \frac{(1225n/36 + 119/6)(n^2 + 23n/7 + 130/49)}{n + 1/7}.$$

Direct cancellation of rising factorials gives $\Delta(a_n r_A(n)) = a_n$. The boundary values are

$$a_0 r_A(0) = \frac{1105}{3}, \qquad \lim_{n\to\infty} a_n r_A(n) = \frac{8575}{36}C.$$

Thus the boundary at infinity must be retained, and summation proves (4.8).

For (4.9), let

$$b_n = \frac{(1)_n(11/7)_n(12/7)_n}{(20/7)_n(17/7)_n n!}, \qquad r_B(n) = \frac{49}{2}n^2 + \frac{161}{2}n + 65.$$

Then $\Delta(b_n r_B(n)) = b_n$, with initial value 65 and limit $49C'/2$ at infinity. This proves (4.9).

For (4.10), use the common term

$$h_n = \frac{(2)_n(3/7)_n(4/7)_n}{(26/7)_n(23/7)_n n!}$$

and the rational certificate

$$r_C(n) = \frac{(7n+16)(7n+19)}{49(n+1)}\left(\frac{1953}{22} + \frac{30779}{264}n + \frac{3087}{88}n^2 - \frac{343}{132}n^3\right).$$

Let c_n and $w_{2,n}$ be the usual summands of F_C and W_2. Then the exact rational identity

$$c_n - \frac{38000}{77}w_{2,n} = \Delta(h_n r_C(n))$$

holds. Its initial boundary is $h_0 r_C(0) = 42408/77$ and its boundary at infinity is zero. Summing proves (4.10).

For (4.11), take

$$\widetilde{h}_n = \frac{(1)_n(3/7)_n(4/7)_n}{(19/7)_n(16/7)_n n!}, \qquad r_E(n) = \frac{(7n+9)(7n+12)}{12}.$$

If $e_n, w_{1,n}$ are the summands of F_E, W_1, respectively, then

$$e_n - 9w_{1,n} = \Delta(\widetilde{h}_n r_E(n)).$$

The initial boundary is 9 and the limit at infinity is zero. This proves (4.11). All the displayed boundary limits follow from $\Gamma(n+a)/\Gamma(n+b) \sim n^{a-b}$; all four series converge.

Substitution into the integral. For transparency, put $g_j = \Gamma(j/7)$ and $s_j = \sin(j\pi/7)$. Zudilin's reduction [11] specializes to

$$J(4/7) = \frac{\pi}{s_1}(AF_A - BF_C), \qquad J(11/7) = -\frac{\pi}{s_1}(A'F_B - B'F_E),$$

where

$$A = \frac{g_8^2 g_{10} g_{12}}{g_1 g_{20} g_{17}}, \quad B = \frac{g_8 g_{10}^2 g_{18}}{g_4 g_{26} g_{23}}, \quad A' = \frac{g_8 g_{10} g_{12}}{g_{20} g_{17}}, \quad B' = \frac{g_8 g_{10}^2}{g_{19} g_{16}}.$$

Substitute (4.8)–(4.13), reduce gamma arguments by recurrence, reflection, and duplication, and use

$$s_2 = u s_1, \qquad s_3 = (1+t)s_1, \qquad s_1^2 = \frac{2-t}{4}, \qquad t^3 + t^2 - 2t - 1 = 0.$$

These substitutions give exactly (4.7); every step is an identity in rational functions, gamma recurrences, or the explicit cubic field defined above.

Transcendence. The common-sum beta representation is

$$R = \frac{B(1/7, 4/7)}{B(2/7, 3/7)}.$$

For rational $a, b > 0$ with $a + b < 1$, write $T_{a,b} = \{v \in (\mathbb{Z}/7\mathbb{Z})^* : \{va\} + \{vb\} < 1\}$. Here

$$T_{1/7,4/7} = \{1, 2, 4\}, \qquad T_{2/7,3/7} = \{1, 4, 5\}.$$

Already at $v = 2$ their fractional-part sums are $3/7$ and $10/7$. Wolfart–Wüstholz's beta-value theorem and classical-relation criterion [3, Satz 4 and Proposition 2] therefore show that R is transcendental. The factor $t^2/(1 + t)$ is nonzero algebraic, so L is transcendental. Finally the determinant of the fractional linear transformation in (4.7) is

$$525 \cdot 462 - (-567)(-425) = 1575 \neq 0.$$

Consequently K_3 is transcendental. Positivity and finiteness of $J(4/7)$ also ensure that the displayed denominator is nonzero. This argument establishes transcendence; it does not establish the quantitative irrationality-measure conjecture accompanying the candidate.

4.6 The fourth denominator-seven target

We consider the constant already listed in the DBKZ collection,

$$K_4 = K\left(\frac{1}{7}, 0, \frac{4}{7}, \frac{2}{7}, \frac{5}{7}\right) = \frac{J_1}{J_0}, \qquad J_j = J_3\left(\frac{1}{7}, 0, \frac{4}{7}, \frac{2}{7} + j, \frac{5}{7}; 0\right) \quad (j = 0, 1).$$

Write $g_r = \Gamma(r/7)$ for $1 \leq r \leq 6$, and put

$$W = \frac{g_4^2 g_3}{g_2 g_1^2}, \qquad P = \frac{7\pi g_1^2}{2970 \sin(2\pi/7) g_3 g_6} > 0.$$

We will prove the exact evaluations

$$J_0 = -P(101 - 2744W), \qquad J_1 = 33P(11 - 294W), \tag{4.14}$$

and hence

$$\boxed{K_4 = \frac{9702W - 363}{101 - 2744W}.} \tag{4.15}$$

The identification of this particular target with a gamma expression was already announced, without its formula, in [11, p. 3]. The purpose of the calculation below is to give the explicit expression and an exact, independently checkable certificate, rather than to claim that its gamma evaluation was previously unknown.

4.6.1 Four exact telescoping certificates

For rational A, B, C, D let

$$T_n = \frac{(A)_n (B)_n}{(C)_n (D)_n}.$$

If polynomials p, q satisfy

$$q(n + 1)(n + A)(n + B) = (q(n) + p(n))(n + C)(n + D), \tag{4.16}$$

then $p(n)T_n = q(n+1)T_{n+1} - q(n)T_n$. In each of the four cases below, $m = C + D - A - B = \deg q$ is a positive integer, and the gamma-ratio asymptotic gives

$$\lim_{N \longrightarrow \infty} q(N)T_N = \mathrm{lc}(q)\frac{\Gamma(C)\Gamma(D)}{\Gamma(A)\Gamma(B)}.$$

Consequently the convergent sum is exactly

$$\sum_{n=0}^{\infty} p(n)T_n = \mathrm{lc}(q)\frac{\Gamma(C)\Gamma(D)}{\Gamma(A)\Gamma(B)} - q(0). \tag{4.17}$$

Here the four parameter tuples and polynomials are

r	A	B	C	D	$p_r(n)$
1	2/7	8/7	18/7	20/7	$-(7n-2)(7n+5)/10$
2	11/7	17/7	27/7	29/7	$(n+1)(n+2)/2$
3	8/7	9/7	18/7	20/7	$(7n+5)/5$
4	10/7	11/7	20/7	22/7	$n+1$

and

$$\begin{aligned}
q_1(n) &= \frac{(7n+11)(7n+13)(2107n^2+2765n+606)}{990},\\
q_2(n) &= -\frac{(7n+20)(7n+22)(43n^2+167n+156)}{198},\\
q_3(n) &= -\frac{(7n+11)(7n+13)(21n+22)}{50},\\
q_4(n) &= -\frac{(3n+4)(7n+13)(7n+15)}{10}.
\end{aligned}$$

For every row, identity (4.16) follows by polynomial expansion over $\mathbb{Q}$. The leading coefficients and initial values are

r	m	$\mathrm{lc}(q_r)$	$q_r(0)$
1	4	103243/990	1313/15
2	4	−2107/198	−1040/3
3	3	−1029/50	−1573/25
4	3	−147/10	−78

All four original summands are $O(n^{-2})$; thus there is no regularized or formally assigned sum in this calculation.

Define the four ordinary hypergeometric values

$$\begin{aligned}
F_1 &= {}_4F_3\left(\begin{matrix}1, 2/7, 8/7, 12/7\\ -2/7, 18/7, 20/7\end{matrix}; 1\right),\\
F_2 &= {}_3F_2\left(\begin{matrix}3, 11/7, 17/7\\ 27/7, 29/7\end{matrix}; 1\right),\\
F_3 &= {}_4F_3\left(\begin{matrix}1, 8/7, 9/7, 12/7\\ 5/7, 18/7, 20/7\end{matrix}; 1\right),\\
F_4 &= {}_3F_2\left(\begin{matrix}2, 10/7, 11/7\\ 20/7, 22/7\end{matrix}; 1\right).
\end{aligned}$$

Their summands are precisely $p_r(n)T_n$ in the respective rows. Formula (4.17) proves all four evaluations:

$$\begin{aligned}
F_1 &= \frac{103243}{990}\frac{\Gamma(18/7)\Gamma(20/7)}{\Gamma(2/7)\Gamma(8/7)} - \frac{1313}{15},\\
F_2 &= \frac{1040}{3} - \frac{2107}{198}\frac{\Gamma(27/7)\Gamma(29/7)}{\Gamma(11/7)\Gamma(17/7)},\\
F_3 &= \frac{1573}{25} - \frac{1029}{50}\frac{\Gamma(18/7)\Gamma(20/7)}{\Gamma(8/7)\Gamma(9/7)},\\
F_4 &= 78 - \frac{147}{10}\frac{\Gamma(20/7)\Gamma(22/7)}{\Gamma(10/7)\Gamma(11/7)}.
\end{aligned}$$

4.6.2 Substitution in the Barnes reduction

The Barnes-to-hypergeometric formula in [11, p. 3], with the canceling upper and lower parameters removed, gives

$$J_0 = -\frac{\pi}{\sin(2\pi/7)}(A_0F_1 - B_0F_2),$$
$$J_1 = \frac{\pi}{\sin(2\pi/7)}(A_1F_3 - B_1F_4),$$

where the gamma recurrence $\Gamma(z+1) = z\Gamma(z)$ reduces its prefactors to

$$A_0 = -\frac{7g_1^2}{2574g_3g_6}, \qquad B_0 = \frac{2401g_4^2}{128700g_2g_6}, \qquad A_1 = \frac{35g_1^2}{2574g_3g_6}, \qquad B_1 = \frac{343g_4^2}{1170g_2g_6}.$$

The same recurrence reduces the four proven evaluations to

$$F_1 = -\frac{13(101g_1g_2 - 1204g_4g_6)}{15g_1g_2},$$
$$F_2 = -\frac{520(43g_1g_6 - 98g_3g_4)}{147g_3g_4},$$
$$F_3 = \frac{143(11g_1g_2 - 126g_4g_6)}{25g_1g_2},$$
$$F_4 = -\frac{78(3g_1g_6 - 7g_3g_4)}{7g_3g_4}.$$

Direct rational arithmetic, with no further gamma identities, now gives

$$A_0F_1 - B_0F_2 = \frac{7(101g_1^2g_2 - 2744g_3g_4^2)}{2970g_2g_3g_6},$$
$$A_1F_3 - B_1F_4 = \frac{7(11g_1^2g_2 - 294g_3g_4^2)}{90g_2g_3g_6}.$$

These identities establish (4.14) and (4.15). Both defining integrals are positive and finite; in particular $101 - 2744W \neq 0$ follows from $J_0 > 0$ and $P > 0$.

4.6.3 Transcendence transfer

The gamma recurrence also gives the exact beta quotient

$$W = \frac{1}{7}\frac{B(4/7, 4/7)}{B(1/7, 2/7)}.$$

For completeness, write

$$\epsilon_u(r, s) = \{ur\} + \{us\} - \{u(r+s)\},$$

where braces denote fractional parts. The beta-quotient algebraicity criterion of Wolfart–Wüstholz [3] implies that algebraicity of the quotient here would require the two corresponding functions ϵ_u to agree for every u relatively prime to 7. Already $u = 1$ contradicts this:

$$\epsilon_1(4/7, 4/7) = 1, \qquad \epsilon_1(1/7, 2/7) = 0.$$

Thus W is transcendental. Finally the rational fractional-linear map in (4.15) has nonzero determinant

$$9702 \cdot 101 - 363 \cdot 2744 = -16170.$$

It cannot send a transcendental argument to an algebraic value: an algebraic K_4 would give

$$W = \frac{101K_4 + 363}{2744K_4 + 9702} \in \overline{\mathbb{Q}}.$$

The denominator in this inverse cannot vanish for a value of the nonconstant map. Therefore K_4 is transcendental.

4.7 A Barnes evaluation and transcendence proof for the fifth denominator-seven target

For $d \in \{4/7, 11/7\}$, define the convergent positive integral

$$J_d = \iiint_{[0,1]^3} \frac{(1-x)^{3/7} y^{5/7} (1-y)^{2/7} z^{2/7} (1-z)^{3/7}}{(1-z+xyz)^d}\, dx\, dy\, dz, \qquad C_5 = \frac{J_{11/7}}{J_{4/7}}.$$

Thus $C_5 = K(2/7, 0, 3/7, 4/7, 5/7)$ in the notation of [11]. Both integrals are finite: near $z = 1$, the z-integral for the larger exponent is bounded by a constant times $(xy)^{-1/7}$, which leaves integrable powers of x and y.

Exact evaluation. Put $g_j = \Gamma(j/7)$ for $1 \le j \le 6$ and

$$T = \frac{g_3^2 g_4}{g_5 g_6^2}.$$

Then

$$\boxed{C_5 = \frac{40(13-3T)}{88T-373}.} \tag{4.18}$$

The approximate values are

$$T = 4.270216179039710293695950611\ldots, \qquad C_5 = 2.725438564625374875727251628\ldots.$$

The proof below uses Barnes' second lemma and explicit residues; the numerical values are checks and play no role in the proof.

Mellin–Barnes reduction. The Mellin–Barnes representation in [11], specialized to $a = 2/7, b = 0, c = 3/7, e = 5/7$, gives

$$J_d = P_d M_d, \qquad P_d = \frac{\Gamma(9/7)\Gamma(10/7)}{\Gamma(d)\Gamma(19/7-d)},$$

$$M_d = \frac{1}{2\pi i} \int_{\Re t = -1/2} \frac{\Gamma(d+t)\Gamma(9/7+t)\Gamma(12/7+t)\Gamma(-2-t)\Gamma(10/7-d-t)}{\Gamma(17/7+t)}\, dt.$$

Here the simplifying identity is

$$\frac{\Gamma(1+t)\Gamma(-t)}{\Gamma(3+t)} = \frac{\Gamma(-t)}{(t+1)(t+2)} = \Gamma(-2-t).$$

The chosen vertical line is valid for both values of d.

A contour identity with all residue corrections. Let $S = A + B + C + E - 2$ and set

$$F(t) = \frac{\Gamma(A+t)\Gamma(B+t)\Gamma(C+t)\Gamma(-2-t)\Gamma(E-t)}{\Gamma(S+t)}.$$

For complex parameters with

$$\Re A, \Re B, \Re C > \frac{1}{2}, \qquad \Re E > -\frac{1}{2},$$

one has, with removable singularities interpreted by continuation,

$$\frac{1}{2\pi i} \int_{\Re t = -1/2} F(t)\, dt = G + R_{-2} + R_{-1}, \tag{4.19}$$

where

$$G=\frac{\Gamma(A-2)\Gamma(B-2)\Gamma(C-2)\Gamma(A+E)\Gamma(B+E)\Gamma(C+E)}{\Gamma(A+B+E-2)\Gamma(A+C+E-2)\Gamma(B+C+E-2)},$$

$$R_{-2}=-\frac{\Gamma(A-2)\Gamma(B-2)\Gamma(C-2)\Gamma(E+2)}{\Gamma(S-2)},\qquad R_{-1}=\frac{\Gamma(A-1)\Gamma(B-1)\Gamma(C-1)\Gamma(E+1)}{\Gamma(S-1)}.$$

For a rigorous justification, first assume $\Re A,\Re B,\Re C>9/4$ and $\Re E>0$. The line $\Re t=-9/4$ separates the left poles from the right poles, so Barnes' second lemma [15] evaluates its integral as G. Shifting that line to $\Re t=-1/2$ crosses precisely the poles $t=-2,-1$ of $\Gamma(-2-t)$. Their residues are respectively R_{-2} and R_{-1}, since the residues of this gamma factor are -1 and $+1$. Stirling's estimate gives exponential decay on the horizontal sides. Both vertical lines are oriented upwards, and the resulting identity is $I(-1/2)=I(-9/4)+R_{-2}+R_{-1}$, with the plus signs displayed in (4.19).

The integral on the fixed line $\Re t=-1/2$ is holomorphic throughout the larger connected parameter domain specified above. No numerator pole meets that line; reciprocal gamma is entire; and Stirling's estimate gives locally uniform integrable decay. The right-hand side is meromorphic there. The meromorphic identity theorem therefore continues the identity to that domain, and any apparent poles of the sum on the right are removable. This establishes the formula at the desired parameters without incorrectly using a separating straight contour at their final values.

Application and exact gamma reduction. Take

$$(A,B,C,E)=\left(d,\frac{9}{7},\frac{12}{7},\frac{10}{7}-d\right),\qquad S=\frac{17}{7}.$$

Both $d=4/7$ and $d=11/7$ satisfy the preceding domain conditions. For example, the Barnes term becomes

$$G_d=\frac{\Gamma(d-2)\Gamma(-5/7)\Gamma(-2/7)\Gamma(10/7)\Gamma(19/7-d)\Gamma(22/7-d)}{\Gamma(5/7)\Gamma(8/7)\Gamma(17/7-d)}.$$

For a compact reduction table, define

$$p=\frac{g_2g_3}{g_1g_4},\qquad u=\frac{g_2g_3g_4^2}{g_6},\qquad v=\frac{g_2g_4g_5g_6}{g_3}.$$

Successive uses of $\Gamma(z+1)=z\Gamma(z)$ give

	$d=4/7$	$d=11/7$
P_d	$\frac{3}{4}p$	$\frac{3}{2}p$
G_d	$\frac{308}{75}u$	$-\frac{14}{5}u$
$R_{-2,d}$	$-\frac{637}{50}v$	$\frac{49}{5}v$
$R_{-1,d}$	$-\frac{14}{3}v$	$\frac{7}{3}v$

Consequently,

$$J_{4/7}=\frac{7g_2^2}{200g_1g_6}(88g_3^2g_4-373g_5g_6^2),$$

$$J_{11/7}=-\frac{7g_2^2}{5g_1g_6}(3g_3^2g_4-13g_5g_6^2).$$

Their quotient proves (4.18). In particular $88T-373>0$, as $J_{4/7}>0$.

Transcendence. The beta-function definition and $\Gamma(11/7) = (4/7)\Gamma(4/7)$ give

$$T = \frac{7}{4}\frac{B(3/7, 3/7)}{B(5/7, 6/7)}.$$

We use the following consequence of Wolfart–Wüstholz [3], Satz 4 and Proposition 2. For positive rational arguments, with the individual arguments and their pair sums nonintegral, an algebraic ratio of two beta values requires the equality of their fractional-part profiles

$$h_{a,b}(r) = \{ra\} + \{rb\} - \{r(a+b)\}$$

for every r coprime to a common denominator. The criterion allows a pair sum larger than one; these are among the first- and second-kind beta periods treated in that theorem. For the present ratio and $r = 1$,

$$h_{3/7,3/7}(1) = 0, \qquad h_{5/7,6/7}(1) = 1.$$

Thus T is transcendental. The rational fractional-linear map in (4.18) has determinant

$$(-120)(-373) - 520 \cdot 88 = -1000 \neq 0.$$

Equivalently,

$$T = \frac{373C_5 + 520}{88C_5 + 120},$$

whose denominator is positive because $C_5 > 0$. If C_5 were algebraic, this inverse formula would make T algebraic. Hence

$$\boxed{K(2/7, 0, 3/7, 4/7, 5/7) \text{ is transcendental.}}$$

Scope and priority. The published discussion in [11] explicitly lists this target among the expressions involving ${}_3F_2$ values that seem not reducible to a simpler form. The argument above supplies a gamma evaluation and a transcendence proof using established theorems. Historical priority of this particular reduction has not been established. No claim is made here to prove the associated proposed irrationality measure or divisibility assertions.

Part IV
Gamma quotient fields and continued fractions

5 Balanced Gamma quotients: classification and exact fields

This section concerns balanced products with two Gamma factors in each numerator and denominator. It determines a finite arithmetic classification and the fields generated by the algebraic expressions. The motivating field-dimension question is recorded in [18]; the unconditional transcendence input is the beta-ratio theorem of Wolfart–Wüstholz [3]. All counts below refer to expressions, with a quotient and its reciprocal identified, rather than to pairwise distinct real numbers.

Theorem 5.1. *Let $N \geq 3$, $1 \leq a \leq b < N$, $1 \leq c \leq d < N$, and $a + b = c + d \neq N$. Put*

$$R_N(a, b; c, d) = \frac{\Gamma(a/N)\Gamma(b/N)}{\Gamma(c/N)\Gamma(d/N)}.$$

For a unit $u \pmod N$, define

$$h_{a,b}(u) = \frac{(ua \bmod N) + (ub \bmod N) - (u(a+b) \bmod N)}{N} \in \{0, 1\}.$$

Then $R_N(a, b; c, d)$ is algebraic if and only if $h_{a,b}(u) = h_{c,d}(u)$ for every unit $u \pmod N$. If one equality fails, the quotient is transcendental.

Proof. The balance condition cancels the common gamma factor and gives

$$R_N(a, b; c, d) = \frac{B(a/N, b/N)}{B(c/N, d/N)}.$$

All four arguments are positive nonintegers and their common sum is a noninteger. The beta-ratio criterion of Wolfart–Wüstholz therefore applies, including when the common sum exceeds one. Its character test is precisely the equality of the displayed fractional-part signatures. □

The enumeration convention identifies a quotient with its reciprocal and removes the identity by requiring $(a, b) < (c, d)$ in lexicographic order. The expressions counted need not be pairwise distinct numbers. An exact integer enumeration gives

	denominator divides N			exact common denominator N		
N	all	algebraic	transcendental	all	algebraic	transcendental
21	570	54	516	560	54	506
33	2480	30	2450	2420	30	2390

Here "exact common denominator" imposes $\gcd(N, a, b, c, d) = 1$. This distinction matters because the published question uses the phrase "overall denominator".

For $N = 21$, the nontrivial equal-signature classes consist of 18 classes of size 2 and 12 classes of size 3, giving $18 + 12\binom{3}{2} = 54$ algebraic quotient expressions. For $N = 33$, there are 30 classes of size 2, giving 30 algebraic quotient expressions. The embedded certificate in Section 8 reconstructs these finite lists using integer arithmetic. For each transcendental expression, the first unit with unequal signature entries is an independently checkable witness.

For example,

$$\frac{\Gamma(1/21)\Gamma(3/21)}{\Gamma(2/21)^2} \quad \text{and} \quad \frac{\Gamma(1/33)\Gamma(3/33)}{\Gamma(2/33)^2}$$

are transcendental. The witnesses are respectively $u = 8$ and $u = 13$: the numerator signature equals 0 and the denominator signature equals 1 in each case.

5.1 The complete denominator-21 field

Let $\mathcal{A}_{21}$ consist of the 54 algebraic balanced quotient representatives

$$r(a,b;c,d)=\frac{\Gamma(a/21)\Gamma(b/21)}{\Gamma(c/21)\Gamma(d/21)},$$

where $1\le a\le b<21$, $1\le c\le d<21$, $a+b=c+d\neq 21$, and the unordered pairs are distinct. We choose one orientation from each quotient and its reciprocal. Let

$$H_{21}=\langle r^7 : r\in\mathcal{A}_{21}\rangle\subset\mathbb{R}_{>0},\qquad K_{21}=\mathbb{Q}(H_{21}).$$

Thus H_{21} includes arbitrary products and integer powers, as in the source question [18]. This section determines the whole field and whole linear span, not just a lower bound from one counterexample. The omitted reflection-only cases $a+b=c+d=21$ do not change the answer: their values are sine quotients $s_c/s_a\in F$, and trivial quotients equal 1.

Theorem 5.2. *Put*

$$y=\left(\frac{\Gamma(1/21)\Gamma(4/21)}{\Gamma(2/21)\Gamma(3/21)}\right)^7,\qquad F=\mathbb{Q}(\zeta_{21})^+.$$

Then

$$\boxed{K_{21}=F(y)=\mathbb{Q}(y),\qquad [K_{21}:\mathbb{Q}]=\dim_{\mathbb{Q}}\operatorname{span}_{\mathbb{Q}}H_{21}=36.}$$

More precisely, among the 54 *chosen seventh powers,* 18 *lie in* F*,* 18 *lie in* yF*, and* 18 *lie in* $y^{-1}F$*.*

5.1.1 The cyclotomic containment lemma

For $1\le k\le 10$, set

$$q_k=2\sqrt{3}\sin(k\pi/21).$$

These numbers belong to F. Indeed $\eta=e^{\pi i/21}$ lies in $\mathbb{Q}(\zeta_{21})$, as does $\sqrt{-3}$, and

$$q_k=-\sqrt{-3}\,(\eta^k-\eta^{-k})$$

is real. Consequently, whenever $C,h,m_1,\dots,m_{10}$ are integers and $h-M$ is even, where $M=\sum_{k=1}^{10}m_k$, one has

$$2^C3^{h/2}\prod_{k=1}^{10}\sin(k\pi/21)^{m_k}=2^{C-M}3^{(h-M)/2}\prod_{k=1}^{10}q_k^{m_k}\in F. \tag{5.1}$$

All the sine factors are nonzero, so negative exponents are harmless.

5.1.2 Exact certificates for every quotient

Write $g_k=\log\Gamma(k/21)$ and $s_k=\sin(k\pi/21)$. The identities used in the verification are exactly

$$\begin{aligned}
&R_k: g_k+g_{21-k}=\log\pi-\log s_k && (1\le k\le 10),\\
&T_k: g_k+g_{k+7}+g_{k+14}-g_{3k}=\log(2\pi)+(1/2-k/7)\log 3 && (1\le k\le 6),\\
&P_k: \sum_{j=0}^{6}g_{k+3j}-g_{7k}=3\log(2\pi)+(1/2-k/3)\log 7 && (k=1,2).
\end{aligned}$$

These are reflection, triplication, and septuplication, with positive real arguments. Thus rational linear combinations of their logarithms are valid without any branch ambiguity.

The exact certificate in Section 8 enumerates the balanced pairs and applies the previously established Wolfart–Wüstholz test

$$\left\{\frac{ua}{21}\right\}+\left\{\frac{ub}{21}\right\}=\left\{\frac{uc}{21}\right\}+\left\{\frac{ud}{21}\right\} \quad \text{for every } u\in(\mathbb{Z}/21\mathbb{Z})^{\times}.$$

Exactly 54 of the 570 nontrivial representatives satisfy it. For each of these 54 targets $z = r(a,b;c,d)^7$, the certificate gives an explicit rational linear combination of the displayed identities and verifies an equality

$$\boxed{z = y^{\varepsilon}2^{C}3^{h/2}\prod_{k=1}^{10}s_k^{m_k},} \tag{5.2}$$

where

$$\varepsilon\in\{-1,0,1\},\quad C,h,m_k\in\mathbb{Z},\quad h-\sum_k m_k\in 2\mathbb{Z}.$$

The certificate constructs these identities by rational row reduction of the displayed reflection and multiplication equations. It checks every Gamma exponent vector, the vanishing of the remaining $\log\pi$ and $\log 7$ coefficients, and the stated parity condition. Its complete code is embedded below; no external coefficient table is required.

For example, if $z = r(1,10;2,9)^7$, the table gives

$$z = y\,2^7\sqrt{3}\,s_4^7\in yF.$$

For $z = r(2,8;4,6)^7$, it gives

$$z = y^{-1}\,2^7 3^{5/2}s_6^7\in y^{-1}F.$$

The exact counts for $\varepsilon = -1,0,1$ are $18,18,18$ respectively. Equation (5.1) therefore proves $K_{21}\subseteq F(y)$.

5.1.3 Equality and the exact dimension

For completeness, the degree-36 argument is recalled here. Put $t = 2\cos(2\pi/21)$ and $R = (t+1)/t$. For an explicit seed certificate, the Gamma exponents cancel in

$$2\log R_{21}(1,4;2,3) = -R_2+R_4-R_6-R_8+T_1-T_2-T_3+P_1.$$

Thus, with $s_k = \sin(k\pi/21)$,

$$R_{21}(1,4;2,3) = 2\,3^{1/28}7^{1/12}\sqrt{s_2s_6s_8/s_4} = 3^{1/28}7^{1/12}\sqrt{(t+1)/t}.$$

Here $s_2/s_4 = 1/t$ and $s_6s_8 = (t+1)/4$. Consequently

$$y = 3^{1/4}7^{7/12}R^{7/2},\qquad a = y^6 = 1029\sqrt{21}\,R^{21}\in F.$$

The identity $\sqrt{21}\in F$ can be verified explicitly from

$$\sqrt{21} = 2t^5+2t^4-10t^3-8t^2+12t+3,\quad t^6-t^5-6t^4+6t^3+8t^2-8t+1=0.$$

If an automorphism σ of F fixes a, then $\sigma(R) = \pm R$. The positive choice fixes t. The negative choice would give $\sigma(t) = -t/(2t+1)\in(-1/2,0)$, while all six conjugates $2\cos(2\pi u/21)$, for $u = 1,2,4,5,8,10$, are positive or less than -1. Hence $\mathbb{Q}(a) = F$.

The norm $\mathrm{N}_{F/\mathbb{Q}}(a) = -3^9 7^{42} < 0$ shows that a is not a square in F. Also

$$a = (7\sqrt{21}R^7)^3/7$$

shows it is not a cube: otherwise the abelian extension F would contain the nonnormal field $\mathbb{Q}(\sqrt[3]{7})$. Let $b = y^3 = \sqrt{a}$. Then $[F(b) : F] = 2$. If $Z^3 - b$ had a root in $F(b)$, taking its norm to F would make $-a$, hence a, a cube in F. Thus $[F(y) : F] = 6$, and $[\mathbb{Q}(y) : \mathbb{Q}] = 36$.

Since $y \in H_{21}$, one has $\mathbb{Q}(y) \subseteq K_{21}$; since $F = \mathbb{Q}(y^6)$, the upper bound already proved gives $K_{21} = F(y) = \mathbb{Q}(y)$. Finally $1, y, \dots, y^{35}$ belong to H_{21} and are linearly independent. Every group member lies in the 36-dimensional field K_{21}. Its linear span therefore has dimension exactly 36 as claimed.

The degree-36 seed is already a counterexample to a proposed six-dimensional span of the multiplicative group of seventh powers: $1, y, \dots, y^{35}$ are linearly independent. This is a statement about this explicitly defined group, not about all Gamma monomials.

5.2 The denominator-33 field and individual degrees

Write $g_j = \Gamma(j/33)$ and let $\mathcal{A}_{33}$ consist of the nonidentity algebraic expressions

$$Q = \frac{g_a g_b}{g_c g_d}, \qquad 1 \le a \le b < 33, \quad 1 \le c \le d < 33, \quad a + b = c + d \ne 33,$$

with a quotient and its reciprocal identified. The previously proved Wolfart–Wüstholz classification shows that there are exactly 30 such expressions. This notation counts expressions, not necessarily distinct real numbers.

Theorem 5.3. *Let $\zeta = \exp(2\pi i/33)$ and $F = \mathbb{Q}(\zeta + \zeta^{-1})$. Then*

$$\mathbb{Q}(Q^{11} : Q \in \mathcal{A}_{33}) = F, \qquad [F : \mathbb{Q}] = 10,$$

and

$$\mathbb{Q}(Q : Q \in \mathcal{A}_{33}) = F(3^{1/11}), \qquad [F(3^{1/11}) : \mathbb{Q}] = 110.$$

In fact every individual $Q \in \mathcal{A}_{33}$ generates the respective field: $\mathbb{Q}(Q^{11}) = F$ and $\mathbb{Q}(Q) = F(3^{1/11})$.

Proof. Put

$$t_j = 2\sqrt{3}\sin(j\pi/33), \qquad 1 \le j \le 16.$$

Since $\exp(\pi i/33) = -\zeta^{17}$ and $\sqrt{-3} = 1 + 2\zeta^{11}$, we have the exact cyclotomic expression

$$t_j = -(1 + 2\zeta^{11})((-\zeta^{17})^j - (-\zeta^{17})^{-j}).$$

It is real, so $t_j \in F$.

The reflection and triplication identities take the form

$$g_j g_{33-j} = \frac{\pi}{\sin(j\pi/33)}, \qquad \frac{g_j g_{j+11} g_{j+22}}{g_{3j}} = 2\pi\, 3^{1/2 - j/11} \quad (1 \le j \le 10).$$

For each of the 30 expressions, the exact certificate in Section 8 gives one triplication identity, with exponent $\epsilon \in \{1, -1\}$, and a product of reflection identities. The powers of π cancel. More explicitly, every expression has a certified identity

$$Q = 3^{m/11} \prod_j t_j^{e_j}, \qquad m \in \{-5, -4, -3, -2, -1, 1, 2, 3, 4, 5\}, \qquad e_j \in \{-1, 0, 1\}.$$

Consequently all eleventh powers belong to F, and all original expressions belong to $F(3^{1/11})$. These 30 identities are verified by comparing their exponent vectors in the free abelian group on $g_1, \dots, g_{32}$, together with their explicit scalar factors.

For the reverse inclusion a single expression suffices:

$$Q_0 = \frac{g_1 g_{12}}{g_3 g_{10}} = \frac{t_{10}}{3^{1/11}}, \qquad y = Q_0^{11} = \frac{t_{10}^{11}}{3}.$$

Its square is

$$y^2 = 3^9\bigl(2 - 2\cos(20\pi/33)\bigr)^{11}.$$

The 10 conjugates of $2 - 2\cos(20\pi/33)$ in F are pairwise distinct positive real numbers, because $\gcd(10, 33) = 1$. Raising them to the eleventh power preserves distinctness. Thus $[\mathbb{Q}(y^2) : \mathbb{Q}] = 10$, and $\mathbb{Q}(y) = F$. Moreover $3^{1/11} = t_{10}/Q_0$ belongs to $\mathbb{Q}(Q_0)$, so $\mathbb{Q}(Q_0) = F(3^{1/11})$. The degrees of F and $\mathbb{Q}(3^{1/11})$ are respectively 10 and 11; the latter follows from Eisenstein's criterion. Their compositum therefore has degree 110.

For the stronger assertion about every individual expression, the embedded certificate performs exact arithmetic in $\mathbb{Q}[z]/(\Phi_{33}(z))$, applies every substitution $z \mapsto z^u$ for $u \in (\mathbb{Z}/33\mathbb{Z})^\times$, and verifies exactly 10 different conjugates of each Q^{11}. This proves $\mathbb{Q}(Q^{11}) = F$. Since $\gcd(m, 11) = 1$, the displayed formula for Q then recovers $3^{1/11}$ by Bézout's identity and proves $\mathbb{Q}(Q) = F(3^{1/11})$ for every expression. □

Exact-computation specification. The modulus used by the exact polynomial calculation is

$$\Phi_{33}(z) = z^{20} - z^{19} + z^{17} - z^{16} + z^{14} - z^{13} + z^{11} - z^{10} + z^9 - z^7 + z^6 - z^4 + z^3 - z + 1.$$

The embedded certificate verifies this polynomial by the identity

$$\Phi_{33}(z) = \frac{(z^{33} - 1)(z - 1)}{(z^{11} - 1)(z^3 - 1)}.$$

The calculation uses rational coefficients, derives all 30 exponent certificates directly from reflection and triplication, reconstructs the complete fractional-part list, and counts exact Galois images. Neither floating-point approximations nor integer-relation guesses are used.

Relation to the published question. This proves the conjectured dimension 10 for the field generated by eleventh powers of the balanced four-gamma expressions at denominator 33. However, the accompanying assertion that 10 of the original expressions have degree 55 is false as written: all 30 have degree 110. No assertion here extends the classification to arbitrary products of gamma values with more than two factors in numerator and denominator.

6 Generalized Brouncker fractions and effective decisions

For $s, r \in \mathbb{Q}$, $s > 0$, $r > 1/2$, let

$$Y(s, r) = s + \mathop{\mathbf{K}}_{n=1}^{\infty} \frac{(2n-1)^2 r^2 - (r-1)^2}{2s}.$$

Kushel's Theorem 1.4 [19] gives the convergent value

$$Y(s, r) = 4r\frac{\Gamma(a + 1/2)\Gamma(b + 1/2)}{\Gamma(a)\Gamma(b)}, \qquad a = \frac{s+1}{4r}, \quad b = \frac{s + 2r - 1}{4r}. \tag{6.1}$$

Notice that $a, b > 0$ and $|a - b| < 1/2$.

Theorem. If $2a \in \mathbb{Z}$ or $2b \in \mathbb{Z}$, then $Y(s, r)$ is transcendental. Otherwise choose any even positive integer N for which $Na, Nb \in \mathbb{Z}$. Then $Y(s, r)$ is algebraic if and only if, for every integer k coprime to N, the two fractional parts $\{ka\}$ and $\{kb\}$ lie in opposite open intervals $(0, 1/2)$ and $(1/2, 1)$. Consequently one same-half witness k is a finite transcendence certificate.

Proof. Write $R(x) = \Gamma(x+1/2)/\Gamma(x)$. For an integer $m \geq 1$, $R(m) \in \mathbb{Q}^\times\sqrt{\pi}$, and for a positive nonintegral half-integer h, $R(h) \in \mathbb{Q}^\times/\sqrt{\pi}$. If both a, b are half-integers, the strict bound $|a-b| < 1/2$ forces $a = b$, so their product is a nonzero rational multiple of π or $1/\pi$ and is transcendental. If precisely one, say a, is a half-integer, there are two possibilities. For integral a,

$$\sqrt{\pi}R(b) = bB(b+1/2, 1/2),$$

and for nonintegral half-integral a,

$$R(b)/\sqrt{\pi} = 1/B(b, 1/2).$$

The beta arguments and their sum are nonintegers. Schneider's beta-value theorem therefore proves transcendence. This also explains why the strict parameter restriction matters: an integer and a nonintegral half-integer can cancel their square-root-of-π factors in unrestricted gamma products.

Now assume neither $2a$ nor $2b$ is integral. The gamma quotient equals

$$Y(s,r) = 4r(a+b)\frac{B(a+1/2, b+1/2)}{B(a,b)}. \tag{6.2}$$

First suppose $a + b \notin \mathbb{Z}$. All arguments and sums needed for the Wolfart–Wüstholz beta-ratio criterion [3] are nonintegral. For rational u, v put $h_{u,v}(k) = \{ku\} + \{kv\} - \{k(u+v)\}$. That criterion says that the quotient in (6.2) is algebraic exactly when its two profiles agree on $(\mathbb{Z}/N\mathbb{Z})^\times$. Every such k is odd. If $x = \{ka\}$ and $y = \{kb\}$, then neither is 0 or $1/2$, and direct calculation gives

$$h_{a+1/2,b+1/2}(k) - h_{a,b}(k) = 1 - \mathbf{1}_{x>1/2} - \mathbf{1}_{y>1/2}.$$

This difference vanishes exactly when x and y lie in opposite halves.

If $a + b \in \mathbb{Z}$, gamma recurrence and reflection directly make the quotient algebraic: writing $t = \{a\} \in (0,1) \setminus \{1/2\}$, one reduces it, up to a nonzero rational factor, to

$$\frac{\Gamma(t+1/2)\Gamma(3/2-t)}{\Gamma(t)\Gamma(1-t)} = (1/2-t)\tan(\pi t) \in \overline{\mathbb{Q}}.$$

For every unit k, the corresponding fractional parts sum to 1; thus they lie in opposite halves, as required. This covers all cases.

For example, the opposite-half test gives

$$Y(2,3) = \frac{3^{1/4}(\sqrt{6}+\sqrt{2})}{2} \in \overline{\mathbb{Q}},$$

whereas $Y(2,4)$ is transcendental: $(a,b) = (3/16, 9/16)$ and $k = 3$ gives the same-half witness $(9/16, 11/16)$. These are applications to a pre-existing family, without a claim of historical priority.

6.1 A terminating minimality and positivity algorithm

The preceding finite classifier can be upgraded from existence of algebraic minimal initial data to a decision procedure for *every specified pair* of real algebraic initial values. No general period-equality conjecture is needed for this particular recurrence family.

Effective classical gamma certificates. Put $G(x) = \Gamma(x)/\sqrt{\pi}$ for $x \in \mathbb{Q}_{>0}$. The following identities have explicitly computable positive real algebraic right-hand sides:

$$\begin{aligned} G(x+1)/G(x) &= x, \\ G(x)G(1-x) &= 1/\sin(\pi x) \quad (0 < x < 1), \\ \frac{\prod_{j=0}^{m-1} G(x+j/m)}{G(mx)} &= 2^{(m-1)/2} m^{1/2-mx} \quad (m \geq 2), \\ G(1/2) &= 1. \end{aligned}$$

For each identity record its finitely supported rational exponent vector in the symbols E_x, together with its algebraic constant. Enumerate the positive rational arguments x and positive integers m, and at every finite stage solve a finite rational linear system to ask whether

$$T = E_{a+1/2} + E_{b+1/2} - E_a - E_b$$

lies in the span of the enumerated exponent vectors. If $T = \sum_j q_j R_j$, with identity constants $c_j > 0$, then

$$\frac{\Gamma(a+1/2)\Gamma(b+1/2)}{\Gamma(a)\Gamma(b)} = \prod_j c_j^{q_j}.$$

All rational powers mean positive real roots, so no branch ambiguity arises. Real algebraic arithmetic computes the right-hand side exactly, for example by a polynomial together with an isolating rational interval.

This search terminates whenever the classifier returns algebraic. Wolfart–Wüstholz, Satz 4, says that an algebraic quotient of two positive rational beta values is classically proportional; their printed page 6 defines classical relations as consequences of the functional, reflection, and distribution relations. Proposition 1 on that page is the finite fractional-part test for these relations. Thus a finite certificate exists. Arguments outside $(0, 1)$ are reduced by recurrence; negative intermediate arguments, if used in a classical derivation, can likewise be shifted to positive ones without changing its validity up to explicit nonzero rational factors. The two cases excluded from a direct nonresonant beta argument are already covered in the classifier: half-integer boundaries are transcendental, and integral $a + b$ has an explicit reflection evaluation. Consequently no unproved general assertion about gamma algebraic relations is being invoked.

Theorem (specified initial data). There is an unconditional algorithm which, given rational $r > 1/2$, $s > 0$ and real algebraic v_{-1}, v_0, decides whether the nonzero solution of

$$(2rn+1)v_n = 2sv_{n-1} + (2rn-1)v_{n-2} \qquad (n \geq 1)$$

is minimal.

Algorithm and proof. To check the continued-fraction normalization, put $P_{-1} = P_0 = 1$, $P_n = \prod_{j=1}^{n}(2rj+1)$ and $u_n = P_n v_n$. Then

$$u_n = 2su_{n-1} + (2rn-1)(2r(n-1)+1)u_{n-2}.$$

The second coefficient equals $(2n-1)^2r^2 - (r-1)^2$, exactly the numerator of the fraction defining $Y(s,r) - s$. Scaling preserves ratios of solutions and therefore minimality. Pincherle's theorem gives the unique minimal initial slope $v_0/v_{-1} = s - Y(s,r)$.

The zero initial pair is separately identified; the theorem concerns nonzero solutions. If $v_{-1} = 0$, answer nonminimal. Otherwise run the finite classifier for $Y(s,r)$. If it returns transcendental, answer nonminimal, since Pincherle's criterion requires $v_0/v_{-1} = s - Y(s,r)$, whereas the left-hand side is algebraic. If it returns algebraic, run the terminating classical-certificate search to obtain an exact algebraic representation of $Y(s,r)$. Decide the equality

$$v_0/v_{-1} = s - Y(s,r)$$

using effective real algebraic arithmetic, answering minimal exactly when it holds. Each branch terminates and the criterion is necessary and sufficient.

This decides every specified real algebraic input pair in the stated two-parameter family. It resolves this restriction of the polynomial-continued-fraction equality/minimality problem discussed by Kenison et al. [20]. It does not settle general second-order holonomic minimality, and the historical priority of this particular corollary has not been established.

A positivity consequence with an elementary proof. The same algorithm decides positivity (all terms, including the specified initial terms, nonnegative) for

$$(2rn+1)w_n = -2sw_{n-1} + (2rn-1)w_{n-2} \qquad (n \geq 1).$$

The zero sequence is positive. Every other positive sequence must have $w_{-1} > 0$: a negative initial term is forbidden, and $w_{-1} = 0 < w_0$ would force $w_1 < 0$. Normalize $w_{-1} = 1$ and put $t = w_0/w_{-1}$. After the same positive scaling as before, let P_n/Q_n be the nth convergent of the positive continued fraction $Y(s,r) - s$, with $P_0 = 0$, $Q_0 = 1$, $P_{-1} = 1$, $Q_{-1} = 0$. Its numerator and denominator recurrences have $b_n = 2s$ and $a_n = (2rn-1)(2r(n-1)+1) > 0$. The scaled solution is

$$W_n = (-1)^n(Q_n t - P_n).$$

Consequently positivity is equivalent to t lying above every even convergent and below every odd convergent. Both subsequences converge to $Y(s,r) - s$, so the condition is exactly

$$w_{-1} > 0, \qquad w_0/w_{-1} = Y(s,r) - s,$$

apart from the separately accepted zero sequence. This is an equality between the same effectively classified constant and an algebraic input, and is decided by the preceding algorithm. The proof uses convergence already supplied by the generalized Brouncker formula and the elementary ordering of positive continued-fraction convergents.

7 Barnes reductions to beta values and logarithms

For $e \in \mathbb{Q}$, $0 < e < \frac{1}{2}$, put $d = \frac{1}{2} - e$ and define

$$J_{p,e} = \int_{[0,1]^3} \frac{\sqrt{x(1-x)(1-z)}\, y^e}{(1-z+xyz)^p}\, dx\, dy\, dz, \qquad K_e = \frac{J_{d+1,e}}{J_{d,e}}.$$

Both integrals are finite and strictly positive. The underlying Mellin–Barnes representation is given in [11]. Every K_e in this family is transcendental. The particular values $e = 1/6$ and $e = 1/3$ belong to this uniform family. No assertion of historical priority for this family is intended.

Here is an explicit formula, including the coefficient that certifies that the fractional-linear transformation is nonconstant. Write

$$B_e = B(2e, \tfrac{1}{2}), \qquad A_e = \frac{2(2e+1)}{(1-e^2)(4e+1)}, \qquad q_e = \frac{(e+1)(4e+1)}{(2e-1)(2e+1)},$$

$$R_{0,e} = \frac{e+3}{e(e-1)(2e+3)}, \qquad R_{1,e} = \frac{2e^2+5e+1}{e(e-1)(2e-1)(2e+1)}.$$

Then the exact evaluations are

$$J_{d,e} = \pi(A_e B_e + R_{0,e}), \qquad J_{d+1,e} = \pi(q_e A_e B_e + R_{1,e}), \tag{7.1}$$

and consequently

$$K_e = \frac{q_e A_e B_e + R_{1,e}}{A_e B_e + R_{0,e}}, \qquad q_e R_{0,e} - R_{1,e} = \frac{1}{(2e-1)(2e+1)(2e+3)} \neq 0. \tag{7.2}$$

Barnes reduction. It is useful first to allow $0 < b < 1$, $e > 0$, $0 < d < 2 - b$, and set

$$\mathcal{J}_d(b,e) = \int_{[0,1]^3} \frac{x^b(1-x)^{1-b}y^e(1-z)^{1-b}}{(1-z+xyz)^d}\,dx\,dy\,dz.$$

For $d \neq 1$ and $e \neq 1$, the Mellin–Barnes representation and

$$\frac{\Gamma(1+s)\Gamma(-s)}{\Gamma(3+s)} = \Gamma(-2-s)$$

give

$$\mathcal{J}_d(b,e) = \frac{\Gamma(2-b)}{\Gamma(d)\Gamma(3-b-d)}\frac{1}{2\pi i}\int_{\Re s=\sigma} H_d(s)\,ds,$$
$$H_d(s) = \frac{\Gamma(d+s)\Gamma(1+b+s)\Gamma(1+e+s)\Gamma(-2-s)\Gamma(2-b-d-s)}{\Gamma(2+e+s)}.$$

For the two exponents $d, d+1$ one may use a common line $\sigma = -\delta$, where

$$\max(0, b+d-1) < \delta < \min(d, 1).$$

This interval is nonempty since $0 < b < 1$ and $0 < d < 2 - b$. Barnes's second lemma, with the residues at $s = -2, -1$ included, evaluates this integral as

$$\mathcal{J}_d(b,e) = \frac{\pi}{\sin \pi b}(R_d + X_d), \tag{7.3}$$

where

$$R_d = \frac{3-b-d}{(d-1)(d-2)(e-1)} + \frac{1-b}{e(d-1)},$$
$$X_d = -\frac{\Gamma(e-1)\Gamma(2-b)\Gamma(3-d)\Gamma(3+e-b-d)}{(d-1)(d-2)\Gamma(3-b-d)\Gamma(1+e-b)\Gamma(2+e-d)}.$$

The same expression, evaluated at $d+1$, is valid even when $d+1 \geq 2-b$: the contour condition just given and the convergence conditions for the original integral are the ones used for that evaluation. Equivalently, this is the continuation of the displayed formula to the second exponent. The two elementary residue contributions after multiplying by the prefactor are exactly

$$\frac{\pi}{\sin \pi b}\frac{3-b-d}{(d-1)(d-2)(e-1)}, \qquad \frac{\pi}{\sin \pi b}\frac{1-b}{e(d-1)}.$$

For completeness, the contour deformation does not presuppose a separating line at the final parameters. Set $(A, B, C, E) = (d, 1+b, 1+e, 2-b-d)$ and $S = A+B+C+E-2$. Initially take $\Re A, \Re B, \Re C > 9/4$ and $\Re E > 0$. Barnes's lemma applies on the separating line $\Re s = -9/4$ to

$$\frac{\Gamma(A+s)\Gamma(B+s)\Gamma(C+s)\Gamma(-2-s)\Gamma(E-s)}{\Gamma(S+s)}.$$

Moving the line to $-\delta \in (-1, 0)$ crosses just $s = -2, -1$; Stirling's formula makes the horizontal sides vanish. With lines oriented upwards the value on the new line is the Barnes value plus these two residues. Both sides then continue meromorphically to the connected domain $\Re A, \Re B, \Re C > \delta$, $\Re E > -\delta$. The fixed-line integral is holomorphic there. This domain contains both required parameter points. The exclusions above avoid the individual gamma poles in the displayed formula; they are automatic in the one-parameter family.

Gamma recurrence gives, identically,

$$X_{d+1} = q_d X_d, \qquad q_d = -\frac{(2-b-d)(1+e-d)}{d(2+e-b-d)},$$
$$q_d R_d - R_{d+1} = \frac{b(b-1)}{d(d-2)(b+d-e-2)}.$$

These expressions also explain why the generic reduction alone does not prove transcendence: a nonconstant fractional-linear expression still requires an arithmetic theorem for its gamma-value parameter.

The beta value on the chosen line. Put $r = 1 - b$ and $t = 2 - d$ and introduce

$$S(e, r, t) = \frac{\Gamma(e)\Gamma(r)\Gamma(t)\Gamma(e + r + t)}{\Gamma(e + r)\Gamma(r + t)\Gamma(t + e)}.$$

Then $X_d = rS(e, r, t)/((e - 1)(d - 1))$. On our line $b = 1/2$, $d = 1/2 - e$, hence $r = 1/2$ and $t = e + 3/2$. Gamma recurrence yields

$$S(e, \tfrac{1}{2}, e + \tfrac{1}{2}) = 2B(2e, \tfrac{1}{2}), \qquad \frac{S(e, \frac{1}{2}, e + \frac{3}{2})}{S(e, \frac{1}{2}, e + \frac{1}{2})} = \frac{(2e + 1)^2}{(e + 1)(4e + 1)}.$$

Substitution gives $X_d = A_e B_e$, $R_d = R_{0,e}$, $R_{d+1} = R_{1,e}$, and $q_d = q_e$, establishing (7.1).

To check convergence directly, it suffices to examine z near 1. Writing $u = 1 - z$ there, the denominator is comparable to $u + xy$. For a general exponent p, integration in u is bounded by a constant, a logarithm, or a multiple of $(xy)^{2-b-p}$. The resulting endpoint integrals are finite when $p < \min(3 - b, 3, 3 + e - b)$. Both $p = d$ and $p = d + 1$ satisfy this condition in the stated domain. Positivity is immediate from the integrands and proves that the denominators in the ratio formula do not vanish.

Transcendence. For rational $0 < e < 1/2$, the positive rational arguments $2e$ and $1/2$ are both nonintegers. Their sum is nonintegral unless $e = 1/4$. Schneider's beta-value theorem therefore proves that B_e is transcendental for $e \neq 1/4$; at $e = 1/4$ it is simply $B(1/2, 1/2) = \pi$. All coefficients in (7.2) are rational, $A_e \neq 0$, and the displayed determinant is nonzero. If K_e were algebraic, inversion of this fractional-linear map would make B_e algebraic, a contradiction. This proves the stated infinite-family result.

As checks on the normalization, at $e = 1/6$, writing $V = B(1/3, 1/2)$, the formula becomes

$$K_{1/6} = \frac{35(17 - 4V)}{2(32V - 133)}.$$

At $e = 1/3$, writing $H = B(2/3, 2/3)/(3\,2^{2/3})$ and using $B(2/3, 1/2) = 6H$, it becomes

$$K_{1/3} = \frac{2002 - 4620H}{825H - 350}.$$

7.1 The resonant line and a two-parameter logarithmic family

The pole collision at $e = 1$ in the gamma-value formula admits a different arithmetic treatment. This section uses Baker's theorem on logarithms [21] rather than an unjustified classification of arbitrary gamma products.

Let $b, d \in \mathbb{Q}$ satisfy

$$0 < b < 1, \qquad 0 < d < 2 - b, \qquad d \neq 1,$$

and define

$$J_p = \int_{[0,1]^3} \frac{x^b(1 - x)^{1-b} y(1 - z)^{1-b}}{(1 - z + xyz)^p}\, dx\, dy\, dz, \qquad K(b, d) = \frac{J_{d+1}}{J_d}.$$

Then $K(b, d)$ is transcendental for every rational pair in this domain. Both integrals are finite and positive. This is an extension proved here; it is not asserted that the members of this family were individually published as open problems or that the general statement has no prior occurrence.

Put $r = 1 - b$, $t = 2 - d$, so that

$$0 < r < 1, \qquad 1 - r < t < 2, \qquad t \neq 1.$$

Write $\psi(z) = \Gamma'(z)/\Gamma(z)$ and

$$\Psi = \psi(1) + \psi(1+r+t) - \psi(1+r) - \psi(1+t).$$

The exact evaluations are

$$\frac{J_d}{\pi/\sin\pi b} = \frac{r + (r+t)\Psi/t}{d-1} = \alpha\Psi + \beta, \tag{7.4}$$

$$\frac{J_{d+1}}{\pi/\sin\pi b} = \frac{r + \dfrac{r+t-1}{t-1}\left(\Psi + \dfrac{r}{t(r+t)}\right)}{d} = \alpha_1\Psi + \beta_1. \tag{7.5}$$

All four coefficients are rational, and their determinant is

$$\alpha_1\beta - \alpha\beta_1 = \frac{r(r-1)}{t^2(t-2)(t-1)} \neq 0. \tag{7.6}$$

Thus it suffices to prove that Ψ is transcendental.

The limiting formula. For e near 1, replace y by y^e and call the resulting integral $J_p(e)$. The preceding Barnes reduction gives

$$J_p(e) = \frac{\pi}{\sin\pi b}\{R_p(e) + X_p(e)\},$$

with R_p, X_p defined in the preceding subsection. Its fixed contour and meromorphic continuation apply to $p = d, d+1$ in the present domain. This explicitly retains the residues at $-2, -1$ before taking the coalescing-pole limit.

Define

$$S(e,r,t) = \frac{\Gamma(e)\Gamma(r)\Gamma(t)\Gamma(e+r+t)}{\Gamma(e+r)\Gamma(r+t)\Gamma(t+e)}.$$

Gamma recurrence rewrites $X_d(e) = rS(e,r,t)/((e-1)(d-1))$. In particular,

$$S(1,r,t) = \frac{r+t}{rt}, \qquad \frac{\partial}{\partial e}\log S(e,r,t)\Big|_{e=1} = \Psi.$$

The pole parts in

$$R_d + X_d = -\frac{r+t}{(d-1)t(e-1)} + \frac{r}{e(d-1)} + \frac{rS(e,r,t)}{(e-1)(d-1)}$$

cancel. Taking the finite part gives (7.4). For the exponent $d+1$, replace t by $t-1$. Gamma recurrence, or the digamma recurrence, gives

$$\Psi(r,t-1) = \Psi(r,t) + \frac{r}{t(r+t)},$$

which proves (7.5). Continuity of the integrals at $e = 1$ follows by domination with $y^{1-\varepsilon}$ for a fixed sufficiently small positive ε. All required convergence inequalities are strict in the stated domain.

Arithmetic of the digamma combination. For rational $0 < x < 1$, Gauss's digamma formula [22] expresses $\psi(x) + \gamma$ as an algebraic multiple of π plus an algebraic linear form in real logarithms of positive algebraic numbers. Integer shifts add rational numbers. Baker's theorem says that a nonzero algebraic linear form in logarithms of algebraic numbers is transcendental; it also says that logarithms which are linearly independent over $\mathbb{Q}$ are linearly independent over $\overline{\mathbb{Q}}$.

Suppose first that $r+t \notin \mathbb{Z}$. The three numbers $r, t, r+t$ are then nonintegral. The coefficient of π in Gauss's expression for Ψ is

$$c = -\frac{1}{2}\big(\cot \pi(r+t) - \cot \pi r - \cot \pi t\big).$$

This is a nonzero algebraic number. In fact, writing $u = \cot \pi r$, $v = \cot \pi t$, we have $u + v \neq 0$ and

$$\cot \pi(r+t) - \cot \pi r - \cot \pi t = -\frac{u^2 + uv + v^2 + 1}{u+v} \neq 0.$$

After a rational constant is subtracted, Ψ is therefore a linear form containing $c\pi = -ic\log(-1)$ with nonzero coefficient, together with logarithms of positive real algebraic numbers. The logarithm $\log(-1) = i\pi$ is linearly independent over $\mathbb{Q}$ of those real logarithms. Choosing a rational basis for the latter and applying Baker's linear-independence theorem proves that this whole linear form cannot vanish. Baker's transcendence theorem then proves that Ψ is transcendental.

If $r + t \in \mathbb{Z}$, the inequalities imply $r + t = 2$. Digamma recurrence shows that Ψ differs by a rational number from

$$F(r) = -2\gamma - \psi(r) - \psi(1-r).$$

The cotangent terms in Gauss's formula cancel, so $F(r)$ is an algebraic linear form in logarithms of positive algebraic numbers, with no added algebraic constant. It is nonzero: ψ is strictly increasing on the positive real axis, hence $\psi(r) < \psi(1) = -\gamma$ and $\psi(1-r) < \psi(1) = -\gamma$, giving $F(r) > 0$. Baker's theorem again proves transcendence. This settles every case in the stated domain. Finally, (7.6) transfers this transcendence to $K(b, d)$.

For example, the choice $b = d = 1/2$ gives

$$\Psi = 4\log 2 - \frac{19}{6}, \qquad J_{1/2} = \frac{\pi}{9}(67 - 96\log 2), \qquad J_{3/2} = \pi(16\log 2 - 11),$$

and therefore

$$K(\tfrac{1}{2}, \tfrac{1}{2}) = \frac{144\log 2 - 99}{67 - 96\log 2} = 1.77603428343791655449\ldots.$$

The excluded line $d = 1$ involves a further collision and is not covered by this digamma argument.

8 Embedded exact certificate for the two Gamma fields

The finite enumeration and field-containment steps used above involve integer arithmetic and rational linear algebra. The following executable certificate is included to make every finite assertion reproducible from this single source. It requires Python 3 and SymPy but reads no input files and makes no numerical approximations. Assertions verify the counts, every classical Gamma exponent identity, all cyclotomic containment conditions, and all thirty degree-10 Galois orbits. The mathematical degree-36 seed proof and the degree-110 extension argument are the conceptual proofs given above; the program does not replace the beta-ratio theorem by a numerical test.

```
# Exact finite certificates; Python 3 with SymPy. No input files.
from collections import defaultdict, Counter
from itertools import combinations
from math import gcd
import sympy as S

def classified(N):
    units=[u for u in range(1,N) if gcd(u,N)==1]
    groups=defaultdict(list)
    for a in range(1,N):
        for b in range(a,N):
            if a+b!=N: groups[a+b].append((a,b))
    def sig(a,b):
```

```
        return tuple(((u*a)%N+(u*b)%N-(u*(a+b))%N)//N
                     for u in units)
    total=exact=0; alg=[]; exact_alg=0
    for group in groups.values():
        for (a,b),(c,d) in combinations(group,2):
            q=(a,b,c,d)
            primitive=gcd(N,gcd(a,gcd(b,gcd(c,d))))==1
            total+=1; exact+=primitive
            if sig(a,b)==sig(c,d):
                alg.append(q); exact_alg+=primitive
    return sorted(alg), (total,len(alg),exact,exact_alg)

def target(q,N,power=1):
    v=[S.S(0)]*(N-1)
    for k in q[:2]: v[k-1]+=power
    for k in q[2:]: v[k-1]-=power
    return v

# N=21: exact row reduction of classical identities.
A21,counts=classified(21)
assert counts==(570,54,560,54)
L=[]; W=[]
for k in range(1,11):
    v=[S.S(0)]*20; w=[S.S(0)]*14
    v[k-1]=v[20-k]=1; w[3]=1; w[3+k]=-1
    L.append(v); W.append(w)
for step,m,K in [(7,3,range(1,7)),(3,7,range(1,3))]:
    for k in K:
        v=[S.S(0)]*20; w=[S.S(0)]*14
        for j in range(m): v[k+step*j-1]+=1
        v[m*k-1]-=1
        w[0]=w[3]=S.Rational(m-1,2)
        w[1 if m==3 else 2]=S.Rational(1,2)-S.Rational(m*k,21)
        L.append(v); W.append(w)
L=S.Matrix(L); W=S.Matrix(W)
def express(q):
    v=S.Matrix(target(q,21,7))
    c,parameters=L.T.gauss_jordan_solve(v)
    c=c.subs({p:0 for p in parameters})
    assert L.T*c==v
    return list(c.T*W)
base=express((1,4,2,3)); split=Counter()
assert base==[7,S.Rational(1,4),S.Rational(7,12),0,
              0,S.Rational(7,2),0,-S.Rational(7,2),0,
              S.Rational(7,2),0,S.Rational(7,2),0,0]
for q in A21:
    v=express(q); epsilon=v[2]/base[2]
    assert epsilon in (-1,0,1)
    rem=[a-epsilon*b for a,b in zip(v,base)]
    C,h=rem[0],2*rem[1]; M=sum(rem[4:])
    assert rem[2:4]==[0,0]
    assert all(x.q==1 for x in [C,h]+rem[4:])
    assert (h-M)%2==0
    split[int(epsilon)]+=1
assert split=={-1:18,0:18,1:18}
print('N21:',counts,'containment classes:',dict(split))

# N=33: derive all identities, then count exact Galois orbits.
A33,counts=classified(33)
assert counts==(2480,30,2420,30)
z=S.Symbol('z'); phi=S.Poly(S.cyclotomic_poly(33,z),z,domain=S.QQ)
assert S.expand(phi.as_expr()*(z**11-1)*(z**3-1)
                -(z**33-1)*(z-1))==0
one=S.Poly(1,z,domain=S.QQ)
def mul(a,b): return (a*b).rem(phi)
def power(a,n):
    if n<0: a=S.invert(a,phi); n=-n
    out=one
    while n:
        if n%2: out=mul(out,a)
        a=mul(a,a); n//=2
    return out
Z=[S.Poly(z**k,z,domain=S.QQ).rem(phi) for k in range(33)]
ts={j:mul(-(-1)**j*(one+2*Z[11]),
```

```
          Z[(17*j)%33]-Z[(-17*j)%33]) for j in range(1,17)}
pows={(j,e):power(t,11*e) for j,t in ts.items() for e in (-1,1)}
units=[u for u in range(1,33) if gcd(u,33)==1]
for q in A33:
    v=target(q,33); found=[]
    for j in range(1,11):
        for sign in (-1,1):
            w=v.copy()
            for k in (j,j+11,j+22): w[k-1]-=sign
            w[3*j-1]+=sign
            if all(w[k-1]==w[32-k] for k in range(1,17)):
                refs={k:w[k-1] for k in range(1,17) if w[k-1]}
                found.append((j,sign,refs))
    assert found
    j,sign,refs=found[0]; m=-sign*j
    assert sum(refs.values())==-sign and 1<=abs(m)<=5
    assert all(v in (-1,1) for v in refs.values())
    y=S.Poly(S.Rational(3)**m,z,domain=S.QQ)
    for k,e in refs.items(): y=mul(y,pows[k,-e])
    images=[]
    for u in units:
        image=S.Poly(0,z,domain=S.QQ)
        for (k,),coef in y.terms(): image+=coef*Z[(u*k)%33]
        images.append(tuple(image.all_coeffs()))
    assert len(set(images))==10
print('N33:',counts,'each Q^11 has 10 images; each Q degree 110')
```

The first output has counts $(570, 54, 560, 54)$ and containment multiplicities $18, 18, 18$. The second has counts $(2480, 30, 2420, 30)$ and ten distinct Galois images for each eleventh power. In each count tuple the entries are (total expressions, algebraic expressions, exact-denominator expressions, exact-denominator algebraic expressions).

Part V
Incomplete periods and arithmetic quantiles

9 Closed periods and the length of polynomial lemniscates

The passage from an integral representation to a transcendence statement requires attention to the integration cycle. In particular, a sum or a limit of transcendental lengths need not be transcendental. The following construction puts the entire length on one connected algebraic curve.

Theorem 9.1 (Polynomial lemniscate length)**.** *Let $P \in \overline{\mathbb{Q}}[z]$ be nonconstant. The Euclidean length of $\Lambda_P = \{z \in \mathbb{C} : |P(z)| = 1\}$ is transcendental. The statement allows several components, critical levels, and repeated polynomial factors. Length is measured as the length of the set, without algebraic multiplicity. Each geometric connected component also has transcendental length.*

The question motivating this statement appears in [23, Remark 8.16]. The proof below is a proposed resolution of that stated mathematical question; historical priority and external refereeing are not asserted. The transcendence input is the established closed-period theorem: a nonzero integral of an algebraic rational differential over a closed integral cycle avoiding its poles, on a smooth connected projective curve over $\overline{\mathbb{Q}}$, is transcendental [2, Theorem 13.9 and Corollary 13.13].

Proof. Write $P = Q^d$, choosing algebraic constants so that the multiplicities $m_1, \dots, m_r$ of the distinct zeros of Q have greatest common divisor one. Then $\Lambda_P = \Lambda_Q$. Put $n = \deg Q$ and let $\overline{Q}$ denote coefficientwise complex conjugation. We first show that

$$Q(z)\overline{Q}(w) = 1 \tag{9.1}$$

is geometrically irreducible. Over $\overline{\mathbb{Q}}(z)$ consider $\overline{Q}(w) - 1/Q(z)$. At a zero of Q of multiplicity m_j, every root ξ in a splitting field has valuation $-m_j/n$: it cannot have nonnegative valuation, and for negative valuation the leading term of $\overline{Q}(\xi)$ uniquely has smallest valuation. A monic factor of degree e would consequently have constant coefficient of valuation $-em_j/n$, an integer. Thus $n \mid em_j$ for every j, whence $n \mid e$. No proper factor exists. Gauss's lemma gives the assertion for (9.1).

Set $z = x + iy$, $w = x - iy$, and normalize the projective closure to obtain a connected smooth projective curve X with a real algebraic model. The highest homogeneous part of its plane equation is $|a_n|^2(x^2 + y^2)^n$. Thus there are no real points at infinity. The real locus $X(\mathbb{R})$ is a finite union of circles. It is nonempty, since a regular value of Q on the unit circle has a nonempty inverse image.

We check that the normalization map is an immersion along this real locus, including above critical points. Near a point z_0 with $|Q(z_0)| = 1$, choose a local logarithm of $Q(z)/Q(z_0)$. A local biholomorphic coordinate $u = h(z)$ makes this logarithm u^m, for some $m \geq 1$. The level set is then $\mathrm{Re}(u^m) = 0$, a union of m distinct real lines. The normalization separates these lines, and h^{-1} has nonzero derivative. Consequently

$$q = (dx)^2 + (dy)^2 = dz\, dw$$

is positive and nonvanishing on every real tangent line of $X(\mathbb{R})$. Normalization is one-to-one off finitely many crossing points, which have length zero, so adding the lengths of these circles counts the original set exactly once.

Choose a nonzero algebraic rational differential τ on X and write $q = f\tau^2$. If f is not a square in $\overline{\mathbb{Q}}(X)$, let Y be the smooth connected curve with function field $\overline{\mathbb{Q}}(X)(\sqrt{f})$ and put $\omega = \sqrt{f}\,\tau$. If f is a square, take $Y = X$ with either global square root. In both cases $\omega^2 = q$ after pullback.

For the nonsquare case, orient a real circle and parametrize it smoothly by $c(t)$. There is a unique lift at each point for which

$$\omega(\dot{\tilde{c}}(t)) = \sqrt{\dot{x}(t)^2 + \dot{y}(t)^2} > 0.$$

The square-root cover is unramified here because q does not vanish. This choice is continuous, and after one circuit both the position and the tangent vector have returned to their starting values. Uniqueness therefore makes the lift closed. In the square case the global square root has a constant sign on the tangent direction of each real circle; orient the circles separately to make their integrals positive. There are no poles on the resulting lifted cycles. Their sum Γ is a closed integral cycle on the single connected curve Y, and

$$\int_{\Gamma} \omega = \mathrm{length}(\Lambda_Q) > 0.$$

The closed-period theorem proves transcendence. Selecting only the normalization circles belonging to a specified geometric component gives the componentwise statement. □

The same conclusion holds for $|P(z)| = \rho$ with algebraic $\rho > 0$, by replacing P with P/ρ. The nonvanishing of the real tangent vector is essential: the algebraic curve parametrized by $z(t) = (1 + e^{it})^2$, $-\pi \le t \le \pi$, has a cusp and length $\int_{-\pi}^{\pi} 4\cos(t/2)\,dt = 16$. It is not a counterexample to Theorem 9.1, because its normalization map is not an immersion at the cusp.

9.1 A length certificate for a cubic lemniscate

For $L_3 = \mathrm{length}\{|z^3 - z| = 1\}$, the theorem yields transcendence. One useful positive series is obtained from

$$y(t)^3 - ty(t) = 1, \quad y(0) = 1, \qquad G(t) = \left(\frac{3}{3y(t)^2 - t}\right)^{1/2} = \sum_{j\ge 0} g_j t^j, \quad G(0) = 1.$$

The coefficients are rational. The map $\phi(u) = u\,y(u^{-2})$ parametrizes the lemniscate once on $|u| = 1$; its three branches above u^3 remain distinct because the discriminant does not vanish there. Since $|\phi'(u)| = |G(u^{-2})|^2$, Fourier orthogonality gives

$$L_3 = 2\pi \sum_{j\ge 0} g_j^2, \qquad \sum_{j>M} g_j^2 \le \frac{1089}{40}(4/9)^{M+1}. \tag{9.2}$$

For completeness, the bound follows by using $|t| = 3/2$. Every root of $y^3 - ty - 1$ has modulus less than $3/2$, hence $|3y_j^2 - t| \le 33/4$, while $\prod_j (3y_j^2 - t) = 27 - 4t^3$ has modulus at least $27/2$. Thus $|G|^2 \le 121/8$, and Cauchy's estimate gives (9.2). It yields

$$L_3 = 6.464706327813477265623126732842916032 78\ldots.$$

This proves neither transcendence of L_3/π nor algebraic independence of different lemniscate lengths. It is unrelated to maximizing length among monic polynomials of a fixed degree.

10 A logarithm lemma for cyclotomic abelian components

We use the analytic subgroup theorem in its strengthened form [2, Theorem 6.2]: if an algebraic linear subspace of the Lie algebra of a connected commutative algebraic group contains a logarithm of an algebraic point, it contains the Lie algebra of a connected algebraic subgroup containing that logarithm. The identity exponential image is allowed. The following formulation also accommodates the additive directions needed for second-kind differentials.

Lemma 10.1 (Abelian projection of logarithm relations)**.** *Let*

$$0 \longrightarrow V \longrightarrow G \xrightarrow{\rho} A \longrightarrow 0$$

be an extension over $\overline{\mathbb{Q}}$ of an abelian variety by a vector group. Suppose an automorphism T of G descends to A, its characteristic polynomial on $H_1(A, \mathbb{Q})$ is the single cyclotomic polynomial Φ_M, and an algebraic invariant differential η on G satisfies $T^*\eta = \xi\eta$, with ξ a primitive Mth root of unity. Assume η has at least one nonzero period. Identify $K = \mathbb{Q}[T|_A]$ with $\mathbb{Q}(\xi)$ using this eigenvalue. Then the following statements hold.

1. *If w is a logarithm of an algebraic point of G and $\eta(w) \in \overline{\mathbb{Q}}$, then $d\rho(w) = 0$.*
2. *If u, v are such logarithms, $d\rho(v) \neq 0$, and $\eta(u) - \lambda\eta(v) = c$ with $\lambda, c \in \overline{\mathbb{Q}}$, then $\lambda \in K$ and*

$$d\rho(u) = da\big(d\rho(v)\big),$$

where $a \in K \subset \mathrm{End}^0(A)$ corresponds to λ. In particular, if $d\rho(v)$ is a rational period, so is $d\rho(u)$.

Proof. The vector group is contractible, so $H_1(G, \mathbb{Q}) = H_1(A, \mathbb{Q})$. The kernel of the period map $\mathcal{P}_\eta$ is T-invariant. By irreducibility of Φ_M and the assumed nonzero period, it is zero. Moreover, since $H_1(A, \mathbb{Q})$ is one-dimensional over K,

$$\operatorname{im} \mathcal{P}_\eta = \Omega K \quad (\Omega \neq 0). \tag{10.1}$$

These are assertions about rational homology, not its complexification.

For the first assertion, apply the analytic subgroup theorem to $(w, 1)$ in $G \times \mathbb{G}_a$ and the hyperplane $\eta(Z) - \eta(w)t = 0$. Let H be the subgroup obtained, and let B be its image in A. The kernel of $H \to B$ is a vector group; hence a nonzero rational cycle of a positive-dimensional B lifts to H. Its image in $H_1(A, \mathbb{Q})$ is nonzero. But the hyperplane equation forces its η-period to vanish, since dt has zero periods. This contradicts injectivity. Thus $B = 0$ and $d\rho(w) = 0$.

For the second assertion apply the theorem to $(u, v, 1)$ in $G^2 \times \mathbb{G}_a$ and the corresponding homogeneous hyperplane. Its subgroup has positive-dimensional second abelian projection because $d\rho(v) \neq 0$. A rational cycle with nonzero second projection gives

$$\lambda = \frac{\mathcal{P}_\eta(\gamma_1)}{\mathcal{P}_\eta(\gamma_2)} \in K$$

by (10.1). Choose $A_0(X) \in \mathbb{Q}[X]$ with $A_0(\xi) = \lambda$. The quasi-endomorphism $A_0(T)$ exists on G and descends to a on A. Clear denominators and apply the first assertion to $mu - d(mA_0(T))v$, whose η-value is mc. This gives the desired abelian projection identity. Quasi-endomorphisms preserve rational periods. □

When $G = A$ and η is a nonzero holomorphic eigendifferential, the nonzero-period hypothesis follows automatically: a holomorphic differential vanishing on every lattice period is zero. Thus in this case $\eta(w) = 0$ forces $w = 0$. If v is a nonzero logarithm of a torsion point, the same lemma also gives the useful equivalence

$$\frac{\eta(u)}{\eta(v)} \in \overline{\mathbb{Q}} \quad \Longleftrightarrow \quad \exp_A(u) \text{ is torsion}, \tag{10.2}$$

provided u is a logarithm of an algebraic point. The reverse implication uses (10.1). An arbitrary algebraic scalar has not been treated as an endomorphism: membership in K was proved first.

11 Incomplete hyperelliptic integrals and leaf division

For $N \geq 1$ put

$$I_N(s) = \int_0^s \frac{dt}{\sqrt{1-t^N}}, \qquad R_N(s) = \frac{I_N(s)}{I_N(1)}, \quad 0 \leq s \leq 1,$$

with the positive real square root. Let $s_{N,q}$ be the unique solution of $R_N(s) = q$ in $(0,1)$. These quantities also satisfy

$$R_N(s) = \mathrm{I}_{s^N}(1/N, 1/2), \tag{11.1}$$

where $\mathrm{I}_x(a,b) = B_x(a,b)/B(a,b)$ is the regularized incomplete beta function.

11.1 The primitive component and rational division

For $N \geq 3$ consider $H_N : y^2 = 1 - x^N$. Its holomorphic differentials are $\omega_k = x^{k-1}dx/y$, $1 \leq k \leq \lfloor (N-1)/2 \rfloor$. Under $T(x,y) = (\zeta_N x, y)$ the eigenvalue is ζ_N^k, and the characteristic polynomial on rational homology is

$$\chi_N(X) = \begin{cases} (X^N-1)/(X-1), & N \text{ odd}, \\ (X^N-1)/(X^2-1), & N \text{ even}. \end{cases}$$

The rational idempotent selecting the factor Φ_N gives an algebraic quotient $\pi : J_N \to A_N$, after clearing denominators, with

$$H_1(A_N, \mathbb{Q}) \simeq \mathbb{Q}(\zeta_N), \qquad \dim A_N = \varphi(N)/2.$$

This is the usual rational-idempotent construction [24]. For $(k,N) = 1$ and $1 \leq k < N/2$, there are differentials η_k on A_N with $\pi^*\eta_k = \omega_k$. No simplicity of A_N is assumed. Lemma 10.1 applies to η_1.

Let $O = (0,1)$ and $P(s) = (s, \sqrt{1-s^N})$. The actual logarithm vector $V(s)$ along the real path satisfies

$$\exp_{A_N} V(s) = \pi([P(s) - O]), \qquad \eta_k(V(s)) = I_{N,k}(s) := \int_0^s \frac{t^{k-1}dt}{\sqrt{1-t^N}}.$$

Theorem 11.1 (All rational division ratios)**.** *For every $q \in \mathbb{Q} \cap (0,1)$,*

$$s_{N,q} \in \overline{\mathbb{Q}} \quad \Longleftrightarrow \quad N \in \{1,2,3,4,6\}.$$

More generally, if $\varphi(N) \geq 4$ and $0 \leq a < b \leq 1$ are algebraic, the interior solution of $\int_a^s (1-t^N)^{-1/2}dt = q\int_a^b (1-t^N)^{-1/2}dt$ is transcendental for every rational $0 < q < 1$.

Proof. Suppose $\varphi(N) \geq 4$ and $q = a_0/b_0$ with integers $0 < a_0 < b_0$. If s were algebraic, the vector $U = b_0V(s) - a_0V(1)$ would exponentiate to an algebraic point, and $\eta_1(U) = 0$. The lemma gives $U = 0$. There is a primitive index $1 < k < N/2$. With $w(t) = (1-t^N)^{-1/2}$, the weighted mean

$$M_k(s) = \frac{\int_0^s t^{k-1}w(t)dt}{\int_0^s w(t)dt}$$

has derivative

$$M_k'(s) = \frac{w(s)}{I_N(s)^2}\int_0^s (s^{k-1} - t^{k-1})w(t)dt > 0.$$

Thus $\eta_k(U) = b_0 I_N(s)(M_k(s) - M_k(1)) < 0$, a contradiction. The same argument with base point $P(a)$ proves the truncated-interval statement.

For $N = 1$, $s_{N,q} = 2q - q^2$. For $N = 2$, $s_{N,q} = \sin(\pi q/2)$ is algebraic by a root-of-unity expression. For $N = 3, 4$ the curves are elliptic, and $[P(1) - O]$ is torsion. One uniform verification, also valid for later use, is to put $O' = (0, -1)$ and $W_r = (r, 0)$, $r^N = 1$, and use

$$\operatorname{div} \frac{y-1}{y+1} = N(O - O'), \qquad \operatorname{div} \frac{x-r}{x} = 2W_r - O - O'. \tag{11.2}$$

Hence $2N[W_r - O] = 0$. A rational multiple of a torsion logarithm on an elliptic curve exponentiates to an algebraic torsion point, proving the claim for $N = 3, 4$. For $N = 6$ use the finite map $(x, y) \mapsto (u, v) = (x^2, xy)$ to $v^2 = u - u^4$, for which $du/v = 2dx/y$ and both endpoint images are branch points. The same elliptic argument and algebraicity of finite fibers finish the proof. □

The primitive-component argument subsumes the earlier prime-leaf argument using the absolute simplicity of the Jacobian of $y^2 = x - x^{p+1}$ for primes $p \geq 5$. In that proof the increasing weight was t^2. Here irreducibility of the rational cyclotomic action replaces simplicity and includes $N = 8$ and all composite exponents with $\varphi(N) \geq 4$.

11.2 Geometric division and its boundary exceptions

For $C_n = \{|z^n - 1| = 1\}$, a half-leaf has polar equation $r^n = 2\cos(n\theta)$ and length from the origin to radius r equal to

$$\ell(r) = 2^{1/n} I_{2n}(r/2^{1/n}). \tag{11.3}$$

This is also the geometric setting of [23, Section 3.4 and Remark 5.7]. For every $n \geq 4$, every interior rational division of a half-leaf has transcendental radius. Each Cartesian coordinate is separately transcendental: if one were algebraic, the equation

$$(X^2 + Y^2)^n - 2\operatorname{Re}(X + iY)^n = 0$$

would be a monic polynomial equation for the other, making the radius algebraic too. The two coordinates are not algebraically independent, since they satisfy this curve equation.

For a traversal starting at the origin and covering all $2n$ half-leaves, the jth point in an M-fold equal-length division is a half-leaf boundary precisely when $M \mid 2nj$. These boundary points are algebraic origins or tips; all other division points have both coordinates transcendental. Consequently all division points are algebraic exactly when $M \mid 2n$. Repeated visits to the origin count as parameter times, not as different geometric points.

11.3 Algebraic ratios and exceptional torsion

Equation (11.2) and (10.2) give, for algebraic $0 < s < 1$,

$$R_N(s) \in \overline{\mathbb{Q}} \quad \Longleftrightarrow \quad \pi([P(s) - O]) \text{ is torsion.} \tag{11.4}$$

It is the projected point that is torsion; the entire Jacobian point need not be torsion for a general composite N.

Theorem 11.2 (Algebraic ratios)**.** *1. If $N = 8$, or $N = p$ or $2p$ for a prime $p \geq 5$, then $R_N(s)$ is transcendental at every algebraic $s \in (0, 1)$. Equivalently, $s_{N,q}$ is transcendental for every algebraic $q \in (0, 1)$.*

2. If $\varphi(N) \geq 4$, the set of algebraic $s \in (0, 1)$ for which $R_N(s)$ is algebraic is finite. At any such exception, $q = R_N(s)$ satisfies

$$\mathbb{Q}(q) = \mathbb{Q}(\zeta_N + \zeta_N^{-1}), \qquad \deg q = \varphi(N)/2,$$

and all its conjugates are distinct and lie in $(0, 1)$.

Proof. For $N = p$, the full Jacobian is the primitive component. The change $(X, Y) = (-x, y)$ gives $Y^2 = X^p + 1$, and $\operatorname{div}(y-1) = p(O - \infty)$ shows that the torsion condition uses the same base point as the known torsion-packet classification [25, Theorem 6.2.21]. In the interval $-1 < X < 0$ there are no torsion points: the branch points and $X = 0$ are absent, and the additional $p = 5$ orbit has $X^5 = 4$.

For $N = 2p$ use instead $y^2 = x - x^{p+1}$ and $P(s) = (s^2, s\sqrt{1-s^{2p}})$. Pullback of dx/y is $2ds/\sqrt{1-s^{2p}}$, and the full homology has polynomial Φ_{2p}. Both reference endpoints are branch points. The map

$$X = -1/x, \qquad Y = i\, y/x^{(p+1)/2}$$

sends the curve to $Y^2 = X^p + 1$, sends the base point to infinity, and places $P(s)$ at $X < -1$. The same torsion list excludes it.

For $N = 8$ use $y^2 = x - x^5$ and $P(s) = (s^2, s\sqrt{1-s^8})$. The automorphism $(x, y) \mapsto (ix, \zeta_8 y)$ acts with characteristic polynomial Φ_8 on the full genus-two Jacobian. If $R_8(s)$ were algebraic, $[P(s) - (0, 0)]$ would be torsion. Put $X = \zeta_8 x$, $Y = \delta y$ with $\delta^2 = \zeta_8$; the curve becomes $Y^2 = X^5 + X$. By [26, Theorem (0.3)], its torsion packet based at infinity consists of infinity and points with X a root of $X^5 + X$ or $X^4 \pm 4X^2 + 1$. The base-point change is torsion. For $0 < x < 1$ these polynomials evaluate to $\zeta_8 x(1 - x^4)$ and $1 - x^4 \pm 4ix^2$, all nonzero. This proves (1).

For (2), Lemma 10.1 gives $V(s) = da(V(1))$ for $a \in \mathbb{Q}(\zeta_N)$. Since the first eigenvalue q is real, it belongs to the maximal real subfield. For each primitive $1 \le k < N/2$, its conjugate is

$$\sigma_k(q) = \frac{I_{N,k}(s)}{I_{N,k}(1)} \in (0, 1).$$

These values strictly decrease with k. Indeed, for $k < l$ the difference of cross products is

$$\int_0^s \int_s^1 \frac{t^{k-1} r^{l-1} - t^{l-1} r^{k-1}}{\sqrt{1-t^N}\sqrt{1-r^N}}\, dr\, dt > 0.$$

Thus q has full degree in the real cyclotomic field. Finally, the curve $Z = \pi([H_N - O])$ contains zero and generates A_N. It cannot be a torsion translate of an elliptic subvariety when $\dim A_N \ge 2$. The Manin–Mumford theorem [27] makes its torsion intersection finite. The nonconstant map from H_N to Z has finite fibers, proving finiteness of the exceptional s. □

For example, $q = (\sqrt{5} - 1)/2$ and $q = 1/\sqrt{2}$ have negative conjugates. The theorem excludes algebraic $s_{N,q}$ for $\varphi(N) \ge 4$. For $N = 3, 4, 6$ an algebraic real ratio would belong to the real part of $\mathbb{Q}(\zeta_N)$, which is $\mathbb{Q}$. For $N = 2$ the conclusion follows from Gelfond–Schneider applied to $e^{i\pi q}$. Thus for each of these two probabilities, $s_{N,q}$ is transcendental for every $N \ge 2$. For $N = 1$ the inverse remains $2q - q^2$ and is algebraic.

12 Local nonvanishing and uniform degree bounds

The finite exceptional set in Theorem 11.2 does not by itself decide an individual composite-exponent input. A local logarithm supplies a different test. We normalize valuations by $v_\ell(\ell) = 1$.

Lemma 12.1 (A ramified local test)**.** *Let $N \ge 3$, let $\ell \ge 5$ be prime with $\ell \nmid N$, and let an algebraic $s \in (0, 1)$ have an ℓ-adic embedding for which it is a unit and*

$$h = v_\ell(1 - s^N) > 2/(\ell - 1).$$

Then $R_N(s)$ is transcendental. The local field containing s may be ramified.

Proof. Hensel's lemma gives $r^N = 1$ with the same residue as s. Set $z = \sqrt{1 - s^N}$, so $v_\ell(z) = h/2$. The curve H_N has good reduction here, and at $W_r = (r, 0)$ the local parameter y gives $x = r(1 - y^2)^{1/N}$. The local abelian integral is

$$\mathcal{L} = \int_{W_r}^{P(s)} \frac{dx}{y} = -\frac{2r}{N} \sum_{j \geq 0} \frac{(1 - 1/N)_j}{j!(2j + 1)} z^{2j+1}. \tag{12.1}$$

The Abel–Jacobi map is integral on the residue disk, and $v_\ell(z) > 1/(\ell - 1)$ places the evaluation in the small convergence domain of the formal logarithm. Differentiation and termwise integration identify (12.1) with its logarithm coordinate; functoriality is the standard one for ℓ-adic abelian logarithms [28].

The coefficients $(1-1/N)_j/j!$ belong to $\mathbb{Z}_\ell$. If $k = v_\ell(2j+1)$, then $\ell^k \leq 2j+1$ and $\ell^k \geq 1+k(\ell-1)$, so $v_\ell(2j + 1) \leq 2j/(\ell - 1)$. Relative to the first term, each subsequent term has valuation at least

$$jh - v_\ell(2j + 1) \geq j\big(h - 2/(\ell - 1)\big) > 0.$$

These differences tend to infinity. The first term uniquely has the smallest valuation, and $v_\ell(\mathcal{L}) = h/2 < \infty$. If the projected point in (11.4) were torsion, then so would be $\pi([P(s) - W_r])$, by (11.2). Its logarithm is zero. Since dx/y is pulled back from A_N, this would force $\mathcal{L} = 0$, a contradiction. □

What transfers between the complex and local arguments is the torsion property of an algebraic point. The real integral has not been identified with a p-adic numerical value.

Theorem 12.2 (All rational radii)**.** *For every integer $N \geq 3$ and every rational $s \in (0, 1)$, $R_N(s)$ is transcendental. If $N = 2^h M$, $M \geq 3$, and $s^{2^h} \in \mathbb{Q} \cap (0, 1)$, the same conclusion holds.*

Proof. Write $s = a/b$ in lowest terms. Bang–Zsigmondy supplies a primitive prime divisor ℓ of $b^N - a^N$ for $N \geq 3$, except at $(b, a, N) = (2, 1, 6)$; see [29] for the classical input. The residue of a/b has order N, hence $N \mid \ell - 1$, $\ell \geq 5$, and $\ell \nmid Nab$. Now $v_\ell(1 - s^N) \geq 1 > 2/(\ell - 1)$, so Lemma 12.1 applies. At the exceptional triple use $\ell = 7$, since $2^6 - 1 = 63$ and the required valuation is one. For the extension put $s^{2^h} = a/b$ and choose the same good prime for $b^M - a^M$. The roots of $X^{2^h} - a/b$ lie in an unramified extension at the odd prime ℓ, and $1 - s^N = 1 - (a/b)^M$ has positive integral valuation. Apply the local lemma again. □

This theorem evaluates rational positions; Theorem 11.1 instead inverts rational ratios. Neither direction should be substituted for the other. Some illustrative transcendental values are

$$\begin{aligned} R_{12}(1/2) &= 0.450017109099808036 11\ldots, \\ R_{12}(2^{-1/4}) &= \mathrm{I}_{1/8}(1/12, 1/2) \\ &= 0.76065803913457167869\ldots. \end{aligned}$$

Theorem 12.3 (Height-independent bounded-degree criterion)**.** *For $D \geq 1$ set*

$$C(D) = \max\{3^D,\ 4000(D \log(3D))^{12}\}.$$

If $s \in \overline{\mathbb{Q}} \cap (0, 1)$, $[\mathbb{Q}(s) : \mathbb{Q}] \leq D$, and $N > C(D)$, then $R_N(s)$ is transcendental. In particular, $D = 2$ and $N > 20\,000\,000\,000$ suffice.

Proof. The precise primitive-divisor input is Voutier's [29, Theorem 2(i)]: for coprime nonzero algebraic integers A, B with A/B not a root of unity and of degree d, a prime ideal dividing $B^N - A^N$ but no earlier $B^j - A^j$ exists when

$$N > \max\{2(2^d - 1), 4000(d \log(3d))^{12}\}.$$

Its bound uses the degree of the ratio, not that of an auxiliary field.

To handle denominators, in $K = \mathbb{Q}(s)$ write $(s) = \mathfrak{a}\mathfrak{b}^{-1}$ with coprime integral ideals. Choose h with $\mathfrak{b}^h = (b_0)$, adjoin B with $B^h = b_0$, and put $A = sB$. In that extension, $(B) = \mathfrak{b}$ and $(A) = \mathfrak{a}$ after extension of ideals, since the fractional ideal group is torsion-free. Thus A, B are coprime algebraic integers and $A/B = s$. Apply Voutier and contract the resulting prime to a prime $\mathfrak{p}$ of K. The residue of s still has exact order N, because the residue-field inclusion is injective.

Let $\mathfrak{p} \mid \ell$ have ramification index e and residue degree f. Then

$$N \mid \ell^f - 1, \qquad \ell \nmid N, \qquad v_\ell(1 - s^N) \geq 1/e, \qquad ef \leq d \leq D.$$

If $\ell \leq 2e + 1$, then $N + 1 \leq \ell^f \leq (2e + 1)^f \leq 3^{ef} \leq 3^D$, a contradiction. Thus $\ell > 2e + 1$ and $1/e > 2/(\ell - 1)$; the local lemma applies. The initial Voutier threshold is covered because $2(2^d - 1) \leq 3^d \leq 3^D$. Finally, $\log 6 < 9/5$ gives $4000(2 \log 6)^{12} < 4000(18/5)^{12} < 20\,000\,000\,000$. □

The bound is explicit but is not claimed to be optimal. For a fixed algebraic s it confines possible exceptional exponents to a finite interval independently of its height. Conversely, any exceptional sequence with $N \to \infty$ must have input degrees tending to infinity.

A small ramified example does not need the large bound. Let $s_0 = 3 - \sqrt{7}$. For $N = 6m$ with $7 \nmid m$, expansion at $\sqrt{7}$ shows

$$v_7(1 - s_0^N) = 1/2 > 1/3 = 2/(7 - 1) :$$

the constant term $1 - 3^N$ has valuation at least one, the linear term $N3^{N-1}\sqrt{7}$ has valuation exactly $1/2$, and all remaining terms have valuation at least one. Therefore every $R_{6m}(s_0)$ in this range is transcendental. For example

$$R_{12}(3 - \sqrt{7}) = 0.31883299555500762110\ldots.$$

13 Second-kind beta integrals and polynomial weights

For positive rational parameters write $B_x(a, b) = \int_0^x t^{a-1}(1 - t)^{b-1}dt$ and $B(a, b) = B_1(a, b)$. The restriction that a beta differential be holomorphic can be removed by preserving its pole data in a generalized Jacobian.

Theorem 13.1 (Affine independence of beta periods)**.** *Let p, r be distinct primes with $\max(p, r) \geq 5$, and let $a = i/p$, $b = j/r$, where $1 \leq i < p$, $1 \leq j < r$. For every algebraic $x \in (0, 1)$, the three numbers*

$$1, \quad B(a, b), \quad B_x(a, b)$$

are linearly independent over $\overline{\mathbb{Q}}$. There is no restriction $a + b < 1$.

Proof. On the smooth projective model of $C : U^p + V^r = 1$, with its unique point ∞ at infinity, consider

$$\eta = U^{i-1}V^{j-r}dU.$$

It is regular at all finite points. At infinity its order is

$$\operatorname{ord}_\infty \eta = pr - ri - pj - 1. \tag{13.1}$$

Since the prime denominators are distinct, $a + b \neq 1$. Thus η is holomorphic if $a + b < 1$, and otherwise has a pole of order at least two. Its residue is zero by the residue theorem, since it has no other pole.

Choose $M \geq 2$ at least its pole order. The generalized Jacobian for the modulus $M\infty$ is an extension

$$0 \longrightarrow \mathbb{G}_a^{M-1} \longrightarrow G \xrightarrow{\rho} J \longrightarrow 0.$$

There is an algebraic Abel map $j_O : C \setminus \{\infty\} \to G$ based at $O = (0, 1)$ and an invariant differential $\widetilde{\eta}$ with $j_O^* \widetilde{\eta} = \eta$. Here is a local check of the differential condition. The pinched curve for this modulus has completed local ring $\overline{\mathbb{Q}} + t^M \overline{\mathbb{Q}}[[t]]$. The Rosenlicht condition is $\mathrm{Res}(f\eta) = 0$ for every element f of this ring. The constant term contributes zero by the zero residue, and the remaining terms remove the pole. Thus η belongs to its dualizing differentials and corresponds to an invariant differential of G. This standard algebraic construction, including the vector extension and Abel-map interpretation, is described in [30, 31].

The rotation $T(U, V) = (\zeta_p U, \zeta_r V)$ preserves the modulus. The holomorphic characters $u/p + v/r < 1$, together with their conjugates, give every primitive prth root exactly once on $H_1(J, \mathbb{Q})$. Its characteristic polynomial is Φ_{pr}, and $T^* \widetilde{\eta} = \xi \widetilde{\eta}$ with $\xi = \zeta_p^i \zeta_r^j$ primitive. Unlike the holomorphic case, nonzeroness of a second-kind differential does not suffice to assert a nonzero period. We exhibit one. Let ℓ be the positive real arc from $O = (0, 1)$ to $W = (1, 0)$, and put $S(U, V) = (\zeta_p U, V)$, $R(U, V) = (U, \zeta_r V)$. Then $(1 - R_*)(1 - S_*)\ell$ is closed and

$$\int_{(1-R_*)(1-S_*)\ell} \eta = (1 - \zeta_r^j)(1 - \zeta_p^i) \frac{B(a, b)}{p} \neq 0. \tag{13.2}$$

Its image in G is a period, so Lemma 10.1 applies.

For $P_x = (x^{1/p}, (1 - x)^{1/r})$, let u_x and v be the logarithms of $j_O(P_x)$ and $j_O(W)$ along these real paths. Then

$$\widetilde{\eta}(u_x) = B_x(a, b)/p, \qquad \widetilde{\eta}(v) = B(a, b)/p.$$

The divisors of $V - 1$ and $U - 1$ are respectively $p(O - \infty)$ and $r(W - \infty)$, so $[W - O]$ is torsion in J. Its path logarithm $d\rho(v)$ is nonzero: evaluate it on the holomorphic differential with indices $i = j = 1$, whose real path integral is strictly positive and whose holomorphicity follows from $1/p + 1/r < 1$.

Suppose $c_0 + c_1 B(a, b) + c_2 B_x(a, b) = 0$ with algebraic coefficients. If $c_2 \neq 0$, the second part of the lemma forces $[P_x - O]$ to be torsion in J, hence $[P_x - \infty]$ is torsion. For distinct odd primes, $(X, Y) = (-U, V)$ identifies the curve with $Y^r = X^p + 1$. The torsion classification [25, Theorem 6.2.21] excludes finite torsion points with both coordinates nonzero. If one prime is two, reorder the variables and write $Y^2 = X^p + 1$ with $X \in (-1, 0)$. The only extra case, $p = 5$, has $X^5 = 4$, also impossible on this interval. Thus $c_2 = 0$. If $c_1 \neq 0$, the first part of the lemma applied to v would force $d\rho(v) = 0$. Hence $c_1 = 0$ and then $c_0 = 0$. □

Two limitations of the proof are material. It only forces torsion after projection to the ordinary Jacobian; additive logarithm directions in G need not vanish. Also, the cyclotomic field was not assumed to embed in the entire endomorphism algebra of G: the proof uses a polynomial in the actual automorphism T and then descends it.

13.1 An exact classifier for all polynomial weights

Let $W(t) = \sum_k w_k t^k \in \overline{\mathbb{Q}}[t]$ and define

$$c_W = \sum_k w_k \frac{(a)_k}{(a + b)_k}, \qquad D(x) = x^a (1 - x)^b. \tag{13.3}$$

There is a polynomial $P_W \in \overline{\mathbb{Q}}[x]$ with

$$\int_0^x W(t) t^{a-1} (1 - t)^{b-1} dt = c_W B_x(a, b) + D(x) P_W(x). \tag{13.4}$$

To compute it, start with $c_0 = 1$, $P_0 = 0$ for $W = 1$, and use

$$c_{k+1} = \frac{a + k}{a + b + k} c_k, \qquad P_{k+1} = \frac{a + k}{a + b + k} P_k - \frac{x^k}{a + b + k}$$

for $W = t^{k+1}$, then combine linearly. These are beta recurrence identities [32]; differentiation gives the useful exact check

$$x(1-x)P'_W + [a-(a+b)x]P_W = W - c_W. \tag{13.5}$$

Because $D(0) = D(1) = 0$, the complete weighted integral is $c_W B(a,b)$.

Corollary 13.2 (Polynomial-weight classification)**.** *Under the hypotheses of Theorem 13.1, and for every algebraic $x \in (0,1)$,*

$$\int_0^x W(t)t^{a-1}(1-t)^{b-1}dt \in \overline{\mathbb{Q}} \quad \Longleftrightarrow \quad c_W = 0.$$

If $c_W \neq 0$, its quotient by the complete weighted integral is transcendental. If W has real algebraic coefficients and is positive on $(0,1)$, every interior algebraic-probability quantile of the resulting density is transcendental. In particular this holds for every Beta$(n+a, m+b)$ *with integers $n, m \geq 0$.*

Proof. At algebraic x, the second term in (13.4) is algebraic. If $c_W = 0$, this proves algebraicity. If $c_W \neq 0$, an algebraic incomplete integral would contradict linear independence of $1, B_x$; an algebraic normalized integral would contradict that of $1, B, B_x$. Positivity gives $c_W B(a,b) > 0$, and strict increase of the cumulative distribution provides its unique interior quantiles. An algebraic quantile would contradict the normalized-value statement. Finally choose $W = t^n(1-t)^m$. □

Thus, in the space of all polynomial weights of degree at most any fixed $D \geq 0$, the algebraic incomplete integrals correspond exactly to the hyperplane $c_W = 0$. The exceptional hyperplane is nonempty. For $a = 2/3$, $b = 2/5$, $W(t) = 8t - 5$ has $c_W = 0$ and

$$\int_0^x (8t-5)t^{-1/3}(1-t)^{-3/5}dt = -\frac{15}{2}x^{2/3}(1-x)^{2/5}.$$

Discarding the nonzero-coefficient condition would therefore give a false statement.

For the positive weight $W = t(1-t)$ at these same parameters, $c_W = 15/124$ and $P_W/c_W = 4x - 3/2$. Consequently

$$\begin{aligned} B(2/3,2/5)\,\mathrm{I}_x(5/3,7/5) = {} & B_x(2/3,2/5) \\ & + (4x-3/2)x^{2/3}(1-x)^{2/5}. \end{aligned} \tag{13.6}$$

The median of Beta$(5/3, 7/5)$ is therefore transcendental; its approximate value is

$$0.55398928402851700562\ldots.$$

The base parameters sum to $16/15 > 1$, and the underlying differential has a double pole. This example specifically uses the second-kind extension rather than a holomorphic argument.

The excluded prime pair $\{2,3\}$ is elliptic and has algebraic rational-probability division points. Equal prime denominators also cannot be admitted indiscriminately, since symmetric distributions have the algebraic median $1/2$. Neither algebraic independence of different quantiles nor a complete arbitrary-composite-denominator extension is concluded here.

13.2 Numerical records and their role

All decimals in these sections denote the exact integral or inverse value specified before the decimal expansion. A finite decimal itself is rational. The previous computation records used exact rational intervals, integer root bounds, and the positive expansions

$$I_{N,k}(s) = \sum_{j\geq 0} \frac{\binom{2j}{j} s^{Nj+k}}{4^j(Nj+k)}, \qquad B_x(a,b) = x^a \sum_{j\geq 0} \frac{(1-b)_j x^j}{j!(a+j)}.$$

For $0 < a, b < 1$, successive terms in the latter series have ratio less than x, so the first omitted term divided by $1 - x$ bounds the tail. Complete integrals are evaluated by the two half-interval identity $B(a, b) = B_{1/2}(a, b) + B_{1/2}(b, a)$. These methods supply reproducible sign tests at rational brackets for quantiles. Their role is numerical certification; the transcendence conclusions use the proofs above and the explicitly cited established theorems, not a finite computation.

14 Arithmetic classifications of statistical quantiles

All probabilities in this section refer to exact real numbers, rather than rounded entries in statistical tables. Write $\overline{\mathbb{Q}}$ for the field of algebraic numbers. The transcendence input is Baker's theorem: an algebraic linear combination of fixed logarithms of nonzero algebraic numbers is either zero or transcendental [33]. We also use Hermite–Lindemann and Gelfond–Schneider in their classical forms; the latter says that every value of α^β is transcendental when $\alpha \neq 0, 1$ is algebraic and β is algebraic irrational [34]. These are established inputs, not new results. The use of logarithmic forms for rational-function integrals also has a classical literature [35].

14.1 Student distributions: a complete integer-degree classification

Let $Q_\nu(p)$ denote the quantile of the standard Student distribution with positive integer degrees of freedom ν.

Theorem 14.1 (Student quantiles)**.** *For $p \in \overline{\mathbb{Q}} \cap (0, 1)$:*

1. *if ν is even, $Q_\nu(p)$ is algebraic;*
2. *if $\nu \geq 3$ is odd, $Q_\nu(p)$ is transcendental unless $p = 1/2$, when it equals zero;*
3. *if $\nu = 1$, $Q_1(p)$ is algebraic if and only if p is rational.*

Proof. For $\nu = 2m + 1$, $m \geq 1$, put $u = x/\sqrt{\nu}$. Integration by parts gives

$$F_{2m+1}(x) = \frac{1}{2} + \frac{\arctan u + R_m(u)}{\pi}, \qquad R_m(u) = \sum_{k=1}^{m} \frac{(2k-2)!!}{(2k-1)!!} \frac{u}{(1+u^2)^k}. \tag{14.1}$$

Every coefficient in this sum is positive, so $R_m(u) \neq 0$ for real $u \neq 0$. If both $x \neq 0$ and $p = F_{2m+1}(x)$ were algebraic, then, with compatible fixed logarithms,

$$\log \frac{1+iu}{1-iu} - (2p-1)\log(-1) = -2iR_m(u), \qquad \log(-1) = i\pi.$$

This contradicts Baker's theorem, since the right side is nonzero algebraic.

For $\nu = 2m$, set $y = x/\sqrt{2m + x^2}$. The transformed density is $c_m(1 - y^2)^{m-1}$, where $c_m = (2m)!/(4^m m!(m-1)!) \in \mathbb{Q}$. Consequently

$$F_{2m}(x) = \frac{1}{2} + c_m \sum_{k=0}^{m-1} (-1)^k \binom{m-1}{k} \frac{y^{2k+1}}{2k+1}.$$

Setting this nonconstant polynomial equal to algebraic p makes y, and hence $x = \sqrt{2m}\, y/\sqrt{1-y^2}$, algebraic. Finally $Q_1(p) = \tan(\pi(p - 1/2))$. Rational p gives an algebraic value by roots of unity. If p were algebraic irrational and $Q_1(p)$ algebraic, then

$$\frac{1 + iQ_1(p)}{1 - iQ_1(p)} = (-1)^{2p-1}$$

would contradict Gelfond–Schneider, using the value defined by $\log(-1) = i\pi$. □

For example, $Q_3(39/40) = 3.182446305\ldots$ is transcendental, whereas $Q_2(39/40) = 19\sqrt{2/39}$ is algebraic. At this particular rational probability, Hermite–Lindemann alone proves the first assertion: if $u = Q_3(39/40)/\sqrt{3}$ were algebraic and $r = u/(1+u^2) > 0$, then $\arctan u + r = 19\pi/40$ would imply $e^{40ir} = -((1+iu)/(1-iu))^{-20}$, a contradiction.

14.2 Half-integer beta quantiles and three possible exceptions

For integers $n, m \geq 0$, let $F_{n,m}$ be the distribution function of $\mathrm{Beta}(n+1/2, m+1/2)$. For $N = n+m \geq 1$, define

$$K_{n,m} = \frac{\pi}{\mathrm{B}(n+1/2, m+1/2)} = \frac{4^{n+m} n!m!(n+m)!}{(2n)!(2m)!} \in \mathbb{Q}.$$

There is a unique $P_{n,m} \in \mathbb{Q}[x]$, of degree at most $N-1$, such that

$$2x(1-x)P'_{n,m} + (1-2x)P_{n,m} = 2K_{n,m}x^n(1-x)^m - 2, \tag{14.2}$$

$$\pi F_{n,m}(x) = 2\arcsin\sqrt{x} + \sqrt{x(1-x)}P_{n,m}(x). \tag{14.3}$$

For completeness, the operator on the left of (14.2) is injective on polynomials of degree at most $N-1$: a leading term $a_d x^d$ is sent to a polynomial with leading term $-(2d+2)a_d x^{d+1}$. Its image has codimension one among polynomials of degree at most N. That image is exactly the kernel of integration against $1/(2\sqrt{x(1-x)})$, because the integrand is the derivative of $\sqrt{x(1-x)}P(x)$. The right side of (14.2) has zero such integral. This proves existence, uniqueness, and rational coefficients; differentiation and the value at zero prove (14.3).

Theorem 14.2 (Three-candidate theorem)**.** *Let $n, m \geq 0$ and $n+m \geq 1$. If $p, x \in \overline{\mathbb{Q}} \cap (0,1)$ and $F_{n,m}(x) = p$, then*

$$(p, x) \in \{(1/3, 1/4), (1/2, 1/2), (2/3, 3/4)\}.$$

A listed pair occurs if and only if $P_{n,m}(x) = 0$. For $n, m \geq 1$ at most one listed pair occurs; if $n = 0$ or $m = 0$, none occurs.

Proof. Put $\theta = \arcsin\sqrt{x}$ and $u = \sqrt{x/(1-x)}$. If x is algebraic, then $A = (1+iu)/(1-iu)$ is algebraic and we can fix $\log A = 2i\theta$. Equation (14.3) becomes

$$p\log(-1) - \log A = i\sqrt{x(1-x)}P_{n,m}(x).$$

Baker's theorem forces $P_{n,m}(x) = 0$. Hence $p = 2\theta/\pi$ and $A = (-1)^p$; Gelfond–Schneider forces $p \in \mathbb{Q}$.

It remains to control the roots of $P_{n,m}$. Subtract the arcsine distribution function $F_{0,0}(x) = 2\arcsin\sqrt{x}/\pi$ and write $H = F_{n,m} - F_{0,0}$. Then

$$H'(x) = \frac{K_{n,m}x^n(1-x)^m - 1}{\pi\sqrt{x(1-x)}}.$$

For $n, m \geq 1$, the numerator has signs $-, +, -$: the function $K_{n,m}x^n(1-x)^m$ is strictly unimodal, vanishes at both ends, and has average one under the arcsine probability measure. Since $H(0) = H(1) = 0$, H is negative just after zero, positive just before one, and crosses zero exactly once in its increasing region. Thus $P_{n,m}$ has exactly one root in $(0,1)$. When one of n, m is zero, the density ratio is strictly monotone; the same derivative argument gives no interior root.

For rational p, put $\zeta = e^{i\pi p}$. Then $x = (2 - \zeta - \zeta^{-1})/4$. The order of ζ is at least three, so every conjugate of x is real and lies strictly between zero and one. Every conjugate is also a root of $P_{n,m} \in \mathbb{Q}[x]$. The unique-root property therefore implies $x \in \mathbb{Q}$. Now $2\cos(\pi p) = 2 - 4x$ is both a rational number and an algebraic integer, lying in $(-2, 2)$. It is consequently -1, 0, or 1. These give precisely the three pairs stated. Their sufficiency follows by substituting $P_{n,m}(x) = 0$ in (14.3). □

This is an effective classification for each specified pair (n, m): only three rational evaluations of its correction polynomial are required. It is not a classification of all integer pairs (n, m) for which $P_{n,m}(1/4) = 0$. The omitted arcsine case $n = m = 0$ has quantile $\sin^2(\pi p/2)$; for algebraic p this is algebraic exactly when p is rational, by the same Gelfond–Schneider argument.

For Beta$(3/2, 5/2)$,

$$P_{1,2}(x) = -\frac{2}{3}(4x - 1)(2x - 3).$$

Its unique algebraic–algebraic interior pair is $F_{1,2}(1/4) = 1/3$. In particular its median and its $1/40$ and $39/40$ quantiles are transcendental.

14.3 Posterior intervals and Fisher distributions

In a binomial sample with $N \geq 1$ observations and s successes, the Jeffreys posterior is Beta$(s + 1/2, N - s + 1/2)$. Theorem 14.2 makes both endpoints of its exact, unmodified 95% equal-tail interval transcendental, since neither $1/40$ nor $39/40$ is a candidate probability. Its posterior median is $1/2$ when $s = N - s$ and transcendental otherwise. For the latter assertion, compare the density at x and $1 - x$ on $(0, 1/2)$ to exclude a median of $1/2$ in the asymmetric case. A convention that replaces an endpoint by 0 or 1 at boundary data changes this conclusion for that replaced endpoint.

By contrast, Clopper–Pearson endpoints are quantiles of beta distributions with positive integer parameters, or boundary values $0, 1$. Their distribution functions are rational-coefficient polynomials, so all such endpoints at algebraic confidence levels are algebraic. These definitions agree with the standard statistical conventions described in [36]; the arithmetic distinction expresses no claim about statistical superiority.

For a standard Fisher $F(d_1, d_2)$ variable Y, the transformation $X = d_1Y/(d_1Y + d_2)$ gives $X \sim$ Beta$(d_1/2, d_2/2)$ and preserves algebraicity in both directions. Thus for positive integer d_1, d_2:

- If either degree is even, every quantile at algebraic probability is algebraic. If exactly one beta parameter is half-integral, a square-root substitution reduces its integral to a polynomial.
- If both degrees are odd and $(d_1, d_2) \neq (1, 1)$, there is at most one algebraic quantile at algebraic probability, determined by Theorem 14.2. In particular the 95% quantile is transcendental.
- For $(d_1, d_2) = (1, 1)$, algebraic probabilities give algebraic quantiles exactly at rational probabilities.

For example the sole algebraic–algebraic interior pair for $F(3, 5)$ is its probability-$1/3$ quantile $5/9$.

15 Shortest beta probability intervals

Let f and F be the density and distribution function of Beta(α, β). We measure interval length in the original probability coordinate x, not after a nonlinear reparameterization. Assume

$$\alpha, \beta \in \mathbb{Q}, \qquad \alpha, \beta > 1, \qquad \alpha + \beta \in \mathbb{Z}, \qquad c \in \overline{\mathbb{Q}} \cap (0, 1). \tag{15.1}$$

The strict unimodality of f gives a unique highest-density interval $[a, b]$ of mass c, with

$$0 < a < \frac{\alpha - 1}{\alpha + \beta - 2} < b < 1, \qquad f(a) = f(b), \qquad F(b) - F(a) = c.$$

Indeed its density level decreases continuously as its mass increases from zero to one. This interval is the unique shortest interval of that mass: at its boundary level λ, replacing any part of it by a set outside it loses density, and shortening it strictly loses mass.

Theorem 15.1 (Arithmetic dichotomy for shortest intervals)**.** *Under* (15.1)*, if α, β are integers, then a, b, and $b - a$ are algebraic. If they are nonintegers, then each of a, b, and $b - a$ is transcendental, and*

$$\operatorname{trdeg}_{\overline{\mathbb{Q}}} \overline{\mathbb{Q}}(a, b) = 1.$$

In the noninteger case the midpoint is $1/2$ for $\alpha = \beta$, and is transcendental for $\alpha \neq \beta$.

15.1 A correction-to-density monotonicity lemma

In the noninteger case write uniquely $\alpha = n + r$, $\beta = m + 1 - r$, where $n, m \geq 1$ and $r \in \mathbb{Q} \cap (0, 1)$. Set $N = n + m$ and

$$w_0(x) = x^{r-1}(1 - x)^{-r}, \quad B_0 = \mathrm{B}(r, 1 - r) = \frac{\pi}{\sin \pi r}, \quad K = \frac{B_0}{\mathrm{B}(\alpha, \beta)} = \frac{N!}{(r)_n (1 - r)_m}.$$

The beta recurrence and reflection identities used here are standard [37]. Exactly as in the polynomial image argument above, there is a unique $P \in \mathbb{Q}[x]$, $\deg P \leq N - 1$, with

$$x(1 - x)P'(x) + (r - x)P(x) = Kx^n(1 - x)^m - 1, \tag{15.2}$$

$$B_0 F(x) = \int_0^x w_0(t)\, dt + D(x), \quad D(x) = x^r(1 - x)^{1-r} P(x). \tag{15.3}$$

For that argument the relevant annihilating functional is integration against $w_0(x)$, and $w_0[x(1 - x)P' + (r - x)P] = (x^r(1 - x)^{1-r}P)'$.

Lemma 15.2. *The rational function $R(x) = P(x)/(x^{n-1}(1 - x)^{m-1})$ satisfies $R'(x) > 0$ for $0 < x < 1$.*

Proof. Expand in the unnormalized Bernstein basis, $P(x) = \sum_{k=0}^{N-1} p_k x^k (1 - x)^{N-1-k}$. With $p_{-1} = p_N = 0$, comparison of degree-N coefficients in (15.2) yields

$$(j + r)p_j - (N - j + 1 - r)p_{j-1} = K\mathbf{1}_{j=n} - \binom{N}{j} \qquad (0 \leq j \leq N).$$

Starting with $p_0 = -1/r$, forward recurrence gives $p_k < 0$ for $k < n$. Starting with $p_{N-1} = 1/(1-r)$, backward recurrence gives $p_k > 0$ for $k \geq n$. The corresponding summand of R is $p_k x^j (1 - x)^{1-j}$, with $j = k - n + 1$, whose derivative equals

$$p_k x^{j-1}(1 - x)^{-j}(j - x).$$

It is strictly positive both for $k < n$ ($p_k < 0$, $j \leq 0$) and for $k \geq n$ ($p_k > 0$, $j \geq 1$). □

Since $D(x) = (B_0/K) f(x) R(x)$, the equal-density condition implies the decisive noncancellation

$$D(b) - D(a) = \frac{B_0}{K} f(a)[R(b) - R(a)] > 0. \tag{15.4}$$

Knowing only the algebraic mass $F(b) - F(a)$ does not make either $F(a)$ or $F(b)$ algebraic. This is why the single-quantile theorem alone would not suffice for shortest intervals.

15.2 Proof of the dichotomy and its geometric consequences

Suppose first that the noninteger parameters satisfy (15.1) and that both endpoints are algebraic. Then $D(b) - D(a)$ is nonzero algebraic. Write $r = p/q$ in lowest terms and make the positive-real substitution $y = (x/(1-x))^{1/q}$. It gives

$$w_0(x)\,dx = \frac{qy^{p-1}}{1+y^q}\,dy.$$

This is a proper rational function with distinct algebraic poles. Its integral between algebraic endpoints is therefore an algebraic linear combination of logarithms of nonzero algebraic numbers; logarithms are chosen continuously along the integration path. The same is true of cB_0, because $\sin \pi r$ is nonzero algebraic and $\pi = \log(-1)/i$. But (15.3) implies

$$D(b) - D(a) = cB_0 - \int_a^b w_0(x)\,dx,$$

contradicting Baker's theorem and (15.4).

Choose a common positive denominator q so that $A = q(\alpha - 1)$ and $B = q(\beta - 1)$ are positive integers, and put $h(X) = X^A(1-X)^B$. Equal density gives the polynomial relation $h(a) = h(b)$. If either endpoint were algebraic, this relation would make the other algebraic too. Thus both are transcendental; the nonzero relation also proves that their joint transcendence degree is exactly one.

If $w = b - a$ were algebraic, the transcendental number a would satisfy $h(a + w) - h(a) = 0$. For $w > 0$ the polynomial $h(X + w) - h(X)$ is nonzero, contradicting transcendence. If the midpoint $z = (a + b)/2$ were algebraic, the same reasoning applied to $h(2z - X) - h(X)$ would force this polynomial to vanish identically. Reflection would then preserve the roots $0, 1$ and their multiplicities. Necessarily $z = 1/2$ and $A = B$, hence $\alpha = \beta$. In this symmetric case uniqueness of the interval indeed gives $z = 1/2$.

For the integer case let $A = \alpha - 1$, $B = \beta - 1$. Now F is a rational-coefficient polynomial. The equal-density pair is parameterized by $t > 1$ as

$$a(t) = \frac{t^A - 1}{t^{A+B} - 1}, \qquad b(t) = t^B a(t).$$

To derive this, use $t = (b/a)^{1/B}$ in $a^A(1-a)^B = b^A(1-b)^B$. The mass $C(t) = F(b(t)) - F(a(t))$ is a rational function in $\mathbb{Q}(t)$, and it is nonconstant because its limits as $t \downarrow 1$ and $t \to \infty$ are zero and one. The equation $C(t) = c$ therefore gives a nonzero algebraic-coefficient polynomial equation for t. Thus t, a, b, and $b - a$ are algebraic, completing the proof of Theorem 15.1.

15.3 Examples and scope

For Beta$(3/2, 5/2)$ and $c = 19/20$, the unique shortest interval is approximately

$$[a, b] = [0.009556870873\ldots,\ 0.770757269257\ldots].$$

Both endpoints, its length, and its midpoint are transcendental. This conclusion also holds for every algebraic mass in $(0, 1)$. The result is not confined to half-integer parameters: for Beta$(4/3, 5/3)$ one has $r = 1/3$, $n = m = 1$, $K = 9$, and $P(x) = -3 + (9/2)x$, so the same four conclusions hold.

With at least one success and one failure, a Jeffreys beta posterior has transcendental shortest-interval endpoints and length at every algebraic credibility level; its midpoint is transcendental unless the counts are equal. A uniform-prior posterior instead has integer parameters and algebraic endpoints and length. The theorem does not cover zero-success or zero-failure HPD boundary cases, U-shaped beta densities, or general rational parameters whose sum is not an integer. No claim of priority for these particular classifications is made here, and no conclusion about the transcendence of a specified normal quantile or Euler's constant follows from them.

16 Finite Gaussian mixtures: an exact endpoint classification

The logarithmic arguments above are complemented by an E-function method for finite Gaussian mixtures. The normalization interval is finite throughout this classification. This restriction is essential to its proof and separates it from a classification of the ordinary normal quantiles.

16.1 Linear independence at distinct positive algebraic endpoints

For a positive integer d, define $J_d(x) = \int_0^x e^{-t^{2d}}\,dt$.

Lemma 16.1 (Endpoint separation)**.** *If $r_1, \dots, r_s$ are distinct positive algebraic numbers, then*

$$1,\ J_d(r_1), \dots, J_d(r_s), \quad e^{-r_1^{2d}}, \dots, e^{-r_s^{2d}}$$

are linearly independent over $\overline{\mathbb{Q}}$.

Proof. We do not regard $J_d(z)$ itself as an E-function. Instead set

$$E_{c,d}(z) = \int_0^1 e^{-czt^{2d}}\,dt = \sum_{n\ge 0} \frac{(-c)^n}{2dn+1}\frac{z^n}{n!}, \qquad c \in \overline{\mathbb{Q}}_{>0}.$$

It satisfies $2dzE'_{c,d}(z) + E_{c,d}(z) = e^{-cz}$. Its normalized coefficients have exponentially bounded conjugates and common denominators: if Dc is an algebraic integer, a common denominator through index M is $D^M \operatorname{lcm}(1, \dots, 2dM+1)$, which grows exponentially. Thus $E_{c,d}$ and e^{-cz} are E-functions forming, together with 1, a differential system whose only possible finite singularity is zero.

For distinct $0 < c_1 < \cdots < c_s$, these functions are linearly independent over $\mathbb{C}(z)$. Indeed, assume a nontrivial rational relation and choose the largest c appearing. Along $z = -x$, $x \to +\infty$, the smaller exponential scales and their derivatives are exponentially smaller than e^{cx}, up to polynomial factors. If the coefficient of $E_{c,d}$ is zero, the nonzero rational coefficient of e^{-cz} would have to be exponentially small, an impossibility. Otherwise the relation yields an $R \in \mathbb{C}(z)$ such that $T = e^{cz}E_{c,d} - R$ and T' are exponentially small there. Applying $\mathcal{L}_c = 2dz\,d/dz + 1 - 2dcz$ shows that the rational function $1 - \mathcal{L}_c R$ is exponentially small, hence zero. But

$$2dzR' + (1 - 2dcz)R = 1$$

has no rational solution. A nonzero finite pole away from zero cannot survive its differentiated highest-order term. A pole of order $m \ge 1$ at zero has nonzero coefficient factor $1 - 2dm$. Thus R would be a polynomial, which is excluded by the highest degree term $-2dczR$ (and $R = 0$ is also excluded).

Beukers's refined Siegel–Shidlovskii theorem [38, Theorem 1.3] lifts any algebraic linear relation among these values at the regular algebraic point $z = 1$ to a polynomial-coefficient linear relation among the functions. The established functional independence rules that out. Finally use $c_i = r_i^{2d}$ and $J_d(r_i) = r_i E_{r_i^{2d},d}(1)$. □

16.2 A finite classification algorithm

Let K be a subfield of the real algebraic numbers. Choose $a < b$, positive weights w_j, positive scales λ_j, and centers μ_j, all in K, for $1 \le j \le N$. Merge components with identical (λ_j, μ_j). On $[a, b]$ normalize the positive density

$$f_d(x) = \sum_{j=1}^N w_j e^{-[\lambda_j(x-\mu_j)]^{2d}}, \qquad S_d(x) = \sum_{j=1}^N \frac{w_j}{\lambda_j} J_d(\lambda_j(x-\mu_j)).$$

Its distribution function is $C_d(x) = (S_d(x) - S_d(a))/(S_d(b) - S_d(a))$. For each positive radius r, write

$$V_r(x) = \sum_{j:\,|\lambda_j(x-\mu_j)|=r} \frac{w_j}{\lambda_j} \operatorname{sgn}(x - \mu_j).$$

Zero radii contribute nothing. Since J_d is odd, endpoint separation gives, for $x \in \overline{\mathbb{Q}} \cap (a, b)$ and $p \in \overline{\mathbb{Q}} \cap (0, 1)$,

$$C_d(x) = p \quad \Longleftrightarrow \quad V(x) - (1 - p)V(a) - pV(b) = 0. \tag{16.1}$$

Theorem 16.2 (Finite endpoint classification)**.** *All algebraic–algebraic interior quantile pairs (x, p) for C_d belong to K^2 and form a finite set. They can be determined by testing at most $5N^2$ possible locations and performing algebraic arithmetic. The set of pairs is independent of the positive integer d.*

Proof. Let $\mathcal{R}$ be the nonzero endpoint radii $|\lambda_j(a - \mu_j)|$ and $|\lambda_j(b - \mu_j)|$; then $|\mathcal{R}| \le 2N$. The possible locations are

$$\mu_j, \qquad \mu_j \pm r/\lambda_j \quad (r \in \mathcal{R}), \qquad \frac{\lambda_j\mu_j - \varepsilon\lambda_k\mu_k}{\lambda_j - \varepsilon\lambda_k} \quad (j < k,\ \varepsilon = \pm 1),$$

with zero denominators, duplicates, and locations outside (a, b) discarded. There are at most $N + 4N^2 + 2\binom{N}{2} = 5N^2$ candidates, all in K. At any other algebraic location, every moving radius is nonzero, distinct from all other moving radii, and different from every endpoint radius. Its coefficient in (16.1) is then uniquely $\pm w_j/\lambda_j \neq 0$, a contradiction.

For a surviving location put $D = V(b) - V(a)$. Positivity of the normalizing integral implies $D \neq 0$. A nonzero coordinate gives the only possible probability, $p = (V_r(x) - V_r(a))/D_r \in K$. Check all remaining coordinates and $0 < p < 1$. The criterion does not contain d, proving invariance. The bound counts candidates; it is not a polynomial bound on bit complexity for algebraic-number computations. □

16.3 Concrete quantiles and the limit of the method

Consider the normalized density

$$e^{-(x+1/2)^{2d}} + 2e^{-(x-1/2)^{2d}}, \qquad -1 \le x \le 1.$$

Writing $J = J_d$ gives $S(x) = J(x + 1/2) + 2J(x - 1/2)$ and the quantile equation

$$S(x) = (-1 + 3p)J(1/2) + (-2 + 3p)J(3/2).$$

If x, p are algebraic and $-1 < x < 1$, no moving radius can equal $3/2$, so endpoint separation forces $p = 2/3$. The only interior location candidates are $-1/2, 0, 1/2$. At these points the residual $S(x) - J(1/2)$ is respectively $-2J(1) - J(1/2)$, $-2J(1/2)$, and $J(1) - J(1/2)$, none zero. Thus *every* quantile at an algebraic probability in $(0, 1)$ is transcendental. At $d = 1$ its median is $0.111102019216\ldots$.

The same algorithm records genuine exceptions. For this mixture on $[-3/2, 3/2]$ instead, the unique algebraic–algebraic interior pair is $(x, p) = (1/2, 2/3)$: the outer endpoint radius forces $p = 2/3$ and the three candidates are again $-1/2, 0, 1/2$, of which exactly $1/2$ satisfies $S(x) = J(1)$. For a single density $e^{-x^{2d}}$ on $[0, 1]$, every interior quantile at algebraic probability is transcendental, directly by endpoint separation. In particular the Gaussian truncated median defined by $J_1(m_0) = J_1(1)/2$ is transcendental, with $m_0 = 0.392697625792\ldots$.

For the untruncated standard normal law, $\Phi(x) = 1/2 + J_1(x/\sqrt{2})/\sqrt{\pi}$, the method proves only that *at most one* positive algebraic location can have an algebraic distribution value. Two distinct such locations would eliminate $\sqrt{\pi}$ and contradict endpoint separation. This does not identify which of two specified quantiles is transcendental, nor prove that a particular one, such as $\Phi^{-1}(39/40)$, is transcendental. The finite-normalization theorem must not be extended to that setting without an additional argument.

Part VI
Transfers to Euler constants and Gamma values

17 Gamma values and Euler's constant: transfer and obstruction

The preceding beta-integral results motivate two different arithmetic questions. One concerns an individual Gamma value, such as $\Gamma(1/5)$; the other concerns Euler's constant

$$\gamma = \lim_{n\to\infty}(H_n - \log n) = -\Gamma'(1), \qquad H_n = \sum_{j=1}^{n}\frac{1}{j}, \quad H_0 = 0.$$

Neither question is the functional differential transcendence of $\Gamma(z)$, established by Hölder. In particular, the latter theorem does not prevent the algebraic values $\Gamma(n) = (n-1)!$. The discussion below proves explicit reductions, cancellation identities, and a quantitative obstruction for one approximation scheme. It does not prove that $\Gamma(1/5)$ is transcendental or that γ is irrational. The individual fifth-denominator question is still identified as open in the 2026 account [39]. No priority claim is attached to the auxiliary results assembled here.

17.1 An exact fifth-denominator reduction

Put

$$G_1 = \Gamma(1/5), \qquad G_2 = \Gamma(2/5), \qquad A = B(1/5, 1/5), \qquad B = B(2/5, 2/5).$$

The beta–Gamma identity and reflection formula give

$$A = \frac{G_1^2}{G_2}, \qquad B = \frac{\sin(\pi/5)G_1G_2^2}{\pi}, \qquad \boxed{G_1^5 = \frac{\pi A^2 B}{\sin(\pi/5)}}. \tag{17.1}$$

Since $\sin(\pi/5)$ is nonzero and algebraic, G_1 is transcendental if and only if $\pi A^2 B$ is transcendental. The transcendence or linear independence of separate beta values does not imply this product statement. In particular, a theorem describing linear relations between ordinary 1-periods cannot be used as a theorem describing all polynomial relations between their products. The known bound

$$\operatorname{trdeg}_{\mathbb{Q}}\mathbb{Q}(\pi, G_1, G_2) \geq 2$$

recorded in [40] also does not select G_1: the bound is consistent with G_1 algebraic and π, G_2 algebraically independent.

17.2 Balanced products and the lost linear term

The beta identity

$$B(a,b) = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}$$

is invariant under the formal modification $\Gamma(z) \mapsto \Gamma_\kappa(z) = e^{\kappa z}\Gamma(z)$. This modification is an algebraic device for comparing formulas, not a second Gamma function satisfying all its original normalization, recurrence, and growth conditions.

Proposition 17.1 (Balanced-product invariance)**.** *Suppose that $a_j(t)$ are holomorphic near 0, their Gamma values avoid poles there, and m_j are integers. If $\sum_j m_j a_j(t)$ is constant, then the normalized function*

$$\frac{\prod_j \Gamma(a_j(t))^{m_j}}{\prod_j \Gamma(a_j(0))^{m_j}}$$

is unchanged when Γ *is replaced by* Γ_κ*. Consequently all its Taylor coefficients are invariant under that modification.*

Proof. The numerator and denominator acquire the respective factors $\exp(\kappa\sum_j m_j a_j(t))$ and $\exp(\kappa\sum_j m_j a_j(0))$, which are equal. □

For affine rational parameters this cancellation can be read directly from the digamma function $\psi = \Gamma'/\Gamma$:

$$\frac{d}{dt}\log B(at+b, ct+d) = a\psi(at+b) + c\psi(ct+d) \\ - (a+c)\psi((a+c)t+b+d). \tag{17.2}$$

At rational arguments away from poles, Gauss's formula and the shift formula write $\psi(r) = -\gamma + L_r$, where L_r is an explicit combination of π, logarithms of algebraic numbers, and algebraic coefficients. The coefficient of γ in (17.2) is $-a-c+(a+c) = 0$. For example,

$$\frac{\partial}{\partial a}\log B(a,b)\Big|_{a=b=1/3} = \psi(1/3) - \psi(2/3) = -\frac{\pi}{\sqrt{3}}.$$

Thus differentiation of the balanced Gamma quotients treated earlier does not by itself recover the linear term $-\gamma z$ in $\log\Gamma(1+z)$. This is a limitation of that information, not a theorem that γ belongs to no field of ordinary periods.

There is also a geometric change when parameters are differentiated. For instance,

$$\partial_h[x(1-x)]^h = [x(1-x)]^h \log(x(1-x)).$$

If f is a nonconstant rational function on a projective algebraic curve, no branch of $\log f$ becomes a meromorphic function on a finite algebraic cover. Otherwise its derivative would be an exact meromorphic differential, which has zero residues, whereas the pullback of df/f has residue $e\,\mathrm{ord}_P(f) \neq 0$ at a point of ramification index e over a zero or pole of f. Hence the old algebraic-differential argument cannot simply be reused after inserting this logarithm.

Likewise, the factor e^{-t} in an Euler integral has an essential singularity at infinity: in the coordinate $u = 1/t$ it is $e^{-1/u}$. A finite algebraic cover does not turn that essential singularity into a finite-order pole. The generalized Jacobians with finite modulus used for the earlier first- and second-kind differentials do not directly represent this expression. This assertion concerns the displayed integrands; it does not exclude every possible different representation of γ.

17.3 Confluence is exact, but transcendence need not survive it

For $\Re s > 0$, set

$$F_N(s) = N^s B(s,N) = \int_0^N t^{s-1}(1-t/N)^{N-1}\,dt.$$

For $N \geq 2$ the factor in parentheses, extended by zero past N, is at most $e^{-t/2}$. Dominated convergence, locally uniformly with the necessary logarithmic factors, gives $F_N(s) \to \Gamma(s)$ and convergence of its derivatives on compact subsets of $\Re s > 0$. At $s = 1$,

$$F_N(1) = 1, \qquad F_N'(1) = \log N - H_N \longrightarrow -\gamma. \tag{17.3}$$

An equivalent finite-part version uses

$$\widetilde{F}_N(s) = N^s B(s, N+1) = \frac{N^s N!}{s(s+1)\cdots(s+N)} = \frac{1}{s} + \log N - H_N + O(s)$$

near $s = 0$. These are genuine analytic connections between beta and Gamma values. For $N \geq 2$, each $H_N - \log N$ is transcendental by Lindemann's theorem, but this property is not preserved by taking limits. A separate arithmetic estimate is required.

17.4 An antisymmetric exponential-integral transfer

A different limit arising in the earlier discussion connects an antisymmetric iterated integral with both Euler's constant and a dilogarithm. For real $r, R > 0$ define

$$H_r(R) = \int_0^R \left(\mathrm{Ein}(rt)\frac{1-e^{-t}}{t} - \mathrm{Ein}(t)\frac{1-e^{-rt}}{t} \right) dt.$$

Here $\mathrm{Ein}(z) = \sum_{n\geq 1}(-1)^{n+1}z^n/(n\,n!)$ is entire. The previously selected finite-point constant was $H_2(1)$. For positive arguments it is convenient to define the real branch of the dilogarithm by $\mathrm{Li}_2(-r) = -\int_0^r \log(1+t)\,dt/t$.

Proposition 17.2 (Renormalized iterated-integral limit)**.** *For every real $r > 0$,*

$$\begin{aligned}\lim_{R\to\infty}\big(H_r(R) - \log r \log R\big) &= \gamma\log r + \log^2 r + 2\,\mathrm{Li}_2(-r) + \frac{\pi^2}{6} \\ &= \gamma\log r - 2\,\mathrm{Li}_2(-1/r) - \frac{\pi^2}{6}. \end{aligned} \tag{17.4}$$

Proof. Differentiation under the finite integral and integration by parts give

$$r\partial_r H_r(R) = (1+e^{-rR})\,\mathrm{Ein}(R) + 2\,\mathrm{Ein}(rR) - 2\,\mathrm{Ein}((r+1)R).$$

For completeness, the first differentiated integral is $\mathrm{Ein}(R) + \mathrm{Ein}(rR) - \mathrm{Ein}((r+1)R)$; the term subtracted from it is $-e^{-rR}\,\mathrm{Ein}(R) + \mathrm{Ein}((r+1)R) - \mathrm{Ein}(rR)$. Using $\mathrm{Ein}(x) = \gamma + \log x + E_1(x)$, where $E_1(x) = \int_x^\infty e^{-t}\,dt/t$, shows that

$$r\partial_r H_r(R) - \log R \longrightarrow \gamma + 2\log r - 2\log(1+r)$$

uniformly for r in any compact subinterval of $(0,\infty)$. Since $H_1(R) = 0$, integration with respect to dr/r from 1 to r gives the first limit formula, using $\mathrm{Li}_2(-1) = -\pi^2/12$. The identity

$$\mathrm{Li}_2(-r) + \mathrm{Li}_2(-1/r) = -\frac{\pi^2}{6} - \frac{1}{2}\log^2 r$$

gives the second form. It follows directly by differentiation in r and evaluation at $r = 1$. □

At $r = 2$ the limit is $\gamma\log 2 - 2\,\mathrm{Li}_2(-1/2) - \pi^2/6$, approximately -0.3480102422994023475. This is an exact analytic transfer, but finite-point E-function transcendence theorems do not automatically extend to this renormalized limit at infinity. Even transcendence of the whole limit would not isolate γ without additional control of its relation with the dilogarithm and π^2 terms. Thus this identity closes neither an individual Euler-constant problem nor a dilogarithm independence problem.

17.5 Polynomial logarithmic moments and a shared tail coefficient

The freedom to introduce a polynomial weight can be analyzed completely for one logarithm and the fixed exponential weight. Let $P(t) = \sum_{k=0}^d p_k t^k \in \overline{\mathbb{Q}}[t]$, and write

$$B_P = \sum_{k=0}^d p_k k!, \qquad A_P = \sum_{k=0}^d p_k k! H_k.$$

Since $\Gamma'(k+1) = k!(H_k - \gamma)$, absolute convergence and linearity give

$$J_P := \int_0^\infty P(t)e^{-t}\log t\,dt = A_P - B_P\gamma. \tag{17.5}$$

If $B_P = 0$, this integral is algebraic. If $B_P \neq 0$, its algebraicity is equivalent to the algebraicity of γ.

To expose the tail, introduce

$$Q_P(t) = \sum_{j=0}^{d} P^{(j)}(t), \qquad R_P(t) = \frac{Q_P(t) - Q_P(0)}{t}, \qquad C_P = \sum_{j\geq 0} R_P^{(j)}(1).$$

The last sum is finite, and is zero when $R_P = 0$. Then

$$Q_P - Q_P' = P, \qquad Q_P(0) = B_P, \qquad \int_0^\infty e^{-t} R_P(t)\, dt = A_P.$$

The last identity follows for $P = t^k$ by summing $k!/(k-j)$ over $j = 0, \ldots, k-1$, and then follows generally by linearity. Define

$$E_1(x) = \int_x^\infty \frac{e^{-t}}{t}\, dt, \qquad \eta = \mathrm{Ein}(1) = \gamma + E_1(1), \qquad \delta = eE_1(1).$$

Integration by parts, using $Pe^{-t} = -(Q_P e^{-t})'$ and $\log 1 = 0$, gives

$$T_P := \int_1^\infty P(t)e^{-t} \log t\, dt = B_P E_1(1) + \frac{C_P}{e}, \tag{17.6}$$

$$L_P := -\int_0^1 P(t)e^{-t} \log t\, dt = B_P \eta - A_P + \frac{C_P}{e}. \tag{17.7}$$

Here the polynomial remainder is integrated by repeated integration by parts. The identity $J_P = T_P - L_P$ provides a consistency check.

$P(t)$	A_P	B_P	C_P	J_P
1	0	1	0	$-\gamma$
t	1	1	1	$1-\gamma$
t^2	3	2	4	$3-2\gamma$
t^3	11	6	17	$11-6\gamma$
$t-1$	1	0	1	1

In a finite linear combination, cancellation of the displayed $E_1(1)$ coefficient in (17.6) forces the same cancellation of the γ coefficient in (17.5). This is a statement about the explicit coefficients and does not assume any unknown independence of $E_1(1)$ and e. In particular,

$$\int_0^\infty (t-1)e^{-t} \log t\, dt = 1$$

is a counterexample to the claim that every nonzero exponential logarithmic integral is transcendental. The conclusion is restricted to finite polynomial weights, one logarithm, this fixed exponential, and linear cancellation. Rational weights or nonlinear constructions are not covered.

17.6 A positive kernel and a quantitative degree obstruction

Binet's formulas [41] yield, for $n \geq 1$,

$$\begin{aligned} J_n &= 2\gamma + 2\log n - H_n - H_{n-1} \\ &= \int_0^\infty e^{-nt} \left(\coth(t/2) - \frac{2}{t} \right) dt = \int_0^1 x^{n-1} K(x)\, dx, \\ K(x) &= \frac{1+x}{1-x} + \frac{2}{\log x}, \qquad 0 < x < 1. \end{aligned} \tag{17.8}$$

The notation J_n here refers to this kernel moment, rather than the polynomial logarithmic moment J_P of the preceding subsection. For $P(x) = \sum_{n=1}^{N+1} c_n x^{n-1} \in \mathbb{Q}[x]$, set

$$C = P(1), \qquad A = \sum_{n=1}^{N+1} c_n(H_n + H_{n-1}), \qquad L(P) = \int_0^1 P(x)K(x)\,dx.$$

Exact logarithm cancellation gives

$$\sum_n c_n \log n = 0 \quad \Longrightarrow \quad L(P) = 2C\gamma - A. \tag{17.9}$$

The cancellation condition can be imposed without using γ: after clearing rational denominators, unique factorization shows that it is equivalent to $\sum_n c_n v_\ell(n) = 0$ for each prime $\ell \le N+1$.

Proposition 17.3 (Pointwise kernel bound)**.** *For $0 < x < 1$, one has $K(x) > (1-x)/6$.*

Proof. Let $f(x) = -\log x - 12(1-x)/(5+8x-x^2)$. Direct differentiation gives

$$f'(x) = -\frac{(x-25)(x-1)^3}{x(x^2-8x-5)^2} < 0, \qquad f(1) = 0.$$

Thus $f(x) > 0$ for $x < 1$. Rewriting this inequality and multiplying only positive quantities gives the stated bound. □

Theorem 17.4 (Positive-polynomial endpoint obstruction)**.** *If P is a real polynomial nonnegative on $[0,1]$ and $P(1) > 0$, then*

$$\frac{L(P)}{P(1)} \ge \begin{cases} \dfrac{1}{3(m+1)^2(m+2)^2}, & \deg P \le 2m, \\ \dfrac{1}{3(m+1)(m+2)^2(m+3)}, & \deg P \le 2m+1. \end{cases}$$

If P has rational coefficients and satisfies (17.9), the rational number $a_P = A/(2P(1))$ satisfies the corresponding bounds

$$\gamma - a_P \ge \begin{cases} \dfrac{1}{6(m+1)^2(m+2)^2}, & \deg P \le 2m, \\ \dfrac{1}{6(m+1)(m+2)^2(m+3)}, & \deg P \le 2m+1. \end{cases}$$

Proof. The Markov–Lukács representation for interval positivity [42] gives, in even degree,

$$P(x) = U(x)^2 + x(1-x)V(x)^2, \quad \deg U \le m, \quad \deg V \le m-1.$$

For $m = 0$ take $V = 0$. The shifted Jacobi polynomials $J_k^*(x) = P_k^{(1,0)}(2x-1)$ satisfy

$$J_k^*(1) = k+1, \qquad \int_0^1 (1-x)J_k^*(x)^2\,dx = \frac{1}{2(k+1)}$$

and are mutually orthogonal for this weight [41]. Expanding U and applying Cauchy–Schwarz therefore yields

$$U(1)^2 \le \left(\sum_{k=0}^{m} 2(k+1)^3\right) \int_0^1 (1-x)U(x)^2\,dx.$$

The sum is $(m+1)^2(m+2)^2/2$. Since $P(1) = U(1)^2$, dropping the nonnegative V contribution gives the required bound after Proposition 17.3.

In odd degree write $P = xU^2 + (1-x)V^2$, with both polynomial degrees at most m. Use instead $P_k^{(1,1)}(2x-1)$, whose value at 1 is $k+1$ and whose squared norm for the weight $x(1-x)$ is

$$\frac{k+1}{(2k+3)(k+2)}.$$

The relevant endpoint sum is now

$$\sum_{k=0}^{m}(k+1)(k+2)(2k+3) = \frac{(m+1)(m+2)^2(m+3)}{2}.$$

Since $\int(1-x)P \geq \int x(1-x)U^2$ and $P(1) = U(1)^2$, the same argument proves the odd-degree bound. The rational approximation statements follow by dividing (17.9) by $2P(1)$. □

The endpoint estimates for the weighted polynomial integrals are optimal: equality in Cauchy–Schwarz is achieved by the corresponding reproducing-kernel polynomial and the other squared term may be set to zero. Optimality of the final estimate with the kernel K is not claimed. The proof also works if interval positivity is stated using sums of squares instead of a single square in each term.

For a fully elementary weaker bound, Markov's inequality $\|P'\|_\infty \leq 2N^2\|P\|_\infty$ on $[0,1]$ shows, when $N \geq 1$, that

$$\frac{L(P)}{P(1)} \geq \frac{1}{384N^4}, \qquad \gamma - a_P \geq \frac{1}{768N^4}. \tag{17.10}$$

Indeed, writing $M = \max P$, an interval of length $\ell = 1/(4N^2)$ fits on one side of a maximum point, and $P \geq M/2$ there. Among intervals of that length the smallest integral of $1-x$ is $\ell^2/2$. Hence $\int_0^1(1-x)P\,dx \geq M/(64N^4) \geq P(1)/(64N^4)$.

These inequalities prohibit exponential-in-degree convergence in this specific positive-polynomial scheme. If $a_P = p/q$ is reduced, the criterion $q|\gamma - p/q| \to 0$ necessarily requires $q/N^4 \to 0$. That condition is not sufficient. Signed polynomials, rational kernels, and other approximation methods are outside the theorem. A factor $(1-x)^k$, $k > 0$, gives $P(1) = 0$ and removes the γ coefficient under exact logarithm cancellation; it cannot be used without checking that loss of the target coefficient.

17.7 A tested rational approximation family

For integers $m \geq 2$, choose

$$P_m(x) = 2x^{m-1} - x^{m^2-1} > 0 \quad (0 < x < 1), \qquad P_m(1) = 1.$$

The identity $2\log m - \log(m^2) = 0$ produces the rational number

$$a_m = \frac{2H_m + 2H_{m-1} - H_{m^2} - H_{m^2-1}}{2}, \qquad \gamma - a_m = \frac{2J_m - J_{m^2}}{2} > 0.$$

Binet's second representation gives

$$J_n = 4\int_0^\infty \frac{t\,dt}{(t^2+n^2)(e^{2\pi t}-1)}.$$

Apply $1-u < (1+u)^{-1} < 1$ and the standard zeta evaluations of the first and third moments to obtain

$$\frac{1}{6n^2} - \frac{1}{60n^4} < J_n < \frac{1}{6n^2}.$$

Consequently the error is bracketed by explicit rational numbers:

$$\frac{1}{6m^2} - \frac{1}{10m^4} < \gamma - a_m < \frac{1}{6m^2} - \frac{1}{12m^4} + \frac{1}{120m^8}. \tag{17.11}$$

m	Digits of reduced q_m	$\gamma - a_m$	$\log_{10}(q_m\lvert\gamma - a_m\rvert)$
2	2	0.0355490	−0.0690
4	7	0.0100280	4.1600
10	41	0.00165667	38.0626
16	110	0.000649516	106.3357
32	446	0.000162665	441.6931

The numerators, reduced denominators, and rational brackets were calculated exactly; the displayed decimals are numerical diagnostics. The examples fail to produce a small integer linear form as m grows through the tested values. The table is not an asymptotic impossibility proof. The uniform proved conclusion is Theorem 17.4.

17.8 Exponential-integral values and the exact rank question

Recall that $\mathrm{Ein}(x) = \gamma + \log x + E_1(x)$ for $x > 0$. The Euler–Gompertz constant also has the equivalent expression

$$\delta = \int_0^\infty \frac{e^{-t}}{1+t}\,dt = \int_0^\infty e^{-t}\log(1+t)\,dt = eE_1(1).$$

Rivoal proves that e and $\eta = \mathrm{Ein}(1)$ are algebraically independent [40, §5]. Thus

$$\begin{gathered} 3 \le \dim_{\overline{\mathbb{Q}}}\langle 1, e, \delta, e\eta\rangle \le 4, \\ \dim_{\overline{\mathbb{Q}}}\langle 1, e, \delta, e\eta\rangle = 3 \quad\Longleftrightarrow\quad \gamma \in \mathrm{span}_{\overline{\mathbb{Q}}}\{1, e^{-1}, \eta\}. \end{gathered} \tag{17.12}$$

Indeed, $1, e, e\eta$ are linearly independent, and a rank drop is equivalent to $\delta = A + Be + Ce\eta$ with algebraic A, B, C. Dividing by e gives $\gamma = (1-C)\eta - B - A/e$, and the converse reverses this calculation. Rank four would prove γ transcendental, but was not established. If γ were algebraic, the rank would be three and e, δ would be algebraically independent, since $\delta = e(\eta - \gamma)$ is an invertible rational change of variables. This is fully consistent with the known transcendence-degree bound.

17.9 Multiscale independence and an algebraic-limit counterexample

Proposition 17.5 (A multiscale consequence of classical theorems)**.** *If $n_1, \dots, n_m$ are distinct positive integers, then*

$$e, \mathrm{Ein}(n_1), \dots, \mathrm{Ein}(n_m)$$

are algebraically independent over $\overline{\mathbb{Q}}$.

Proof. Work in $K = \mathbb{C}(z, t)$ with $t = e^{-z}$ and derivation $D = \partial_z - t\partial_t$. The functions z, t are algebraically independent. Put $y_j = \mathrm{Ein}(n_j z)$; then

$$Dy_j = \frac{1 - t^{n_j}}{z}.$$

The primitive case of the Kolchin–Ostrowski theorem [43, Theorem 2.5] says that algebraic dependence of the y_j over K would give a nonzero constant linear combination $\sum_j c_j y_j = R \in K$. There are no new differential constants: the fields in question embed in the common field of meromorphic functions, whose constants are $\mathbb{C}$. Differentiating gives

$$DR = \frac{\sum_j c_j(1 - t^{n_j})}{z}.$$

The numerator is a nonzero polynomial in the independent variable t. Expand R at the divisor $z = 0$ over $\mathbb{C}(t)$. If R has a pole of order $d \geq 1$, $\partial_z R$ has a nonzero pole of order $d+1$, while $-t\partial_t R$ has order at most d. If R is regular, DR is regular. A derivative therefore cannot have precisely the simple pole displayed above. This contradiction proves independence of the primitives over K.

All y_j and t are E-functions. Adding 1 and $e^{-n_j z}$ gives a homogeneous first-order system over $\mathbb{Q}(z)$ whose only finite singularity is 0. The extra exponentials are powers of t, so they do not increase the functional transcendence degree. The Siegel–Shidlovskii theorem at $z = 1$, in the form recalled in [44], gives degree $m + 1$ for the values. Replacing e^{-1} by its inverse finishes the proof. □

In particular, for

$$Y_n = (n+1)\,\mathrm{Ein}(2^n) - n\,\mathrm{Ein}(2^{n+1}), \qquad n \geq 1,$$

every finite collection $e, Y_1, \ldots, Y_N$ is algebraically independent. The linear transformation of the corresponding Ein values has full row rank and can be completed to an invertible one. Logarithm cancellation gives

$$\begin{aligned} Y_n &= \gamma + \varepsilon_n, \\ \varepsilon_n &= (n+1)E_1(2^n) - nE_1(2^{n+1}) \sim \frac{n+1}{2^n} e^{-2^n} > 0. \end{aligned}$$

Positivity follows from $E_1(2x) < E_1(x)$, and the asymptotic follows from $E_1(x) \sim e^{-x}/x$. These conclusions still do not determine the arithmetic nature of the limit.

Proposition 17.6 (Matching algebraic limits)**.** *For each real algebraic a, let p_n be the nth prime and define*

$$Z_n = a + \frac{n+1}{2^n} \exp\left(-2^n + \frac{\sqrt{p_n}}{2^n}\right).$$

Then Z_n decreases strictly to a, $Z_n - a \sim (n+1)2^{-n}e^{-2^n}$, and every finite collection $e, Z_1, \ldots, Z_N$ is algebraically independent.

Proof. Bertrand's postulate implies $p_n \leq 2^n$, so the correction $\sqrt{p_n}/2^n$ tends to zero. The prefactors $(n+1)/2^n$ decrease strictly, and the difference of successive exponents is less than $-2^n + 1 < 0$. This proves the analytic assertions.

The algebraic numbers $1, -2^1 + \sqrt{p_1}/2^1, \ldots, -2^N + \sqrt{p_N}/2^N$ are linearly independent over $\mathbb{Q}$: apply the independent square-root sign changes in the multiquadratic field. Distinct monomials in their exponentials consequently have distinct algebraic exponents. Lindemann–Weierstrass makes those exponentials linearly independent over $\overline{\mathbb{Q}}$, ruling out any nonzero polynomial relation. Algebraic shifts and nonzero rational scalings preserve algebraic independence. □

This counterexample matches finite algebraic independence, positivity, monotonicity, and the leading error scale. It excludes the proposed inference from those properties alone, not a method using stronger arithmetic information about the actual Y_n. The related "critical slope" identity

$$\lim_{n\to\infty} \frac{-\log|P(Y_n)|}{2^n} = \mathrm{ord}_{X=\gamma} P \qquad (0 \neq P \in \mathbb{C}[X])$$

is just Taylor expansion at the limit; the same identity holds for Z_n and a. It supplies no independent exclusion of an algebraic limit.

17.10 Two arithmetic blocks and a fixed differential equation

The following reduction makes the connection constant explicit. Set

$$A(z) = e^z \operatorname{Ein}(z) = \sum_{n\geq 0} \frac{H_n}{n!} z^n, \qquad \widehat{D}(z) = \sum_{n\geq 0} (-1)^n n! z^{-n-1}.$$

The first is an E-function. The second is divergent for every fixed nonzero complex z, but has the positive-direction Borel–Laplace sum

$$D_0(z) = \int_0^\infty \frac{e^{-t}}{z+t}\, dt = e^z E_1(z), \qquad z > 0.$$

We use the mixed arithmetic-Gevrey framework of Fischler–Rivoal [45]; in particular, formal statements about its two parts are distinguished from statements about one analytic sum. For algebraic r define

$$\widehat{\Psi}_r = A - re^z - \widehat{D}.$$

Its positive sum satisfies the exact identity

$$\Psi_{r,0}(z) = e^z(\gamma - r + \log z), \qquad \Psi_{r,0}(1) = e(\gamma - r). \tag{17.13}$$

Since

$$A' = A + \frac{e^z - 1}{z}, \qquad D_0' = D_0 - \frac{1}{z},$$

all these sums satisfy

$$\Psi_{r,0}' = \Psi_{r,0} + \frac{e^z}{z}, \qquad zY'' + (1-2z)Y' + (z-1)Y = 0. \tag{17.14}$$

The second-order equation has local solution basis $e^z, e^z \log z$. The number $\gamma - r$ is a connection constant selecting a particular arithmetic solution; it is not a coefficient in this rational differential equation. Existence of the equation therefore proves no arithmetic property of that constant.

Proposition 17.7 (Formal division obstruction)**.** *For algebraic r, the formal expression $\widehat{\Psi}_r$ cannot be written as*

$$(z-1)\left(F(z) + \sum_{n\geq 1} b_n z^{-n}\right)$$

where F is an E-function and every b_n is algebraic. In particular, no such mixed arithmetic-Gevrey quotient exists.

Proof. The nonnegative powers would give

$$A(z) - re^z = (z-1)F(z) + b_1.$$

Since F is entire, evaluation at 1 gives $b_1 = e\eta - re$. This contradicts the algebraic independence of e, η: otherwise $XY - rX - b_1$ would be a nonzero algebraic polynomial relation between them. □

What remains unproved is the lifting implication

$$\Psi_{r,0}(1) = 0 \quad \Longrightarrow \quad \widehat{\Psi}_r = (z-1) \times \text{a mixed arithmetic-Gevrey expression.} \tag{17.15}$$

An analytic zero permits division on that analytic branch, but does not ensure the arithmetic coefficient and denominator conditions of the quotient. A general zero-removal assertion of this kind is Conjecture 3 of [45]. For the displayed family, requiring (17.15) for all algebraic r is already equivalent to γ being transcendental: one direction follows from Proposition 17.7 and (17.13); in

the other direction its antecedent never occurs. Thus this implication is not an independently proved lemma that lowers the difficulty of the target.

There is a concrete branch issue as well. Under the hypothetical choice $r = \gamma$, the quotient of the positive sum is $e^z \log z/(z-1)$, with a removable singularity at 1. Continuation around 0 adds $2\pi i e^z/(z-1)$, which has a pole at 1 on the new branch. This observation explains why one-branch removability must not be confused with a global arithmetic assertion; it is not itself a transcendence proof.

17.11 Disjoint direct ℓ-adic convergence domains

The natural two formal blocks do not admit simultaneous direct evaluation at a nonzero ℓ-adic input. This can be checked exactly, including their common boundary.

Proposition 17.8 (Complementary convergence domains)**.** *Fix a prime ℓ, normalize $v_\ell(\ell) = 1$, and fix an algebraic ℓ-adic embedding of r. For nonzero z in a complete extension, the respective convergence conditions are exactly*

$$\sum_{n\geq 0} \frac{H_n - r}{n!} z^n : \quad v_\ell(z) > \frac{1}{\ell - 1}, \qquad \widehat{D}(z) : \quad v_\ell(z) < \frac{1}{\ell - 1}.$$

Both series diverge on the boundary.

Proof. Legendre's formula and the elementary harmonic-number estimate give

$$v_\ell(n!) = \frac{n - s_\ell(n)}{\ell - 1}, \qquad v_\ell(H_n) \geq -\lfloor \log_\ell n \rfloor.$$

These imply convergence in the indicated open domains: the linear term in n dominates the logarithmic correction. To obtain the reverse statements and test the boundary, take $n = \ell^k$. The unique term of smallest valuation in H_{ℓ^k} is $1/\ell^k$, so $v_\ell(H_{\ell^k}) = -k$ and, for all sufficiently large k, $v_\ell(H_{\ell^k} - r) = -k$. When $v_\ell(z) = 1/(\ell - 1)$, the corresponding E-series term has valuation

$$-k - \frac{\ell^k - 1}{\ell - 1} + \frac{\ell^k}{\ell - 1} = -k + \frac{1}{\ell - 1}.$$

The negative-power factorial term has valuation

$$\frac{\ell^k - 1}{\ell - 1} - \frac{\ell^k + 1}{\ell - 1} = -\frac{2}{\ell - 1}.$$

Neither sequence of terms tends to zero. On the wrong side of either boundary, the same subsequence has a strictly negative linear trend, so convergence also fails there. □

This result rules out direct simultaneous evaluation of these two particular series. It does not rule out ℓ-adic analytic continuation, Frobenius structures, overconvergence, or another local construction. A local nonvanishing computation needs an independently proved comparison theorem before it can yield a conclusion about the complex connection constant γ.

17.12 The Euler exponential motive and an exact lifting equivalence

Fresán–Jossen construct an exponential motive $M(\gamma)$ and prove [46, Proposition 12.8.7] that

$$0 \longrightarrow \mathbb{Q}(0) \longrightarrow M(\gamma) \longrightarrow \mathbb{Q}(-1) \longrightarrow 0$$

is nonsplit, with $\mathrm{Hom}_{\mathrm{mot}}(M(\gamma), \mathbb{Q}(0)) = 0$. Their Corollary 12.8.8 identifies its motivic Galois group with $\mathbb{G}_a \rtimes \mathbb{G}_m$ and makes the algebraic independence of γ, π conditional on the exponential period conjecture. Geometric nonsplitting alone is not that conjecture.

The arithmetic content of a proposed lift can be measured directly. Consider the unfiltered de Rham–Betti comparison category, with a $\overline{\mathbb{Q}}$ de Rham space, a rational Betti space, and the comparison matrix

$$P_\gamma = \begin{pmatrix} 1 & \gamma \\ 0 & 2\pi i \end{pmatrix}, \qquad \mathrm{comp}(d_0) = b_0, \quad \mathrm{comp}(d_1) = \gamma b_0 + 2\pi i b_1.$$

A compatible morphism to $\mathbb{Q}(0)$ has rational Betti row (a, b) and de Rham row $(a, a\gamma + 2\pi i b)$. It exists exactly when the second entry is algebraic. Since γ is real, subtracting the complex conjugate of that entry forces $b = 0$ by the transcendence of π. A nonzero map then has $a \neq 0$, forcing γ algebraic. Conversely, algebraic γ gives the compatible rows $(1, 0)$ and $(1, \gamma)$. Hence

$$\boxed{\mathrm{Hom}_{\mathrm{dR,B}}(M(\gamma), \mathbb{Q}(0)) = 0 \quad \Longleftrightarrow \quad \gamma \text{ is transcendental.}} \tag{17.16}$$

Because the motivic Hom space is zero, surjectivity of the single map

$$\mathrm{Hom}_{\mathrm{mot}}(M(\gamma), \mathbb{Q}(0)) \longrightarrow \mathrm{Hom}_{\mathrm{dR,B}}(M(\gamma), \mathbb{Q}(0))$$

is already equivalent to the desired transcendence.

Equivalently, for algebraic r the line $\overline{\mathbb{Q}}(d_1 - r d_0)$ compares to $\mathbb{C}b_1$ exactly when $r = \gamma$. If this pair lifted to a submotive, it would split the extension. Nonsplitting excludes a motivic complement, not every numerically compatible pair. Full exponential Hodge realization has additional filtration and sheaf data; a theorem about that realization must not be replaced by a claim about the bare numerical matrix. The ordinary 1-period theorems used earlier establish no such lift for this exponential motive.

17.13 Conditional closure and verified scope

Let $\mathcal{E}$ be the ring of values of E-functions at algebraic points and $\mathcal{D}$ the ring of the appropriate sums of arithmetic Gevrey series of order $+1$, with the definitions of [45]. The assertion

$$\mathcal{E} \cap \mathcal{D} = \overline{\mathbb{Q}} \tag{17.17}$$

is a conjecture. Its implication for γ is transparent: $\delta \in \mathcal{D}$, while $e, \eta \in \mathcal{E}$. If γ were algebraic, $\delta = e\eta - e\gamma$ would also belong to $\mathcal{E}$. Equation (17.17) would make δ algebraic, contradicting the algebraic independence of e, η. This is a conditional argument, not a proof of the intersection assertion. Fischler–Rivoal's Theorem 2 already gives the broader conditional transcendence of $\Gamma^{(m)}(a)$ at positive rational a, except the factorial case $m = 0$, a a positive integer.

Results about infinitely many Gamma derivatives do not select $-\Gamma'(1)$. For example,

$$\Gamma''(1) = \gamma^2 + \frac{\pi^2}{6}$$

would be transcendental even under the hypothesis that γ is algebraic. The 2026 derivative results of Powers [47] are therefore not treated as a solution of this individual target and are not assumptions in the proofs above.

Exact verification of the degree argument checked the kernel derivative, both Jacobi norm formulas and endpoint extremizers for $m = 0, \ldots, 5$, and the rational denominators and brackets in (17.11). The separate arithmetic-block audit checked the polynomial identities for 58 weights, 81 convolution coefficients of A, 81 first-order coefficients, and 484 second-order Laurent coefficient identities over four rational choices of r. Boundary valuations were checked for primes $2, 3, 5, 7$ and $k = 1, 2, 3, 4$. Floating-point integral comparisons are diagnostic checks, not interval certificates. The all-degree and all-prime conclusions rely on the proofs, not these finite tests.

The completed conclusions are the explicit Gamma-value reduction, the antisymmetric iterated-integral limit, balanced and weighted cancellations, the positive-polynomial degree bound, the multiscale independence consequence, the matching algebraic-limit counterexample, the formal division obstruction, the disjoint direct ℓ-adic domains, and the exact comparison lifting equivalence. The missing assertions are arithmetic control of the Gamma product, a sufficiently strong new approximation mechanism, or an applicable exponential-period relation theorem. None is replaced here by an unproved assumption disguised as a lemma.

18 Concluding status and reproducibility

The positive results above are specific transcendence statements and classifications obtained from cited arithmetic theorems and explicit calculations. The Euler-constant chapter has a different outcome: it identifies exact cancellations, counterexamples to proposed limit arguments, and precise arithmetic conditions still missing from a proof. Changing the name of a condition equivalent to Euler transcendence does not turn it into an established intermediate theorem.

The finite catalogues, polynomial identities, Frobenius polynomials, cyclotomic calculations, and proof certificates needed to identify the asserted cases are incorporated into this document. Decimal values are orientation data, not evidence for transcendence. Printed algorithms are finite exact procedures; where their results enter a proof, the required inputs and arithmetic tests are specified. Independent internal checking does not constitute a formal proof-assistant verification or external peer review.

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